

Module 2: Physical Chemistry

- **Mole Concept & Eudiometry**
- **Concentration Terms**
- **Redox Reactions and Equivalent Concepts**

MOLE CONCEPT

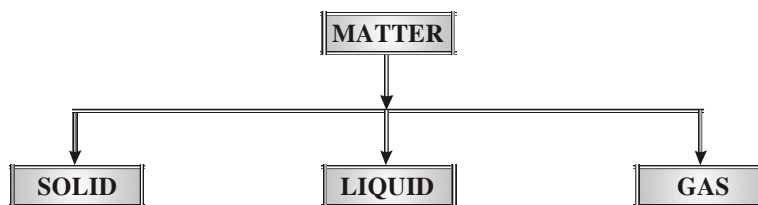
1. **MATTER** : Matter is anything that has mass and occupies space.

Two ways of classifying matter :

I. Physical classification

II. Chemical classification

1.1 Physical classification :

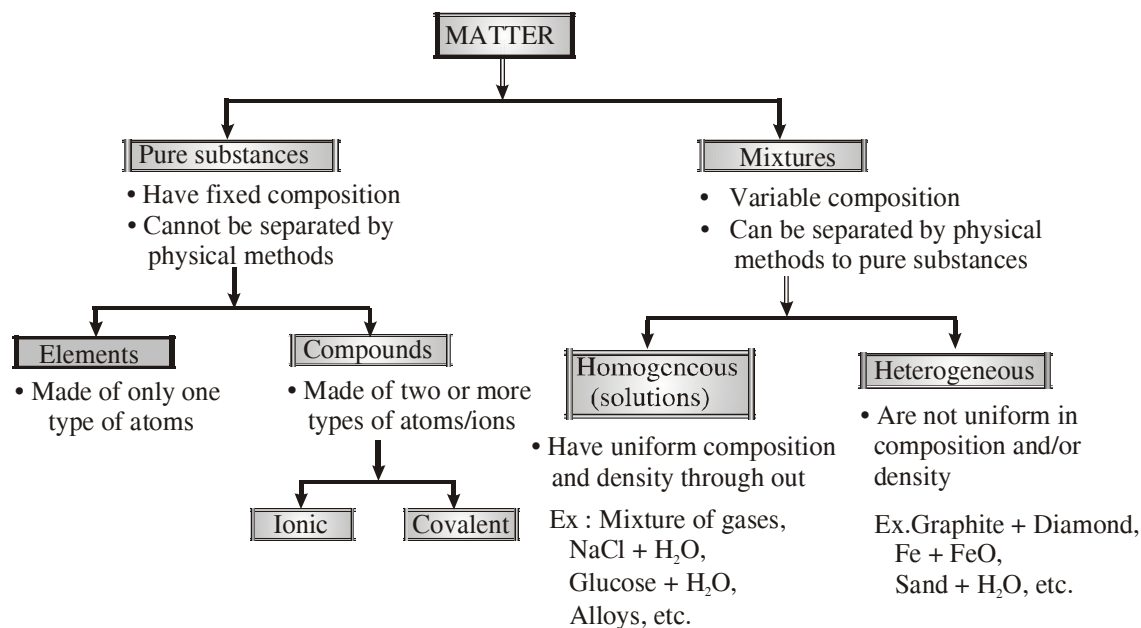


- | | | |
|--|---|--|
| <p>(i) Particles held very closely packed in ordered manner.</p> | <p>Particles are less closely packed.</p> | <p>Particles are farthest apart</p> |
| <p>(ii) No freedom of movement of particles</p> | <p>Particles can move around to some extent</p> | <p>Movement of particles is very easy and fast</p> |
| <p>(iii) Definite shape and volume</p> | <p>Definite volume, indefinite shape</p> | <p>indefinite shape and volume</p> |
| <p>(iv) Exists at low T and high P</p> | <p>Exists at intermediate P & T</p> | <p>Exists at high T and low P</p> |

Note : For same substance :

- Solid and Liquid co-exist at **MELTING POINT**.
- Liquid and gas co-exist at **BOILING POINT**.
- Solid and gas co-exist at **SUBLIMATION POINT**.
- Solid, liquid and gas co-exist at **TRIPLE POINT**.

1.2 Chemical classification :



- Note**
- **PHASE :** It is the state of matter uniform in density and composition.
 - Homogeneous mixtures have single phase while heterogeneous mixtures are multi-phase.
- Ex :** $\text{NaCl} + \text{H}_2\text{O}$ mixture has one phase
Ex : Graphite + Diamond mixture has 2 phases.
-

2. SOME SPECIFIC PROPERTIES OF SUBSTANCES

2.1 Deliquescence :

The property of certain compounds of taking up the moisture present in atmosphere and becoming wet when exposed, is known as deliquescence. These compounds are known as deliquescent. Sodium hydroxide, potassium hydroxide, anhydrous calcium chloride, anhydrous magnesium chloride, anhydrous ferric chloride, etc., are the examples of deliquescent compounds.

2.2 Hygroscopicity :

Certain compounds combine with the moisture of atmosphere and are converted into hydroxides or hydrates. Such substances are called hygroscopic. Anhydrous copper sulphate, quick lime (CaO), anhydrous sodium carbonate, etc., are of hygroscopic nature.

2.3 Efflorescence :

The property of some crystalline substances of losing their water of crystallisation on exposure and becoming powdery on the surface is called efflorescence and such salts are known as efflorescent. The examples are : Ferrous sulphate ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$), sodium carbonate ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$), sodium sulphate ($\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$), potash alum [$\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$], etc.

2.4 Effervescence :

Effervescence is the escape of gas from an aqueous solution and the foaming or fizzing that results from that release. Effervescence can be observed when opening a bottle of carbonated soft drinks. The visible bubbles are produced by the escape from solution of the dissolved gas (which itself is not visible while dissolved in the liquid.)

2.5 Malleability :

This property is shown by metals. When metallic solid is being beaten, it does not break but is converted into thin sheet. It is said to possess the property of malleability. Copper, gold, silver, aluminium, lead, etc., can be easily hammered into sheets. Gold is the most malleable metal.

2.6 Ductility :

The property of metal to be drawn into wires is termed ductility. Copper, silver, gold, aluminium, iron, etc., are ductile in nature. Platinum is the most ductile metal.

2.7 Brittleness :

The solid materials which break into small pieces on hammering are called brittle. The solids of non-metals and ionic solid are generally brittle in nature.

Ex: Ice, Diamond etc.

3. THE LAW OF CHEMICAL COMBINATION

Antoine Lavoisier, John Dalton and other scientists formulated certain laws concerning the composition of matter and chemical reactions. These laws are known as the law of chemical combination.

3.1 Law of indestructibility of matter or conservation of Mass :

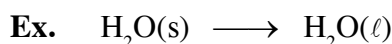
- This law was proposed by *Lavoisier in 1789*.
- In all physical or chemical changes, the total mass of the system remains constant. In a physical or chemical change, mass is neither created nor destroyed. Thus, in a physical or chemical change.



Antoine Lavoisier
(1743-1794)

Antoine-Laurent de Lavoisier, the "father of modern chemistry," was a French nobleman prominent in the histories of chemistry and biology. He named both oxygen and hydrogen and predicted silicon.

Total mass of reactant reacted = Total mass of products formed



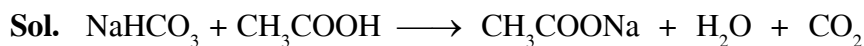
Above reaction shows the physical change and the mass of $\text{H}_2\text{O (s)} = \text{mass of } (\text{H}_2\text{O})(\text{l})$

In case the reacting materials are not completely consumed, the relationship will be.

Total masses of reactants = Total masses of product + masses of unreacted reactants

Note : In nuclear reactions (Mass + energy) is conserved, not the mass separately.

Ex.1 When 4.2 g NaHCO_3 is added to a solution of CH_3COOH weighing 10.0 g, it is observed that 2.2 g CO_2 is released into atmosphere. The residue is found to weigh 12.0 g. Show that these observations are in agreement with the law of conservation of mass.



Initial total mass = 4.2 + 10 = 14.2

Final total mass = 12 + 2.2 = 14.2

Thus, during the course of reaction, law of conservation of mass is obeyed.

3.2. Law of constant or definite proportion :

- This law was given by **Joseph Louis Proust** in 1799.
- Chemical composition of a compound remains constant whether it is obtained by any method or any source.



Joseph Proust
(1754 - 1826)

Proust was born the son of an apothecary at Angers in north-west France. He studied in Paris. He lived in poverty for some years before being awarded a pension by Louis XVIII.

♦ **Example :**

In water (H_2O), Hydrogen and Oxygen combine in 1 : 8 mass ratio, the ratio remains constant whether it is tap water, river water or sea water or produced by any chemical reaction.

Note: This law is not applicable in case of isotopes.

Ex.2 1.80 g of a certain metal burnt in oxygen gave 3.0 g of its oxide. 1.50 g of the same metal heated in steam gave 2.50 g of its oxide. Show that these results illustrate the law of constant proportion.

Sol. In the first sample of the oxide,

$$\text{wt. of metal} = 1.80 \text{ g}, \quad \text{wt. of oxygen} = (3.0 - 1.80) \text{ g} = 1.2 \text{ g}$$

$$\therefore \frac{\text{wt. of metal}}{\text{wt. of oxygen}} = \frac{1.80 \text{ g}}{1.2 \text{ g}} = 1.5$$

In the second sample of the oxide,

$$\text{wt. of metal} = 1.50 \text{ g}, \quad \text{wt. of oxygen} = (2.50 - 1.50) \text{ g} = 1 \text{ g}$$

$$\therefore \frac{\text{wt. of metal}}{\text{wt. of oxygen}} = \frac{1.50 \text{ g}}{1 \text{ g}} = 1.5$$

Thus, in both samples of the oxide the proportions of the weights of the metal and oxygen are fixed. Hence, the results follow the law of constant proportion.

3.3. The law of multiple proportion :

- This law was given by Dalton in 1803.
- If two elements combine to form more than one compound, then the different masses of one element which combine with a fixed mass of the other element, bear a simple ratio to one another.

Ex. Nitrogen and oxygen combine to form the following stable oxides –

N_2O	Nitrogen 28 parts	Oxygen 16 parts
N_2O_2	Nitrogen 28 parts	Oxygen 32 parts
N_2O_3	Nitrogen 28 parts	Oxygen 48 parts
N_2O_4	Nitrogen 28 parts	Oxygen 64 parts
N_2O_5	Nitrogen 28 parts	Oxygen 80 parts

The masses of oxygen which combine with same mass of nitrogen in the five compounds bear a ratio 16 : 32 : 48 : 64 : 80 or 1 : 2 : 3 : 4 : 5.

Note: This law is not applicable in case of isotopes.

Ex.3 Carbon forms two oxides. One contains 27.27 % carbon & another contains 42.86% carbon. Show that the data illustrate the law of multiple proportion.

Sol. The mass ratio of C : O in first oxide = 27.27 : 72.73 = 3 : 8

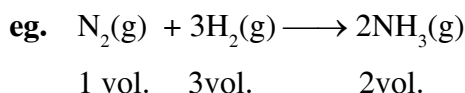
The mass ratio of C : O in second oxide = 42.86 : 57.14 = 3 : 4

Hence for each 3 gm of carbon the masses of oxygen combined is in 2 : 1 ratio.

Hence the data is according the law of multiple proportion.

3.4 Law of Gaseous volumes :

- This law was given by **Gay-Lussac**. in 1808.
- According to this law, gases react with each other in the simple ratio of their volumes and if products are also gaseous then they are also formed in simple ratio of volume provided that all volumes are measured at same temp. & pressure.

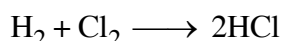


Joseph Louis
Gay Lussac
(1778 - 1850)

Joseph Louis Gay-Lussac also ; 6 December 1778 – 9 May 1850) was a French chemist and physicist. He is known mostly for two laws related to gases, and for his work on alcohol-water mixtures , which led to the degrees Gay-Lussac used to measure alcoholic beverages in many countries.

Ex.4 For the gaseous reaction , $\text{H}_2 + \text{Cl}_2 \longrightarrow 2\text{HCl}$. If 40 ml of hydrogen completely reacts with chlorine then find out the required volume of chlorine and volume of produced HCl ?

Sol. According to Gay Lussac's Law :



∴ 1 ml of H_2 will react with 1 ml of Cl_2 and 2 ml of HCl will be produced.

∴ 40 ml of H_2 will react with 40 ml of Cl_2 and 80 ml of HCl will be produced.

required vol. of Cl_2 = 40 ml, produced vol. of HCl = 80 ml

3.5. Avogadro's law (1811) :

Equal volumes of all gases under similar conditions of temperature and pressure contain equal number of molecules.



Lorenzo Romano
Amedeo Carlo
Avogadro di
Quarega ed i
Carreto
(1776–1856)

4. DALTON'S ATOMIC THEORY

Ancient Indian and Greek philosophers have always wondered about the unknown and unseen form of matter. The idea of divisibility of matter was considered long back in India, around 500 BC. An Indian philosopher **Maharishi Kanad**, postulated that if we go on dividing matter (padarth), we shall get smaller and smaller particles. Ultimately, a time will come when we shall come across the smallest particle beyond which further division will not be possible. He named these particles **Parmanu**. Another Indian philosopher, Pakudha Katayama, elaborated this doctrine and said that these particles normally exist in a combined form which gives us various forms of matter. Around the same era, the Greek philosopher Democritus expressed the belief that all matter consists of very small, indivisible particles, which he named **atomos** (meaning uncuttable or indivisible).



John Dalton (1766 - 1844) , an Englishman, began teaching at a Quaker school when he was 12. His fascination with science included an intense interest in meteorology (he kept careful daily weather records for 46 years), which led to an interest in the gases of the air and their ultimate components, atom. Dalton is best known for his atomic theory, in which he postulated that the fundamental differences among atoms are their masses. He was the first to prepare a table of relative atomic weight.

Although Democritus' ideal was not accepted by many of his contemporaries (notably Plato and Aristotle), some how it endured. Experimental evidence from early scientific investigations provided support for the notion of "atomism" and gradually gave rise to the modern definitions of elements and compounds. It was in **1808**, **John Dalton**, formulated a precise definition of the indivisible building blocks of matter that we call atoms. Dalton's work marked the beginning of the modern era of chemistry. The hypotheses about the nature of matter on which Dalton's atomic theory is based can be summarized as follows :

- (i) Matter consists of indivisible atoms.
- (ii) All atoms of a given element are identical, having the same size, mass and chemical properties. The atoms of one element are different from the atoms of all other elements.
- (iii) Compounds are formed when atoms of different elements combine in a fixed ratio.
- (iv) A chemical reaction involves only the separation, combination or rearrangement of atoms. It does not result in their creation or destruction.

5. ATOMIC AND MOELCULAR MASSES :

The mass of an atom depends on the number of electrons, protons, and neutrons it contains. Knowledge of an atom's mass is important in laboratory work. But atoms are extremely small particles - even the smallest speck of dust that our unaided eyes can detect contains as many as 1×10^{16} atoms ! Clearly we cannot weigh a single atom, but it is possible to determine the mass of one atom relative to another experimentally. The first step is to assign a value to the mass of one atom of a given element so that it can be used as a standard.

5.1 RELATIVE ATOMIC MASS :

Hydrogen, being lightest atom was arbitrarily assigned a mass of 1 (without any units) and other elements were assigned masses relative to it. However, the present system of atomic masses is based on carbon - 12 as the standard and has been agreed upon in 1961. Here, Carbon - 12 is one of the isotopes of carbon and can be represented as ^{12}C . In this system, ^{12}C is assigned a mass of exactly 12 atomic mass unit (**amu**) and masses of all other atoms are given relative to this standard.

Relative Atomic Mass is defined as the number which indicates how many times the mass of one atom of an element is heavier in comparison to 1/12th part of the mass of one atom of C-12.

$$\begin{aligned}\text{Relative atomic mass of an element} &= \frac{\text{mass of one atom of an element}}{\frac{1}{12}[\text{mass of one C-12 atom}]} \\ &= \frac{\text{Mass of one atom of an element}}{1 \text{ amu}}\end{aligned}$$

5.1.1 ATOMIC MASS UNIT (a.m.u. or u) : The quantity 1/12th mass of an atom of C^{12} is known as atomic mass unit.

Since mass of 1 atom of C - 12 = $1.992648 \times 10^{-23} \text{ g}$

$$\therefore 1/12^{\text{th}} \text{ part of the mass of 1 atom} = \frac{1.992648 \times 10^{-23} \text{ g}}{12} = 1.67 \times 10^{-24} \text{ g} = \frac{1}{6.022 \times 10^{23}} \text{ g}$$

It may be noted that the atomic masses as obtained above are the relative atomic masses and not the actual masses of the atoms. These masses on the atomic mass scale are expressed in terms of atomic mass units (abbreviated as amu). Today, 'amu' has been replaced by 'u' which is known as **unified mass**.

5.1.2 GRAM ATOMIC MASS OR MASS OF 1 G-ATOM :

When numerical value of atomic mass of an element is expressed in grams then the value becomes gram atomic mass or GAM.

gram atomic mass (GAM) = mass of 1 **g-atom** = mass of 1 **mole atoms**

$$= \text{mass of } N_A \text{ atoms} = \text{mass of } 6.022 \times 10^{23} \text{ atoms.}$$

Ex. GAM of oxygen = mass of 1 **g-atom** of oxygen = mass of 1 **mol atoms** of oxygen.

$$= \text{mass of } N_A \text{ atoms of oxygen} = \left(\frac{16}{N_A} \text{ g} \right) \times N_A = 16 \text{ g}$$

Ex. Mass of one atom of Oxygen = 16 amu or $16 \times 1.66 \times 10^{-24} \text{ g}$

$$\text{Mass of } N_A \text{ atoms of Oxygen} = 16 \times 1.66 \times 10^{-24} \times 6.022 \times 10^{23} \text{ g} = 16 \text{ g}$$

Now see the table given below and understand the definition given before.

Element	R.A.M. (Relative Atomic Mass)	Atomic mass (mass of one atom)	Gram Atomic mass or weight
N	14	14 amu	14 gm
He	4	4 amu	4 gm
C	12	12 amu	12 gm

5.1.3 AVERAGE ATOMIC MASS :

If an element exists in different isotopic forms having relative abundance $X_1\%$, $X_2\%$ $X_n\%$, with relative atomic masses M_1 , M_2 M_n respectively then ,

$$\text{Avg. Atomic mass of element} = \frac{X_1}{100}(M_1) + \frac{X_2}{100}(M_2) + \dots + \frac{X_n}{100}(M_n) = \sum_{i=1 \text{ to } n} \frac{X_i}{100}(M_i)$$

Ex.5 The atomic mass of an element is 50

(i) Calculate the mass of one atom, in amu

(ii) Calculate the mass of 6.022×10^{23} atoms, in gm

(iii) Calculate the number of atoms in its 10 gm

(iv) What mass of the element contains 3.011×10^{20} atoms

Sol. (i) 50 amu

(ii) 50 gm

(iii) $\therefore$ 50 gm of element contains 6.022×10^{23} atoms

$\therefore$ 10 gm of element will contain $\frac{6.022 \times 10^{23}}{50} \times 10 = 1.2044 \times 10^{22}$ atoms

(iv) $\therefore$ 6.022×10^{23} atoms weighs 50 gm

$\therefore$ 3.011×10^{20} atoms weighs $\frac{50}{6.022 \times 10^{23}} \times 3.011 \times 10^{20} = 0.025$ gm

Ex.6 An element exist in nature in two isotopic forms : X^{30} (90%) and X^{32} (10%). What is the average atomic mass of element ?

Sol. Av. atomic mass = $\frac{\Sigma(\% \text{abundance} \times \text{atomic mass})}{100} = \frac{90 \times 30 + 10 \times 32}{100} = 30.2$

5.2 RELATIVE MOLECULAR MASS :

The number which indicates how many times the mass of one molecule of a substance is heavier in comparison to 1/12th part of the mass of an atom of C-12.

OR

The molecular mass of a substance is the sum of atomic masses of all the atoms present in a molecule. It is obtained by multiplying the atomic mass of each element by the number of its atoms and adding them together.

Ex.	molecular mass of oxygen (O_2)	=	32
	molecular mass of (O_3)	=	48
	molecular mass of HCl	=	$1 + 35.5 = 36.5$
	molecular mass of H_2SO_4	=	$2 + 32 + 64 = 98$

5.2.1 GRAM MOLECULAR MASS (MASS OF 1 G-MOLECULE) :

When numerical value of molecular mass of the substance is expressed in grams then the value becomes gram molecular mass or GMM.

$$\begin{aligned} \text{gram molecular mass (GMM)} &= \text{mass of 1 g-molecule} = \text{mass of 1 mole molecules} \\ &= \text{mass of } N_A \text{ molecules} = \text{mass of } 6.022 \times 10^{23} \text{ molecules} \end{aligned}$$

Ex. Molecular Mass of $N_2 = 28 \text{ amu} = 28 \times 1.66 \times 10^{-28} \text{ g}$
 Mass of N_A molecules of $N_2 = 28 \times 1.66 \times 10^{-24} \times 6.022 \times 10^{23} \text{ g} = 28 \text{ g}$

$$M_{\text{avg.}} = \frac{\text{Total mass of mixture}}{\text{Total mole}} = \frac{\Sigma(\% \text{vol} \times \text{molecular mass})}{100} = \frac{100}{\Sigma \left(\frac{\% \text{mass}}{\text{molecular mass}} \right)}$$

(iv) *How many molecules are in its 2.5 mg*

$$\therefore 2.5 \times 10^{-3} \text{ gm will contain } \frac{6.022 \times 10^{23}}{75} \times 2.5 \times 10^{-3} = 2.007 \times 10^{19} \text{ molecules.}$$

Sol. $M_{av} = \frac{\Sigma(\% \text{ by vol.} \times \text{molecular mass})}{100} = \frac{40 \times 2 + 60 \times 4}{100} = 3.20$

6 INTRODUCTION TO MOLE

Atoms and molecules are extremely small in size and their numbers in even a small amount of any substance is really very large. To handle such large numbers, a unit of similar magnitude is required. The 14th Geneva conference on weight and measures adopted **mole** as a *seventh basic SI unit of the amount of a substance*. Mole concept is essential tool for the fundamental study of chemical calculations. This concept is simple but its application requires a thorough practice. There are many ways of measuring the amount of substance, weight and volume being the most common, but basic unit of chemistry is the atom or a molecule and measuring the number of molecule is more important.

6.1 DEFINITION OF MOLE AND MOLAR MASS :

- A mole is the amount of a substance that contains as many entities (Atoms, Molecules, Ions or any other particles) as there are atoms in exactly 12 g of C-12 isotope.
- A mole of a substance contains Avogadro's number (6.022×10^{23}) of particles.

The term mole, like a dozen or a gross, thus refers to a particular number of things. A dozen eggs equals 12 eggs, a gross of pencils equals 144 pencils, and a mole of ethanol equal 6.022×10^{23} ethanol molecules.

- The **molar mass** of a substance is the mass of one mole of the substance. Carbon-12 has a molar mass of exactly 12 g/mol, by definition.
- 1 g-atom = 1 mole atoms = N_A atoms
- 1 g-molecule = 1 mole molecules = N_A molecules
- 1 g-ion = 1 mole ions = N_A ions

6.2 Methods to calculate moles :

- (i) If number of particles (molecules or atoms) is given then,

$$\text{mole} = \frac{\text{Given number of molecule / atom}}{N_A}$$

- (ii) If mass is given then, number of mole = $\frac{\text{Given mass of substance (in gm)}}{\text{GAM / GMM}}$

- (iii) If volume of gas is given then, mole

$$= \frac{\text{Volume of gas at STP}}{22.7 \text{ L}} = \frac{\text{Volume of gas at } 0^\circ\text{C and } 1 \text{ atm}}{22.4 \text{ L}}$$

* Standard molar volume is the volume occupied by 1 mole of an ideal gas at STP (Standard temperature and pressure which is 273.15 K & 1 bar respectively), which is equal to 22.7 L.

* 1 mole of an ideal gas occupy 22.4 L at 0°C and 1 atm.

- (iv) Under any condition of temperature and pressure, moles of gases may be calculated using **IDEAL GAS EQUATION :**

$$\boxed{PV = nRT}$$

where, R = Universal Gas Constant

$$= 0.082 \text{ L-atm/K-mol}$$

$$= 8.314 \text{ J/K-mol}$$

$$\approx 2 \text{ cal/K-mol}$$

Units of pressure and their relation:

$$1 \text{ atm} = 76 \text{ cm Hg}$$

$$= 760 \text{ mm Hg}$$

$$= 760 \text{ torr} \quad (1 \text{ torr} = 1 \text{ mm Hg})$$

$$= 1.01325 \times 10^6 \text{ dyne/cm}^2$$

$$= 1.01325 \times 10^5 \text{ N/m}^2 \text{ or Pa}$$

$$= 1.01325 \text{ bar} \quad (1 \text{ bar} = 10^5 \text{ Pa})$$

$$1 \text{ bar} = 75 \text{ cm Hg}$$

Units of Volume and their relation:

$$1 \text{ ml} = 1 \text{ cm}^3 = 1 \text{ c.c.}$$

$$1 \text{ Litre} = 1000 \text{ ml} = 1 \text{ dm}^3$$

$$1 \text{ m}^3 = 1000 \text{ L}$$

Units of Temperature and their relation:

$$T = 273.15 + t$$

where, T = Absolute temperature (in Kelvin) and t = temperature in °C

(Normally, we take 273K in calculation)

- (v) Sometimes gas is collected over water. In this case, the measured pressure is sum of pressure of gas and the vapour pressure of water (also called Aqueous Tension). In order to calculate moles of gas, the vapour pressure of water should be deducted from the measured pressure.

Ex.9 Calculate the number of g-molecules (mole of molecules) in the following : $[N_A = 6 \times 10^{23}]$

(i) 3.2 gm CH_4

(ii) 70 gm nitrogen

(iii) 4.5×10^{24} molecules of ozone

(iv) 2.4×10^{21} atoms of hydrogen

(v) 11.2 L ideal gas at 0°C and 1 atm

(vi) 4.54 ml SO_3 gas at STP

(vii) 8.21 L C_2H_6 gas at 400K and 2atm

(viii) 164.2 ml He gas at 27°C and 570 torr

Sol. (i) 3.2 gram CH_4

$$\text{number of moles } (\text{CH}_4) = \frac{w}{M} = \frac{3.2}{16} = 0.2 \text{ moles}$$

(ii) 70 gram N_2

$$\text{Number of moles} = \frac{w}{M} = \frac{70}{28} = 2.5$$

(iii) 4.5×10^{24} molecules of O_3

$$\text{Number of moles} = \frac{\text{no. of molecules}}{N_A} = \frac{4.5 \times 10^{24}}{6 \times 10^{23}} = 7.5$$

(iv) 2.4×10^{21} atoms of hydrogen

$$\text{Number of gram molecules of } H_2 = \frac{\text{no. of molecules}}{N_A} = \frac{2.4 \times 10^{21}}{2 \times 6 \times 10^{23}} = 0.002$$

(v) 11.2 litre ideal gas at 0°C and 1 atm

$$\text{Number of moles} = \frac{\text{Volume at } 0^\circ\text{C \& 1 atm}}{22.4 \text{ litre}} = \frac{11.2}{22.4} = 0.5$$

(vi) 4.54 ml SO_3 gas at STP

$$\text{Number of moles} = \frac{V_{\text{STP}}(\text{ml})}{22700 \text{ ml}} = \frac{4.54}{22700} = 2 \times 10^{-4}$$

(vii) 8.21 litre C_2H_6 at 400 K and 2 litre

$$n = \frac{PV}{R.T} = \frac{2 \times 8.21}{0.0821 \times 400} = 0.5$$

(viii) 164.2 ml He gas at 27°C and 570 torr

$$n = \frac{PV}{RT} = \left(\frac{570}{760} \text{ atm} \right) \times \frac{164.2 \times 10^{-3} \text{ litre}}{0.0821 \times 300} = 0.005$$

Ex.10 Find no. of protons in 180 ml H_2O . Density of water = 1 gm/ml.

Sol. Mass of water = density $\times$ volume = 180 g

$$\text{Moles of water} = \frac{180}{18} = 10$$

1 mol water has 10 mol protons

$$\begin{aligned} \therefore 10 \text{ mol water has } 100 \text{ mol protons} &= 100 \times 6.022 \times 10^{23} \text{ protons} \\ &= 6.022 \times 10^{25} \text{ protons.} \end{aligned}$$

Ex.11 What mass of $\text{Na}_2\text{SO}_4 \cdot 7\text{H}_2\text{O}$ contains exactly 6.022×10^{22} atoms of oxygen ?

Sol. Molar mass of $\text{Na}_2\text{SO}_4 \cdot 7\text{H}_2\text{O} = 268 \text{ gm}$.

1 mole $\text{Na}_2\text{SO}_4 \cdot 7\text{H}_2\text{O}$ has 11 mol O-atoms.

$\Rightarrow 11 N_A$ O – atoms are in 268 g $\text{Na}_2\text{SO}_4 \cdot 7\text{H}_2\text{O}$

$\Rightarrow 6.022 \times 10^{22}$ O – atoms are in $= \frac{268}{11 \times 6.022 \times 10^{23}} \times 6.022 \times 10^{22} \text{ g} = 2.44 \text{ g}$

Ex.12 What is number of atoms and molecules in 112 L of $\text{O}_3(\text{g})$ at 0°C and 1atm ?

Sol. Moles of molecules $= \frac{112}{22.4} = 5$

Moles of atoms $= 5 \times 3 = 15$

No. of molecules $= 5 N_A$

No. of atoms $= 15N_A$.

7 DENSITY :

It is of two types.

I. Absolute density

II. Relative density

7.1 For liquids and solids :

Absolute density $= \frac{\text{mass}}{\text{volume}}$

Relative density or specific gravity $= \frac{\text{density of the substance}}{\text{density of water at } 4^\circ\text{C} (1\text{gmml}^{-1})}$

7.2 For gases :

Absolute density $= \frac{\text{mass}}{\text{volume}} = \frac{PM}{RT}$

where P is pressure of gas, M = molar mass of gas, R is the gas constant, T is absolute temperature.

Vapour Density :

Vapour density is defined as the density of the gas with respect to hydrogen gas at the same temperature and pressure.

Vapour density $= \frac{d_{\text{gas}}}{d_{\text{H}_2}} = \frac{PM_{\text{gas}}/RT}{PM_{\text{H}_2}/RT}$

$V.D. = \frac{M_{\text{gas}}}{M_{\text{H}_2}} = \frac{M_{\text{gas}}}{2} \Rightarrow \boxed{M_{\text{gas}} = 2 \times V.D.}$

Ex.13 A gaseous mixture of H_2 and NH_3 gas contains 68 mass % of NH_3 . The vapour density of the mixture is –

Sol. No. of moles of NH_3 in 100g mixture $= \frac{68}{17} = 4$

No. of moles of H_2 in 100g mixture $= \frac{32}{2} = 16$

$M_{\text{average}} = \frac{\text{Total mass}}{\text{Total moles}} = \frac{100}{4 + 16} = 5$

$V.D. = \frac{5}{2} = 2.5$

8. STOICHIOMETRY

Stoichiometry is the calculation of amounts of reactants and products involved in a reaction. Stoichiometric calculations require a balanced chemical equation of the reaction.

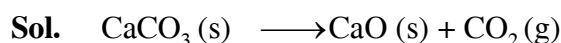
A balanced chemical equation is one which contains an equal number of atoms of each element on both sides of equation.

8.1 SIGNIFICANCE OF STOICHIOMETRIC COEFFICIENTS :

Stoichiometric coefficients of chemical equation tells us about the ratio in which moles of reactants react and moles of products form.

Ex.	$2\text{H}_2(\text{g})$	+	$\text{O}_2(\text{g})$	$\longrightarrow$	$2\text{H}_2\text{O}(\text{g})$
1 st interpretation	2 moles		1 mole		2 moles
2 nd interpretation	2N_A molecules		N_A molecules		2N_A molecules
3 rd interpretation	2 molecules		1 molecules		2 molecules

Ex.14 What mass of CaO is formed by heating 50 g CaCO_3 in air ?

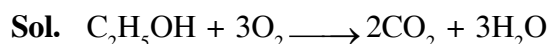


50 gm

$$= \frac{50}{100} \text{ mol}$$

$$= \frac{1}{2} \text{ mol} \quad \frac{1}{2} \text{ mol} = \frac{1}{2} \times 56 = 28 \text{ gm}$$

Ex.15 If 1 mole of ethanol ($\text{C}_2\text{H}_5\text{OH}$) completely burns to form carbon dioxide and water, mass of carbon dioxide formed is about

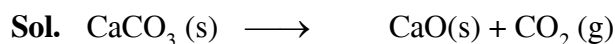


1 mole

2 mole

$$\therefore \text{mass of } \text{CO}_2 \text{ formed} = 2 \times 44 = 88 \text{ gm}$$

Ex.16. What volume of CO_2 at 0°C and 1 atm is formed by heating 200 g CaCO_3 ?



200 gm

$$= \frac{200}{100} \text{ mol} = 2 \text{ mol} \quad 2 \text{ mol}$$

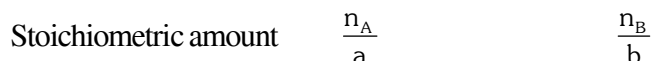
$$\text{Volume of gas at } 0^\circ\text{C and 1 atm} = \text{No. of moles} \times 22.4 \text{ L} = 2 \times 22.4 = 44.8 \text{ L.}$$

8.2 LIMITING REAGENT (L.R.) :

- (i) The reactant which is completely consumed when a reaction goes to completion is called **Limiting Reactant or Limiting reagent**.
- (ii) The reactant whose stoichiometrically taken amount is least, is limiting reactant.

$$\text{stoichiometrically taken amount} = \frac{\text{Given moles of reactant}}{\text{Stoichiometric coefficient of reactant in balance Reaction}}$$

- (iii) When amounts of two or more than two reactants are given :



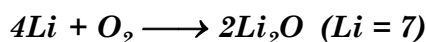
If $\frac{n_A}{a} < \frac{n_B}{b} \Rightarrow A \text{ is limiting reagent.}$

If $\frac{n_A}{a} = \frac{n_B}{b}$ then reaction occurs to completion & no reactant is left at the end.

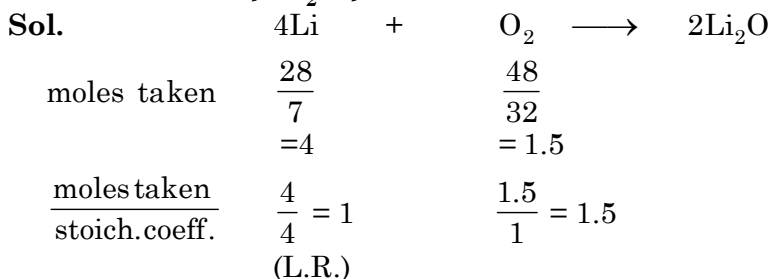
If $\frac{n_A}{a} > \frac{n_B}{b} \Rightarrow B \text{ is limiting reagent.}$

For calculation of moles of product, LR should be used.

Ex.17. 28 gm Lithium is mixed with 48 gm O₂ to reacts according to the following reaction.



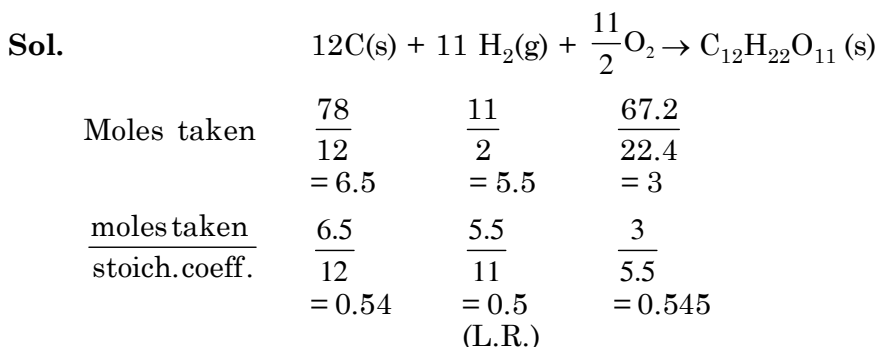
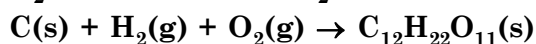
The mass of Li₂O formed is



$$\text{Moles of Li}_2\text{O formed} = \frac{2}{4} \times 4 = 2$$

$$\text{Mass of Li}_2\text{O formed} = 2 \times 30 = 60 \text{ gm}$$

Ex.18 Calculate the mass of sucrose C₁₂H₂₂O₁₁ (s) produced by mixing 78 g of C(s), 11 g of H₂(g) & 67.2 litre of O₂(g) at 0°C and 1 atm according to given reaction (unbalanced) ?



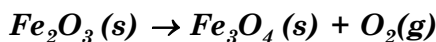
$$\therefore \text{Moles of C}_{12}\text{H}_{22}\text{O}_{11} \text{ formed} = \frac{5.5}{11} = 0.5$$

$$\text{Mass of sucrose obtained} = 0.5 \times 342 = 171 \text{ grams.}$$

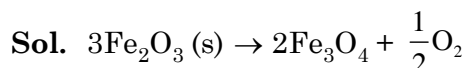
8.3 PROBLEMS BASED ON MIXTURE :

The composition of any mixture may be determined by reacting the mixture with some substance, by which either one or more component of mixture may react.

Ex.19 1.5 gm mixture of SiO_2 and Fe_2O_3 on very strong heating leave a residue weighing 1.46 gm. The reaction responsible for loss of weight is



What is the percentage by mass of Fe_2O_3 in original sample.



$$3 \times 160 \qquad \frac{1}{2} \times 32$$

$$= 480 \text{ gm} \qquad = 16 \text{ gm}$$

$$\text{loss of } 16 \text{ gm} \rightarrow 480 \text{ gm } \text{Fe}_2\text{O}_3$$

$$\text{loss of } 0.04 \text{ gm} \rightarrow 0.04 \times \frac{480}{16} = 1.2 \text{ gm } \text{Fe}_2\text{O}_3$$

$$\% \text{ by mass} = \frac{1.2}{1.5} \times 100 = 80\%$$

8.4 PERCENTAGE YIELD :

In general, when a reaction is carried out in the laboratory we do not obtain the theoretical amount of the product, in reality. The amount of the product that is actually obtained is called the **actual yield**. Knowing the actual yield and theoretical yield the percentage yield can be calculate as :

$$\% \text{ yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100$$

The percentage yield of any product is always equal to the percentage extent of that reaction.

Ex.20 Aluminium reacts with sulphur to form aluminium sulphide. If 5.4 gm of aluminium reacts with 12.8gm sulphur & gives 12gm of aluminium sulphides, then the percent yield of the reaction is. ($\text{Al} = 27, \text{S} = 32$)



$$\begin{array}{l} \text{Mole taken} \quad \frac{5.4}{27} \text{ gm} \quad \frac{12.8}{32} \text{ gm} \\ \qquad \qquad \qquad = 0.2 \quad \quad = 0.4 \end{array}$$

$$\begin{array}{l} \frac{\text{moles taken}}{\text{stoich. coeff.}} \quad \frac{0.2}{2} \quad \frac{0.4}{3} \\ \qquad \qquad \qquad = 0.1 \quad \quad = 0.133 \\ \qquad \qquad \qquad (\text{L.R.}) \end{array}$$

$$\text{moles of } \text{Al}_2\text{S}_3 \text{ formed} = \frac{1}{2} \times 0.2 = 0.1$$

$$\text{mass of } \text{Al}_2\text{S}_3 \text{ formed} = 0.1 \times 150 = 15 \text{ gm}$$

But, from equation, only 12 gm Al_2S_3 is formed.

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100 = \frac{12}{15} \times 100 = 80 \%$$

8.5 DEGREE OF DISSOCIATION, α :

It represents the mole of substance dissociated per mole of the substance taken.

$$A \rightarrow n \text{ particles; } \alpha = \frac{M_o - M}{(n - 1).M}$$

where, n = number of product particles per particle of reactant

M_o = Molar mass of 'A'

M = Molar mass of final mixture

Dissociation decreases the average molar mass of system while association increases it.

Same formula is applicable for association, taking the correct value of 'n'.

Ex.21 For the reaction $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$

Calculate degree of dissociation (α) if observed molar mass of mixture is 13.6

Sol. $M_o = M_{NH_3} = 17$, $M = 13.6$ (given), $n = \frac{4}{2} = 2$

$$\alpha = \frac{M_o - M}{(n - 1).M} = \frac{17 - 13.6}{(2 - 1) \times 13.6} = 0.25$$

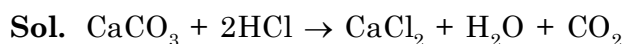
8.6 PERCENTAGE PURITY :

The percentage of a specified compound or element in an impure sample may be given as

$\% \text{purity} = \frac{\text{Actual mass of pure substance}}{\text{Total mass of sample}} \times 100$
--

If impurity is unknown, it is always considered as inert (unreactive) material.

Ex.22 A chalk sample exactly requires 17.52 gram HCl for complete reaction with all $CaCO_3$ present in it. If the chalk sample is 72% pure, the mass of sample taken is



$$\text{Moles of HCl} = \frac{17.52}{36.5}$$

$$\text{Moles of } CaCO_3 = \frac{1}{2} \times \frac{17.52}{36.5}$$

$$\text{Weight of } CaCO_3 \text{ required} = \frac{1}{2} \times \frac{17.52}{36.5} \times 100$$

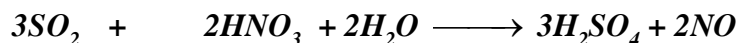
Mass of sample taken :

$$= \frac{1}{2} \times \frac{17.52}{36.5} \times \frac{100 \times 100}{72} = 33.33 \text{ gm}$$

8.7 PROBLEMS RELATED WITH SEQUENTIAL REACTION :

When one of products formed in previous reaction is consumed in the next one.

Ex.23 How many grams H_2SO_4 can be obtained from 1320 gm PbS as per reaction sequence ?



[At. mass : $Pb = 208$, $S = 32$]

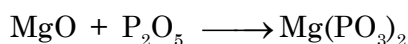
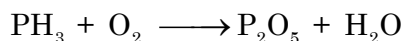
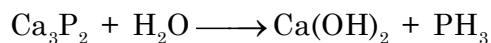
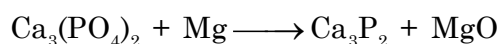
Sol. Moles of $PbS = \frac{1320}{240} = 5.5 \text{ mol}$

Moles of $SO_2 = 5.5 \text{ mol} = \text{moles of } H_2SO_4$

Mass of $H_2SO_4 = 5.5 \times 98 = 539 \text{ gm}$

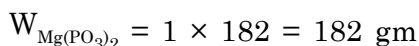
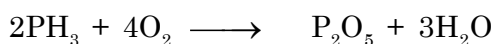
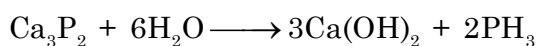
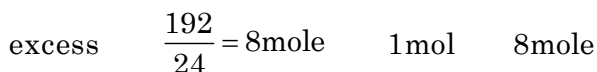
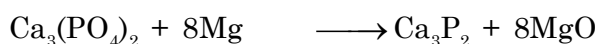
[When amount of only one reactant is given generally other is assumed in excess.]

Ex.24 Calcium phosphide Ca_3P_2 formed by reacting magnesium with excess calcium orthophosphate $Ca_3(PO_4)_2$ was hydrolysed by excess water. The evolved phosphine PH_3 was burnt in air to yield phosphorous pentoxide (P_2O_5). How many gram of magnesium metaphosphate would be obtain if 192 gram Mg were used (Atomic weight of $Mg = 24$, $P = 31$)



magnesium metaphosphate.

Sol. Balanced chemical reaction :



8.8 PROBLEM RELATED WITH PARALLEL REACTION :

When same two reactants form two or more products by independent reactions.

Ex.25 Carbon reacts with oxygen forming carbon monoxide and/or carbon dioxide depending on availability of oxygen. Find moles of each product obtained when 160 gm oxygen reacts with (a) 12 g carbon (b) 120 g carbon (c) 72 g carbon.

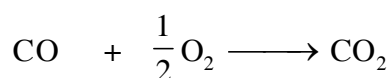
Sol. (a) $C + \frac{1}{2} O_2 \longrightarrow CO$ [initially use a reaction using lesser amount of oxygen]

$t = 0$ 1mol 5mol

$t = \infty$ 0 5-0.5 = 1mol

(LR) 4.5mol

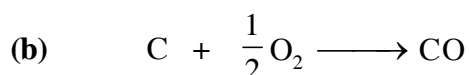
Since CO & O₂ are left, CO₂ will also formed.



$t = 0$ 1mol 4.5mol 0

$t = \infty$ 0 4mol 1mol

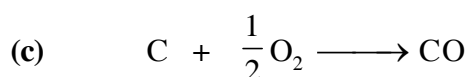
At end, 1 mole CO₂ & no CO present



$t = 0$ 10mol 5mol 0

$t = \infty$ 0 0 10mol

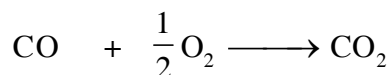
At end only 10 mol CO present.



$t = 0$ 6mol 5mol 0

$t = \infty$ 0 2mol 6mol

[LR]



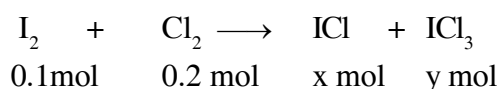
$t = 0$ 6mol 2mol 0

$t = \infty$ 2mol 0 [LR] 4mol

At end [2mol CO + 4mol CO₂] left.

Ex.26 25.4 gm of iodine and 14.2 gm of chlorine are made to react completely to yield mixture of ICl and ICl₃. Ratio of moles of ICl & ICl₃ formed is (Atomic mass : I = 127, Cl = 35.5)

Sol.



Applying conservations of I and Cl -atom

$$0.2 = x + y \quad \Rightarrow \quad x = 0.1$$

$$0.4 = x + 3y \quad \Rightarrow \quad y = 0.1$$

$$\therefore n_{\text{ICl}} : n_{\text{ICl}_3} = x : y = 1 : 1$$

8.9 PRINCIPLE OF ATOM CONSERVATION (POAC)

POAC is nothing but the conservation of atoms of reactants and products involved in a chemical reaction. And if atoms are conserved, moles of atoms shall also be conserved. The principle is fruitful for the students when they don't get the idea of balanced chemical equation. In the problem using POAC we do not need to balance a reaction and we can even add two or more reactions. This principle can be understood by the following example.

Consider the decomposition of $\text{KClO}_3(\text{s}) \rightarrow \text{KCl}(\text{s}) + \text{O}_2(\text{g})$ (unbalanced chemical reaction)

Apply the principle of atom conservation (POAC) for K atoms.

or moles of K atoms in KClO_3 = moles of K atoms in KCl

Now, since 1 molecule of KClO_3 contains 1 atom of K

Thus, moles of K atoms in $\text{KClO}_3 = 1 \times \text{moles of } \text{KClO}_3$

and moles of K atoms in KCl = $1 \times \text{moles of KCl}$

$$\therefore \text{moles of } \text{KClO}_3 = \text{moles of KCl} \quad \text{or} \quad \frac{\text{wt. of } \text{KClO}_3 \text{ in g}}{\text{mol. wt. of } \text{KClO}_3} = \frac{\text{wt. of KCl in g}}{\text{mol. wt. of KCl}}$$

- The above equation gives the mass-mass relationship between KClO_3 and KCl which is important in stoichiometric calculations. Again, applying the principle of atom conservation for O atoms,

moles of O in $\text{KClO}_3 = 3 \times \text{moles of } \text{KClO}_3$

moles of O in $\text{O}_2 = 2 \times \text{moles of } \text{O}_2$

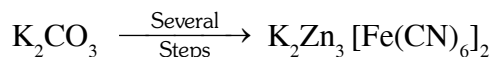
$$\therefore 3 \times \text{moles of } \text{KClO}_3 = 2 \times \text{moles of } \text{O}_2$$

$$\text{or} \quad 3 \times \frac{\text{wt. of } \text{KClO}_3}{\text{mol. wt. of } \text{KClO}_3} = 2 \times \frac{\text{vol. of } \text{O}_2 \text{ at 1atm and } 0^\circ\text{C}}{\text{Molar vol. (22.4 lt)}}$$

- The above equations thus gives the mass-volume relationship of reactants and products.

Ex.27 27.6 g K_2CO_3 was treated by a series of reagents so as to convert all of its carbon to $K_2Zn_3[Fe(CN)_6]_2$. Calculate the weight of the product. [mol. wt. of $K_2CO_3 = 138$ and mol. wt. of $K_2Zn_3[Fe(CN)_6]_2 = 698$]

Sol. Here we have no knowledge about series of chemical reactions but we know about initial reactant and final product, accordingly.



Since C atoms are conserved, applying POAC for C atoms,

moles of C in K_2CO_3 = moles of C in $K_2Zn_3[Fe(CN)_6]_2$

$1 \times \text{moles of } K_2CO_3 = 12 \times \text{moles of } K_2Zn_3[Fe(CN)_6]_2$

$$\frac{\text{wt. of } K_2CO_3}{\text{mol. wt. of } K_2CO_3} = 12 \times \frac{\text{wt. of the product}}{\text{mol. wt. of product}}$$

$$\text{wt. of } K_2Zn_3[Fe(CN)_6]_2 = \frac{27.6}{138} \times \frac{698}{12} = 11.63 \text{ g}$$

9. PERCENTAGE COMPOSITION OF AN ELEMENT IN A COMPOUND :

Percentage of particular element present in a compound given as ,

Mass % of an element

$$= \frac{\text{mass of that element in the compound}}{\text{molar mass of the compound}} \times 100$$

$$= \frac{\text{no. of atoms of that element} \times \text{Atomic mass}}{\text{molar mass of the compound}} \times 100$$

Let us understand it by taking the example of water (H_2O). Since water contains hydrogen and oxygen, the percentage composition of both these elements can be calculated as follows :

$$\text{Mass \% of an element} = \frac{\text{mass of that element in the compound} \times 100}{\text{molar mass of the compound}}$$

$$\text{Molar mass of water} = 18 \text{ g}$$

$$\text{Mass \% of hydrogen} = \frac{2 \times 1}{18} \times 100 = 11.11$$

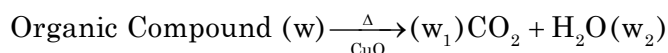
$$\text{Mass \% of oxygen} = \frac{16}{18} \times 100 = 88.89$$

9.1 PERCENTAGE DETERMINATION OF ELEMENTS IN ORGANIC COMPOUNDS :

All these methods are applications of POAC

Do not remember the formulae, derive them using the concept, its easy.

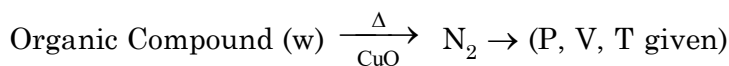
(a) **Liebig's method :** (for Carbon and hydrogen)



$$\% \text{ of C} = \frac{w_1}{44} \times \frac{12}{w} \times 100$$

$$\% \text{ of H} = \frac{w_2}{18} \times \frac{2}{w} \times 100$$

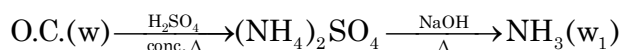
(b) **Duma's method :** (for nitrogen)



use $PV = nRT$ to calculate moles of N_2 , n .

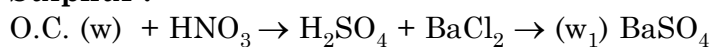
$$\therefore \% \text{ of N} = \frac{n \times 28}{w} \times 100$$

(c) **Kjeldahl's method :** (for nitrogen)



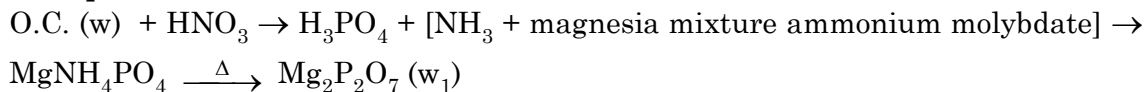
$$\% \text{ of N} = \frac{14}{17} \times \frac{w_1}{w} \times 100$$

(d) **Sulphur :**



$$\Rightarrow \% \text{ of S} = \frac{w_1}{233} \times \frac{1 \times 32}{w} \times 100.$$

(e) **Phosphorus :**



$$\% \text{ of P} = \frac{w_1}{222} \times \frac{2 \times 31}{w} \times 100$$

(f) **Carius method :** (Halogens)



If X is Cl then colour = curdy white

If X is Br then colour = dull yellow

If X is I then colour = bright yellow

Flourine can't be estimated by this

$$\% \text{ of X} = \frac{w_1}{(\text{M. weight of AgX})} \times \frac{1 \times (\text{At. wt. of X})}{w} \times 100$$

Ex.28 Which of the following compound have same % composition of carbon :



Sol. HCHO, CH₃COOH, C₆H₁₂O₆

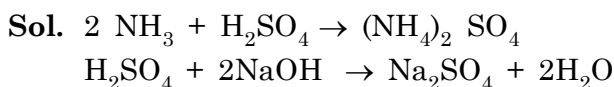
$$\% \text{ C(HCHO)} = \frac{1 \times 12}{30} \times 100 = 40\%$$

$$\% \text{ C(CH}_3\text{COOH)} = \frac{2 \times 12}{60} \times 100 = 40\%$$

$$\% \text{ C(C}_6\text{H}_{12}\text{O}_6) = \frac{6 \times 12}{180} \times 100 = 40\%$$

$$\% \text{ C(CH}_3\text{COCH}_3) = \frac{3 \times 12}{58} \times 100 = 62.07\%$$

Ex.29 A sample of 0.5 gm of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed by 2.45 gm of H_2SO_4 . The residual acid required solution containing 1.2 gm. NaOH for neutralisation. Find the percentage composition of nitrogen in the compound ?



$$\text{Moles of } \text{H}_2\text{SO}_4 \text{ taken} = \frac{2.45}{98} = 0.025$$

$$\text{Moles of NaOH} = \frac{1.2}{40} = 0.03$$

$$\therefore \text{Moles of } \text{H}_2\text{SO}_4 \text{ reacted with NaOH} = \frac{0.03}{2} = 0.015$$

$$\text{Remaining mol of } \text{H}_2\text{SO}_4 = 0.025 - 0.015 = 0.01$$

$$\text{mol of } \text{NH}_3 \text{ evolved} = 0.01 \times 2 = 0.02$$

$$\% \text{ N in sample} = \frac{0.02 \times 14}{0.5} \times 100 = 56\%$$

10. EMPIRICAL AND MOLECULAR FORMULA

We have just seen that knowing the molecular formula of the compound we can calculate percentage composition of the elements. Conversely if we know the percentage composition of the elements initially, we can calculate the relative number of atoms of each element in the molecules of the compound. This gives us the empirical formula of the compound. Further if the molecular mass is known then the molecular formula can be easily determined.

Thus, the **empirical** formula of a compound is a chemical formula showing the relative number of atoms in the simplest ratio, the **molecular formula** gives the actual number of atoms of each element in a molecule.

i.e. **Empirical formula** : Formula depicting constituent atoms in their simplest ratio.

Molecular formula : Formula depicting actual number of atoms in one molecule of the compound.

The molecular formula is generally an integral multiple of the empirical formula.

i.e. $\text{molecular formula} = \text{empirical formula} \times n$

$$\text{where } n = \frac{\text{molecular formula mass}}{\text{empirical formula mass}}$$

Example :

Molecular Formula	H_2O_2 2 : 2	C_6H_6 6 : 6	C_2H_6 2 : 6	$\text{C}_2\text{H}_4\text{O}_2$ 2 : 4 : 2
Simplest ratio	1 : 1	1 : 1	1 : 3	1 : 2 : 1
Empirical Formula	H O	C H	CH_3	CH_2O

10.1 DETERMINATION OF EMPIRICAL FORMULA :

Following steps are involved in determining the empirical formula of the compounds –

- (i) First of all find the % by wt. of each element present in the compound.
- (ii) The % by wt of each element is divided by its atomic weight. It gives atomic ratio of elements present in the compounds.
- (iii) Atomic ratio of each element is divided by the minimum value of atomic ratio so as to get simplest ratio of atoms.
- (iv) If the value of simplest atomic ratio is fractional then raise the value to the nearest whole number or multiply with suitable coefficient to convert it into nearest whole number
- (v) Write the Empirical formula as we get the simplest ratio of atoms.

10.2 DETERMINATION OF MOLECULAR FORMULA :

- (i) Find out the empirical formula mass by adding the atomic masses of all the atoms present in the empirical formula of compound.
- (ii) Divide the molecular mass (determined experimentally by some suitable method) by the empirical formula mass and find out the value of n.
- (iii) Multiply the empirical formula of the compound with 'n' so as to find out the molecular formula of the compound.

Ex.30. *An organic compound contains 49.3% carbon, 6.84% hydrogen and its vapour density is 73. Molecular formula of compound is :-*

Sol. V.D. = 73 $\Rightarrow$ M = 2 $\times$ 73 = 146

$$C = 146 \times \frac{49.3}{100} = 71.978 \text{ g} = \frac{71.978}{12} \simeq 6 \text{ mole}$$

$$H = 146 \times \frac{6.84}{100} = 9.9864 \text{ g} = \frac{9.9864}{1} \simeq 10 \text{ mole}$$

$$O = 146 \times \frac{100 - (49.3 + 6.84)}{100} = \frac{64.86}{16} = 4.05375 \approx 4 \text{ mol}$$

$$\text{M.F.} = \text{C}_6\text{H}_{10}\text{O}_4$$

Ex.31 *The empirical formula of an organic compound containing carbon & hydrogen is CH₂. The mass of 1 litre of organic gas is exactly equal to mass of 1 litre N₂ therefore molecular formula of organic gas is.*

Sol. Empirical Mass of CH₂ = 12 + 2 = 14

$\therefore$ Mass of 1 litre of organic gas = Mass of 1 litre of N₂

Since V, P, T, n are same.

Therefore from $PV = \frac{m}{M}RT$ implies that molar mass should also be same.

$\therefore$ Molecular mass of organic compound will be 28 g

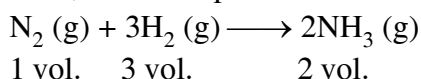
$$n = \frac{\text{Molecular mass}}{\text{Empirical mass}} = \frac{28}{14} = 2$$

So molecular formula = 2 $\times$ CH₂ = C₂H₄

11. EUDIOMETRY :

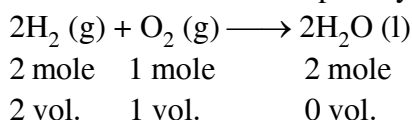
Eudiometry or gas analysis involves the calculations based on gaseous reactions or the reactions in which at least two components are gaseous, in which the amounts of gases are represented by their volumes, measured at the same pressure and temperature. Some basic assumptions related with calculations are:

- (i) Gay-Lussac's law of volume combination holds good. According to this law, the volumes of gaseous reactants reacted and the volumes of gaseous products formed, all measured at the same temperature and pressure, bear a simple ratio.



Problem may be solved directly in terms of volume, in place of mole. The stoichiometric coefficients of a balanced chemical reactions gives the ratio of volumes in which gaseous substances are reacting and products are formed, at same temperature and pressure.

- (ii) The volumes of solids or liquids is considered to be negligible in comparison to the volume of gas. It is due to the fact that the volume occupied by any substance in gaseous state is even more than thousand times the volume occupied by the same substance in solid or liquid states.



- (iii) Air is considered as a mixture of oxygen and nitrogen gases only. It is due to the fact that about 99% volume of air is composed of oxygen and nitrogen gases only.
- (iv) Nitrogen gas is considered as a non- reactive gas. It is due to the fact that nitrogen gas reacts only at very high temperature due to its very high thermal stability. Eudiometry is performed in an eudiometer tube and the tube can not withstand very high temperature. This is why, nitrogen gas can not participate in the reactions occurring in the eudiometer tube.
- (v) The total volume of non-reacting gaseous mixture is equal to sum of partial volumes of the component gases (**Amagat's law**).

$$V = V_1 + V_2 + \dots$$

Partial volume of gas in a non-reacting gaseous mixture is its volume when the entire pressure of the mixture is supposed to be exerted only by that gas.

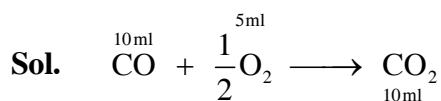
- (vi) The volume of gases produced is often given by certain solvent which absorb contain gases.

Solvent	Gases absorb
KOH	CO ₂ , SO ₂ , Cl ₂
Ammonical Cu ₂ Cl ₂	CO
Turpentine oil	O ₃
Alkaline pyrogallol	O ₂
water	NH ₃ , HCl
CuSO ₄ /CaCl ₂	H ₂ O

(vii) EUDIOMETER

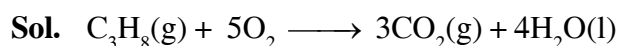
An eudiometer is a laboratory device that measures the change in volume of a gas mixture following a physical or chemical change.

Ex.32 10 ml of CO is mixed with 25 ml air (20% O₂ by volume). Find final volume (in ml) after complete combustion.

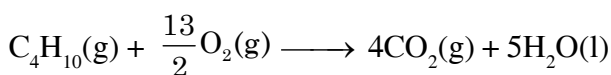


$$V_f = V_{\text{CO}_2} + \text{Volume of remaining air} = 10 + 20 = 30 \text{ ml}$$

Ex.33 A 3 L gas mixture of propane (C₃H₈) and butane (C₄H₁₀) on complete combustion at 25°C produced 10 L CO₂. Assuming constant P and T conditions what was volume of butane present in initial mixture ?



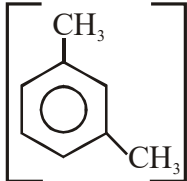
$$x \text{ L} \qquad \qquad \qquad 3x \text{ L}$$



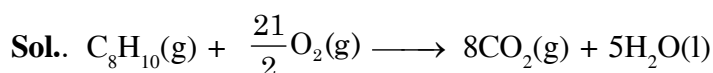
$$(3-x) \text{ L} \qquad \qquad \qquad 4(3-x) \text{ L}$$

$$\text{from question } 3x + 4(3-x) = 10 \Rightarrow x = 2$$

$$\therefore \text{Volume of butane, C}_4\text{H}_{10} = (3-x) = 1 \text{ L}$$

Ex.34 100 ml gaseous meta Xylene  undergoes combustion with excess of oxygen at room

temperature and pressure. Volume contraction / expansion (in ml) during reaction is

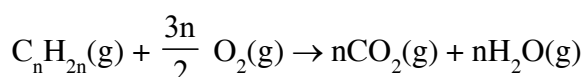


$$\begin{array}{ccccccc} 100 \text{ ml} & \frac{21}{2} \times 100 & & 800 \text{ ml} & & 0 & \\ & = 1050 \text{ ml} & & & & & \end{array}$$

$$\therefore \text{Contraction in volume} = (100 + 1050) - 800 = 350 \text{ ml}$$

Ex.35 An alkene upon combustion produces CO₂(g) and H₂O(g). In this combustion process if there is no volume change occurs then the no. of C atoms per molecule of alkene will be :

Ans.(2)

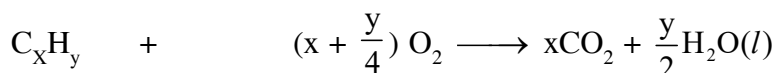


if there no volume changes i.e. $\Delta_{\text{ng}} = 0$

$$(n + n) - \left(1 + \frac{3n}{2}\right) = 0 \Rightarrow n = 2$$

Ex.36 A gaseous hydrocarbon (C_xH_y) requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times of its volume of CO_2 . Find out the value of $(x + y)$

Ans. (012)



$$a \qquad a\left(x + \frac{y}{4}\right) \qquad ax$$

Given that : $a(x + y/4) = 6a$

and $ax = 4(a)$

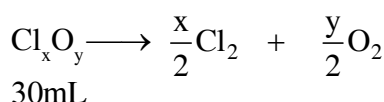
$$x = 4 \text{ and } y = 8 \qquad \dots(2)$$

$$\therefore x + y = 4 + 8 = 12$$

Ex.37 On heating 60 ml mixture containing equal volume of chlorine gas and it's gaseous oxide, volume becomes 75 ml due complete decomposition of oxide. On treatment with KOH volume becomes 15 ml. What is the formula of oxide of chlorine ?

Ans. Cl_2O

Sol. Let oxide of Cl is Cl_xO_y
So in 60 mL $\Rightarrow$ 30 mL Cl_xO_y and 30 mL Cl_2 .
Now,



$$\frac{30.x}{2} \text{ mL} \qquad \frac{30.y}{2} \text{ mL}$$

Given :

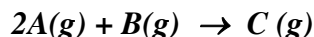
$$75 = 30 + \frac{30x}{2} + \frac{30y}{2} \qquad \Rightarrow \qquad x + y = 3 \dots\dots\dots(i)$$

KOH absorbs Cl_2 and volume becomes 15 mL so,

$$(75 - 15) = V_{Cl_2} = 30 + \frac{30x}{2} \qquad \Rightarrow \qquad x = 2 \text{ and } y = 1$$

So the oxide : Cl_2O

Ex.38 5 L of A (g) & 3 L of B(g) measured at same T & P are mixed together which react as follows



What will be the total volume (in litre) after the completion of the reaction at same T & P.

Ans. (3)

Sol. $2A(g) + B(g) \longrightarrow C(g)$

5L 3L

L.R. is A

$$\text{So, volume of C produced} = \frac{1}{2} \times 5 = 2.5 \text{ L}$$

$$\text{and, volume of B reacted} = \frac{1}{2} \times 5 = 2.5 \text{ L}$$

$$\text{So, volume fo B remained} = 3 - 2.5 = 0.5 \text{ L}$$

$$\text{Hence, } V_{\text{total}} = V_C + V_B = 2.5 + 0.5 = 3 \text{ L}$$

SELF STUDY

12. SIGNIFICANT FIGURES

Every experimental measurement has some amount of uncertainty associated with it. However, one would always like the results to be precise and accurate. Precision and accuracy are often referred to while we talk about the measurement.

Precision refers to the closeness of various measurements for the same quantity. However, **accuracy** is the agreement of a particular value to the true value of the result. For example, if the true value for a result is 2.00 g and a student 'A' takes two measurements and reports the results as 1.95 g and 1.93 g. These values are precise as they are close to each other but are not accurate. Another student repeats the experiment and obtains 1.94 g and 2.05 g as the results for two measurements. These observations are neither precise nor accurate. When a third student repeats these measurements and reports 2.01 g and 1.99 g as the result. These values are both precise and accurate. This can be more clearly understood from the data given in Table

Data to Illustrate Precision and Accuracy

Measurements/g				
	1	2	Average(g)	Remarks
Student A	1.95	1.93	1.940	Precise but not accurate
Student B	1.95	2.05	1.995	Neither precise nor accurate
Student C	2.01	1.99	2.000	Precise as well as accurate

The uncertainty in the experimental or the calculated values is indicated by mentioning the number of significant figures. **Significant figures** are meaningful digits which are known with certainty. The uncertainty is indicated by writing the certain digits and the last uncertain digit. Thus, if we write a result as 11.2 mL, we say the 11 is certain and 2 is uncertain and the uncertainty would be ± 1 in the last digit. Unless otherwise stated, an uncertainty of ± 1 in the last digit is always understood.

There are certain rules for determining the number of significant figures. These are stated below:

- (1) All non-zero digits are significant. For example in 285 cm, there are three significant figures and in 0.25 mL, there are two significant figures.
- (2) Zeros preceding to first non-zero digit are not significant. Such zero indicates the position of decimal point.

Thus, 0.03 has one significant figure and 0.0052 has two significant figures.

- (3) Zeros between two non-zero digits are significant. Thus, 2.005 has four significant figures.
- (4) Zeros at the end or right of a number are significant provided they are on the right side of the decimal point. For example, 0.200 g has three significant figures.

But, if otherwise, the terminal zeros are not significant if there is no decimal point. For example, 100 has only one significant figure, but 100. has three significant figures and 100.0 has four significant figures. Such numbers are better represented in scientific notation. We can express the number 100 as 1×10^2 for one significant figure, 1.0×10^2 for two significant figures and 1.00×10^2 for three significant figures.

- (5) Counting numbers of objects, for example, 2 balls or 20 eggs, have infinite significant figures as these are exact numbers and can be represented by writing infinite number of zeros after placing a decimal i.e., $2 = 2.000000$ or $20 = 20.000000$

In numbers written in scientific notation, all digits are significant e.g., 4.01×10^2 has three significant figures, and 8.256×10^{-3} has four significant figures.

Addition and Subtraction of Significant Figures

The result cannot have more digits to the right of the decimal point than either of the original numbers.

$$\begin{array}{r} 12.11 \\ 18.0 \\ 1.012 \\ \hline 31.122 \end{array}$$

Here, 18.0 has only one digit after the decimal point and the result should be reported only up to one digit after the decimal point which is 31.1.

Multiplication and Division of Significant Figures

In these operations, the result must be reported with no more significant figures as are there in the measurement with the few significant figures.

$$2.5 \times 1.25 = 3.125$$

Since 2.5 has two significant figures, the result should not have more than two significant figures, thus, it is 3.1.

While limiting the result to the required number of significant figures as done in the above mathematical operation, one has to keep in mind the following points for rounding off the numbers

1. If the rightmost digit to be removed is more than 5, the preceding number is increased by one. for example, 1.386
If we have to remove 6, we have to round it to 1.39
2. If the rightmost digit to be removed is less than 5, the preceding number is not changed.
For example, 4.334 if 4 is to be removed, then the result is rounded upto 4.33.
3. If the rightmost digit to be removed is 5, then the preceding number is not changed if it is an even number but it is increased by one if it is an odd number. For example, if 6.35 is to be rounded by removing 5, we have to increase 3 to 4 giving 6.4 as the result. However, if 6.25 is to be rounded off it is rounded off to 6.2.

SOLVED EXAMPLES

Ex.1 The weight of 2.01×10^{23} molecules of CO is- [AIEEE 2002]

- (1) 9.3 g (2) 7.2 g (3) 1.2 g (4) 3 g

Ans. (1)

Sol. $\text{Mass} = \frac{2.01 \times 10^{23}}{6.02 \times 10^{23}} \times 28 = 9.3 \text{ gm}$

Ex.2 How many moles of magnesium phosphate, $\text{Mg}_3(\text{PO}_4)_2$ will contain 0.25 mole of oxygen atoms? [AIEEE 2006]

- (1) 3.125×10^{-2} (2) 1.25×10^{-2} (3) 2.5×10^{-2} (4) 0.02

Ans. (1)

Sol. $x \times 8 = 0.25 \Rightarrow x = 3.125 \times 10^{-2}$

Ex.3 The ratio of number of oxygen atoms (O) in 16.0 g ozone (O_3), 28.0 g carbon monoxide (CO) and 16.0 g oxygen (O_2) is :-

(Atomic mass : C = 12, O = 16 and Avogadro's constant $N_A = 6.0 \times 10^{23} \text{ mol}^{-1}$)

[AIEEE 2012 (Online)]

- (1) 3 : 1 : 1 (2) 1 : 1 : 2 (3) 3 : 1 : 2 (4) 1 : 1 : 1

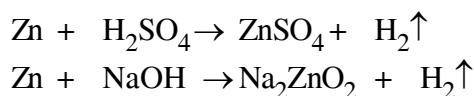
Ans. (4)

$$\begin{aligned} & (\text{O-atoms})_{\text{O}_3} : (\text{O-atoms})_{\text{CO}} : (\text{O-atoms})_{\text{O}_2} \\ &= \left(\frac{16}{48} \times N_A \right) \times 3 : \left(\frac{28}{28} \times N_A \right) \times 1 : \left(\frac{16}{32} \times N_A \right) \times 2 \\ &= 1 : 1 : 1 \end{aligned}$$

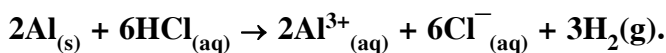
Ex.4 When the same amount of zinc is treated separately with excess of sulphuric acid and excess of sodium hydroxide, the ratio of volume of hydrogen evolved is. [JEE-1979]

- (A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 9 : 4

Ans. (A)



Ex.5 In the reaction [AIEEE-2007]



Select the correct information(s)

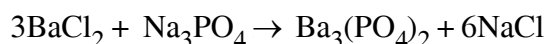
- (1) 6L $\text{HCl}_{(aq)}$ is consumed for every 3L $\text{H}_{2(g)}$ produced
 (2) 33.6 L $\text{H}_{2(g)}$ is produced regardless of temperature and pressure for every mole of Al that reacts.
 (3) 67.2 L $\text{H}_{2(g)}$ at STP is produced for every mole of Al that reacts
 (4) 11.2 L $\text{H}_{2(g)}$ at STP is produced for every mole of $\text{HCl}_{(aq)}$ consumed

Ans. (4)

Ex.6 If 0.50 mole of BaCl_2 is mixed with 0.20 mol of Na_3PO_4 , the maximum number of moles of $\text{Ba}_3(\text{PO}_4)_2$ that can be formed is - [JEE-1981]

- (A) 0.70 (B) 0.50 (C) 0.20 (D) 0.10

Ans. (D)



0.5 mole 0.20 mole 0.1 mole

(L.R.)

Ex.7 What weight of AgCl will be precipitated when a solution containing 4.77 g of NaCl is added to a solution of 5.77 g of AgNO_3 . ($\text{Ag} = 108$)

Ans. (4.879 gm)



5.77 gm 4.77 gm

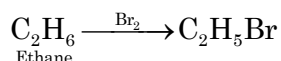
0.034 mole (LR) 0.082 mole

wt. of $\text{AgCl} = 0.034 \times 143.5 = 4.879 \text{ gm}$

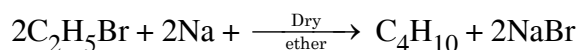
Ex.8 n-butane is product by the monobromination of ethane followed by the Wurtz reaction. Calculate the volume of ethane at NTP required to produce 55g n-butane, if the bromination takes place with 90% yield and the Wurtz reaction with 85% yield. [JEE-1984]

Ans. (55.50 litre)

(i) Monobromination of ethane



(ii) Wurtz reaction :



Molecular weight of

$$\text{C}_4\text{H}_{10} = (12 \times 4) + (10 \times 1) = 58$$

$$\therefore \text{Amount of n-butane to be produced} = \frac{55}{58} \text{ mol} = 0.948 \text{ mol}$$

$$\therefore \text{Amount of } \text{C}_2\text{H}_5\text{Br} \text{ required to obtain } 0.948 \text{ mol}$$

But the conversion is only 85%

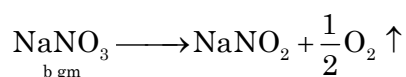
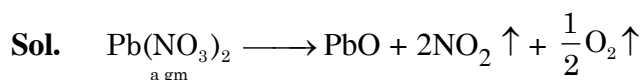
$$\text{Hence, the amount of } \text{C}_2\text{H}_5\text{Br} \text{ required} = \frac{1.896}{85} \times 100 = 2.23 \text{ mol}$$

To obtain $\text{C}_2\text{H}_5\text{Br}$ from C_2H_6 , the same amount of C_2H_6 would be required. But the percent conversion of C_2H_6 to required = $\frac{2.23}{90} \times 100 = 2.478 \text{ mol}$.

Thus, required volume of ethane of NTP = $22400 \times 2.478 = 55507.2 \text{ ml} = 55.50 \text{ L}$.

Ex.9 A solid mixture (5.0 g) consisting of lead nitrate and sodium nitrate was heated below 600°C until the weight of the residue was constant. If the loss in weight is 20% find the amount of lead nitrate and sodium nitrate in the mixture. [JEE-1990]

Ans. $\text{Pb}(\text{NO}_3)_2 = 3.32 \text{ gm}$, $\text{NaNO}_3 = 1.68 \text{ gm}$



$\therefore a + b = 5 \quad \dots (1)$

The loss in weight for 5 g mixture $= 5 \times \frac{28}{100} = 1.4 \text{ g}$

$\therefore$ Residue left $= 5 - 1.4 = 3.6 \text{ g}$

The residue contain $\text{PbO} + \text{NaNO}_2$

$\therefore 331 \text{ g Pb}(\text{NO}_3)_2 \text{ gives } = 223 \text{ g PbO}$

$\therefore a \text{ g Pb}(\text{NO}_3)_2 \text{ gives } = \frac{223 \times a}{332} \text{ g PbO}$

Similarly,

$\therefore 85 \text{ g NaNO}_3 \text{ gives } = 69 \text{ g NaNO}_2$

$\therefore b \text{ g NaNO}_3 \text{ gives } = \frac{69 \times b}{85} \text{ g NaNO}_2$

Solving equation, (1) and (2)

$a = 3.32 \text{ g}$ and $b = 1.68 \text{ g}$

Ex.10 In an organic compound of molar mass 108 g mol^{-1} C, H and N atoms are present in 9 : 1 : 3.5 by weight. Molecular formula can be : [AIIEEE 2002]



Ans. (1)

Sol.	C	H	N
	9	1	3.5
	:	:	:
Mole	$\frac{9}{12}$	$\frac{1}{1}$	$\frac{3.5}{14}$
	:	:	:
	0.75	1	0.25
	C ₃ H ₄ N = emp. formula		
	mol. formula = C ₆ H ₈ N ₂		

Ex.11 A transition metal M forms a volatile chloride which has a vapour density of 94.8. If it contains 74.75% of chlorine the formula of the metal chloride will be [AIEEE 2012 (Online)]

- (1) MCl_2 (2) MCl_4 (3) MCl_5 (4) MCl_3

Ans. 2

Let the formula of metal chloride is MCl_V .

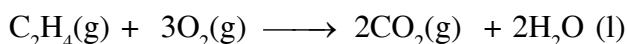
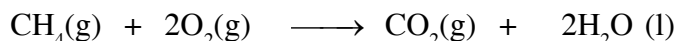
Now, molecular weight, $2 \times 94.8 = A + V \times 35.5$ (1)

$$\text{and } (2 \times 94.8) \times \frac{74.75}{100} = V \times 35 \quad \text{.....(2)}$$

From (2) ; $V = 3.99 \approx 4$

$\therefore$ formula is MCl_4

Ex.12 30 ml gaseous mixture of methane and ethylene in volume ratio X : Y requires 350 ml air containing 20% of O_2 by volume for complete combustion. If ratio of methane and ethylene changed to Y : X. What will be volume of air (in ml) required for complete reaction under similar condition of temperature and pressure.



For given data : $V_1 + V_2 = 30$

$$\text{and } 2V_1 + 3V_2 = 350 \times \frac{20}{100} = 70$$

$$\therefore V_1 = 20, V_2 = 10$$

For required data : $V_1 = 10$ and $V_2 = 20$

$$\therefore \text{Volume of } \text{O}_2 \text{ required} = 2V_1 + 3V_2 = 80 \text{ ml and volume of air required} = 80 = \frac{100}{20} = 400 \text{ ml}$$

Ex.13 5 ml of gas containing only carbon and hydrogen was mixed with an excess of oxygen (30 ml) and the mixture exploded by means of an electric spark. After the explosion, the volume of the mixed gases remaining was 25 ml. On adding a concentrated solution of potassium hydroxide, the volume further diminished to 15 ml the residual gas being pure oxygen. All volumes have been reduced to NTP. Calculate the molecular formula of the hydrocarbon. [JEE-1979]

Ans. C_2H_4

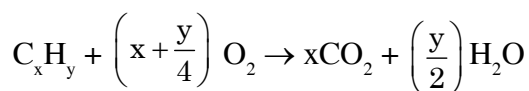
Sol. Volume of oxygen taken = 30 ml

Volume of unused oxygen = 15 ml

Volume of O_2 used = Volume of O_2 added - volume of O_2 left = $30 - 15 = 15$ ml.

and volume of CO_2 produced = $25 - 15 = 10$ ml

General equation of the combustion of a hydrocarbon is as following lows.



$$5 \text{ ml } 5\left(x + \frac{y}{4}\right) \text{ ml } \quad 5x$$

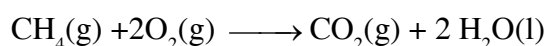
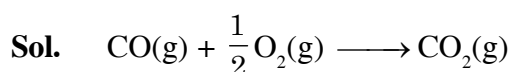
$$\therefore 5x = 10 \Rightarrow x = 2$$

$$\text{and } 5\left(x + \frac{y}{4}\right) = 15$$

$$\therefore \text{Molecular formula of hydrocarbon} = C_2H_4$$

Ex.14 A 20.0 ml mixture of CO, CH₄ and He gases is exploded by an electric discharge at room temperature with excess of oxygen. The volume contraction is found to be 13.0 cm³. A further contraction of 14.0 cm³ occurs when the residual gas is treated with KOH solution. Find out the composition of the gaseous mixture in terms of volume percentage. [JEE-1995]

Ans. (% CH₄ = 20, % CO = 50, % He = 30)



'x' is the volume of CO and y is the volume of CH₄

$$\text{Thus, } \frac{1}{2}x + 2y = 13 \quad (1)$$

$$x + y = 14 \quad (2)$$

$$x = 10 \text{ and } y = 4$$

$$\text{Thus, \% } CH_4 = 20, \% CO = 50, \% He = 30$$

EXERCISE # S- I

ATOMIC MASS, MOLECULAR MASS, MOLE

1. Find :
 - (i) No. of moles of Cu atom in 10^{23} atoms of Cu. MC0001
 - (ii) Mass of 200 $^{16}_8\text{O}$ atoms in amu MC0001
 - (iii) Mass of 100 atoms of $^{14}_7\text{N}$ is $y \times 10^{-22}$ in gm, then value of y is ($1 \text{ amu} = 1.67 \times 10^{-24} \text{ gm}$) MC0001
 - (iv) No. of molecules & atoms in 54 gm H_2O is $y \times 10^{23}$ and $z \times 10^{23}$ respectively then value of y & z is ($N_A = 6.022 \times 10^{23}$) MC0002
 - (v) No. of molecules in 88 gm CO_2 is $y \times 10^{23}$, then value of y is ($N_A = 6.022 \times 10^{23}$) MC0002
2. If mass of one ^{12}C atom is $y \times 10^{-23}$ gm, then value of y is ? MC0003
3. Calculate mass (in gm) of O atoms in 6 gm CH_3COOH ? MC0004
4. Calculate mass of water (in gm) present in 499 gm $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$?
(Atomic mass : Cu = 63.5, S = 32, O = 16, H = 1) MC0005
5. What mass (in gm) of $\text{Na}_2\text{SO}_4 \cdot 7\text{H}_2\text{O}$ contains exactly 6.022×10^{22} atoms of oxygen ?
($N_A = 6.022 \times 10^{23}$) MC0006
6. The weight (in gram) of pure potash Alum ($\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$) which contains 6.4 gm oxygen is. (Atomic weight of K = 39, S = 32, Al = 27) MC0007
7. The Kohinoor diamond was the largest diamond ever found. How many moles of carbon atom were present in it, if it is weigh 3300 carat. [Given: 1 carat = 200 mg] MC0008
8. Calculate volume (in litre) of H_2 gas kept at STP if it contains as many H atoms as in 98 gm H_3PO_4 .
[Atomic mass of P = 31] MC0009
9. 80gm of SO_x gas occupies 14 litre at 2atm & 273K. The value of x is : MC0010
10. 40 mg of gaseous substance (X_2) occupies 4.8 mL of volume at 1 atm and 27°C . Atomic mass of element X is : (R : 0.082 atm L/mole-K) MC0011

STOICHIOMETRY

11. How many gm of HCl is needed for complete reaction with 21.75 gm MnO_2 ?
(Mn = 55, Cl = 35.5)
 $\text{HCl} + \text{MnO}_2 \longrightarrow \text{MnCl}_2 + \text{H}_2\text{O} + \text{Cl}_2$ MC0012
12. Nitric acid is manufactured by the Ostwald process, in which nitrogen dioxide reacts with water.
 $3 \text{NO}_2 (\text{g}) + \text{H}_2\text{O} (\text{l}) \rightarrow 2 \text{HNO}_3 (\text{aq}) + \text{NO} (\text{g})$
How many grams of nitrogen dioxide are required in this reaction to produce 25.2 gm HNO_3 ? MC0013
13. Fluorine reacts with uranium to produce uranium hexafluoride, UF_6 , as represented by this equation
 $\text{U} (\text{s}) + 3\text{F}_2 (\text{g}) \rightarrow \text{UF}_6 (\text{g})$
If no. of fluorine molecules are required to produce 7.04 mg of uranium hexafluoride, UF_6 , from an excess of uranium is $y \times 10^{19}$ then value of y is ? The molar mass of UF_6 is 352 gm/mol.
($N_A = 6.022 \times 10^{23}$) MC0014

14. What total volume, in litre at 627°C and 82.1 atm, could be formed by the decomposition of 16 gm of NH_4NO_3 ? Reaction : $2 \text{NH}_4\text{NO}_3 \rightarrow 2\text{N}_2 + \text{O}_2 + 4\text{H}_2\text{O}_{(\text{g})}$. (R : 0.0821 atm L/mole-K) **MC0015**
15. Calculate mass of phosphoric acid (in gm) required to obtain 53.4g pyrophosphoric acid.
 $2\text{H}_3\text{PO}_4 \rightarrow \text{H}_4\text{P}_2\text{O}_7 + \text{H}_2\text{O}$ **MC0016**

LIMITING REACTANT

16. Carbon reacts with chlorine to form CCl_4 . 36 gm of carbon was mixed with 142 g of Cl_2 . If ratio of mass of CCl_4 produced and the remaining mass of excess reactant is y : 1, then value of y is **MC0017**
17. Potassium superoxide, KO_2 , is used in rebreathing gas masks to generate oxygen :
 $\text{KO}_2(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{KOH}(\text{s}) + \text{O}_2(\text{g})$
 If a reaction vessel contains 0.158 mol KO_2 and 0.10 mol H_2O , how many moles of O_2 can be produced? **MC0018**
18. A chemist wants to prepare diborane (B_2H_6) by the reaction
 $6 \text{LiH} + 8\text{BF}_3 \longrightarrow 6\text{LiBF}_4 + \text{B}_2\text{H}_6$
 If he starts with 2.0 moles each of LiH & BF_3 . How many moles of B_2H_6 can be prepared. **MC0019**
19. Sulphuric acid is produced when sulphur dioxide reacts with oxygen and water in the presence of a catalyst : $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l}) \rightarrow 2 \text{H}_2\text{SO}_4$. If 5.6 mol of SO_2 reacts with 4.8 mol of O_2 and a large excess of water, what is the maximum number of moles of H_2SO_4 that can be obtained? **MC0020**

PROBLEMS RELATED WITH MIXTURE

20. One gram of an alloy of aluminium and magnesium when heated with excess of dil. HCl forms magnesium chloride, aluminium chloride and hydrogen. The evolved hydrogen collected at 0°C has a volume of 1.12 litres at 1 atm pressure. Calculate the composition of (% by mass) of the alloy.
 (Al = 27, Mg = 24) **MC0021**
21. A sample containing only CaCO_3 and MgCO_3 is ignited to CaO and MgO . The mixture of oxides produced weight exactly half as much as the original sample. Calculate the percentages of CaCO_3 and MgCO_3 (by mass) in the sample. **MC0022**
22. Determine the percentage composition (by mass) of a mixture of anhydrous sodium carbonate and sodium bicarbonate from the following data:
 wt. of the mixture taken = 2g
 Loss in weight on heating = 0.124 gm. **MC0023**
23. A sample of mixture of CaCl_2 and NaCl weighing 2.22 gm was treated to precipitate all the Ca as CaCO_3 which was then heated and quantitatively converted to 0.84 gm of CaO . Calculate mass fraction of CaCl_2 in the mixture. **MC0024**
24. When 4 gm of a mixture of NaHCO_3 and NaCl is heated, 0.66 gm CO_2 gas is evolved. If ratio of the percentage composition (by mass) of the NaHCO_3 and NaCl is y : 1 then value of y is. **MC0025**

PERCENTAGE YIELD, DEGREE OF DISSOCIATION, PERCENTAGE

25. A power company burns approximately 500 tons of coal per day to produce electricity. If the sulphur content of the coal is 1.20 % by weight, how many tons SO_2 are dumped into the atmosphere per hour ? **MC0026**

26. Calculate the percent loss in weight after complete decomposition of a pure sample of potassium chlorate. ($K = 39$, $Cl = 35.5$)

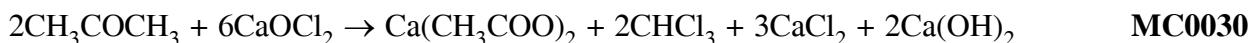


27. A sample of calcium carbonate is 80% pure, 25 gm of this sample is treated with excess of HCl . How much volume (in litre) of CO_2 will be obtained at 1 atm & 273 K? **MC0028**

28. Cyclohexanol is dehydrated to cyclohexene on heating with conc. H_2SO_4 . If the yield of this reaction is 75%, how much cyclohexene (in gm) will be obtained from 100 g of cyclohexanol ?



29. If the yield of chloroform obtainable from acetone and bleaching powder is 58%. What is the weight of acetone (in gm) required for producing 239 mg of chloroform ?



30. Calculate % yield of the reaction if 200g KHCO_3 produces 22g of CO_2 upon strong heating. ($K = 39$) **MC0031**

31. The vapour density of a sample of N_2O_4 gas is 35. What percent of N_2O_4 molecules are dissociated into NO_2 ? **MC0032**

32. If a sample of pure SO_3 gas is heated to 600°C , it dissociates into SO_2 and O_2 gases upto 50%. If the average molar mass of the final sample is M_{avg} find value of $\left(\frac{M_{\text{av}}}{100}\right)$. **MC0033**

33. When silent electric discharge is passed through O_2 gas, it converts into O_3 . If the density of final sample is 20 times the density of hydrogen gas under similar conditions, calculate the mass percent of O_2 in the final sample. **MC0034**

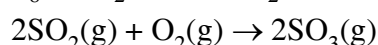
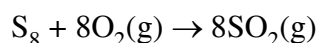
34. When acetylene (C_2H_2) gas is passed through red hot iron tube, it trimerises into benzene (C_6H_6) vapours. If the average molar mass of vapours coming out through the tube is 50, calculate the degree of trimerisation of acetylene. **MC0035**

SEQUENTIAL & PARALLEL REACTIONS

35. $\text{Br}_2(\text{l})$ reacts with $\text{Cl}_2(\text{g})$ to form BrCl and BrCl_3 , simultaneously. How many moles of $\text{Cl}_2(\text{g})$ reacts completely with 0.03 moles of $\text{Br}_2(\text{l})$ to form BrCl and BrCl_3 in 5 : 1 molar ratio **MC0036**

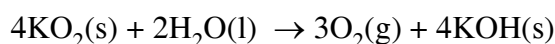
36. When 80 gm CH_4 is burnt, CO and CO_2 gases are formed in 1 : 4 mole ratio. If the mass of O_2 gas used in combustion is w gm then find value of (w/100). **MC0037**

37. Sulphur trioxide may be prepared by the following two reactions :

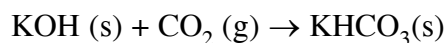


If amount of SO_3 will be produced from 1 mol of S_8 is w gm, then value of w/100 is. **MC0038**

38. Potassium superoxide, KO_2 , is utilised in closed system breathing apparatus. Exhaled air contains CO_2 and H_2O , both of which are removed and the removal of water generates oxygen for breathing by the reaction



The potassium hydroxide removes CO_2 from the apparatus by the reaction :



(a) What mass of KO_2 (in gm) generates 0.24 gm of oxygen ?

(b) What mass of CO_2 (in gm) can be removed from the apparatus by 100 gm of KO_2 ? ($K = 39$)

MC0039

PRINCIPLE OF ATOM CONSERVATION

39. In a determination of P, an aqueous solution of NaH_2PO_4 is treated with a mixture of ammonium and magnesium ions to precipitate magnesium ammonium phosphate $\text{Mg}(\text{NH}_4)\text{PO}_4 \cdot 6\text{H}_2\text{O}$. This is heated and decomposed to magnesium pyrophosphate, $\text{Mg}_2\text{P}_2\text{O}_7$ which is weighed. A solution of NaH_2PO_4 yielded 1.11 g of $\text{Mg}_2\text{P}_2\text{O}_7$. What weight of NaH_2PO_4 (in gm) was present originally ? ($P = 31$)

MC0040

40. 6 gm nitrogen on successive reaction with different compounds gets finally converted into 30 gm $[\text{Cr}(\text{NH}_3)_x\text{Br}_2]$. Value of x is [Atomic mass : $\text{Cr} = 52$, $\text{Br} = 80$]

MC0041

41. A 5.00 gm sample of a natural gas, consisting of methane (CH_4), and ethylene, (C_2H_4) was burned

in excess oxygen, yielding $\frac{44}{3}$ gm of CO_2 and some H_2O as products. What mole percent of the sample was ethylene?

MC0042

42. All carbon atom present in $\text{KH}_3(\text{C}_2\text{O}_4)_2 \cdot 2\text{H}_2\text{O}$ weighing 7.62 gm is converted to CO_2 . How many gm of CO_2 were obtained. ($K = 39$)

MC0043

PERCENTAGE COMPOSITION, EMPIRICAL AND MOLECULAR FORMULA

43. Haemoglobin contains 0.25% iron by mass. The molecular mass of Haemoglobin is 89600. The number of iron atoms per molecule of Haemoglobin (Atomic mass of $\text{Fe} = 56$) -

MC0044

44. 1.6 g of an organic compound containing sulphur, when treated with series of reagents, produces H_2SO_4 which on reaction with BaCl_2 produces 0.233 g of BaSO_4 . Calculate % by mass of S in the organic compound. (Given : Atomic weight of $\text{Ba} = 137$)

MC0045

45. When 2.0 gm of an organic compound is burnt completely, 150 ml N_2 gas at 27°C and 0.821 atm is obtained. The mass fraction of nitrogen in the compound is.

($R = 0.0821 \text{ atm L/mole} \cdot \text{K}$)

MC0046

46. A polystyrene of formula $\text{Br}_3\text{C}_6\text{H}_2(\text{C}_8\text{H}_8)_n$ was prepared by heating styrene with tribromobenzyl peroxide in the absence of air. It was found to contain 10.48% bromine by weight. Find the value of n. ($\text{Br} = 80$).

MC0047

47. A moth repellent has the composition 49% C, 2.7% H and 48.3% Cl. Its molecular weight is 147 gm. Determine its molecular formula

MC0048

48. 0.5 g of NaOH is required by 0.4 gm of a polybasic acid H_nA (Molecular weight = 96gm) for complete neutralization. Value of 'n' would be : (Assume all H atom are replaced)– **MC0049**
49. The empirical formula of a compounds is CH_2O . 0.25 mole of this compound contains 1 gm hydrogen. The molecular formula of compound is - **MC0050**
50. A compound has 62 % carbon, 10.4 % hydrogen and 27.6 % oxygen. If molar mass of compound is 58, if ratio of no. of C atom to H-atom is y : 1 then value of y is. **MC0051**

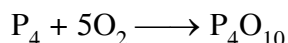
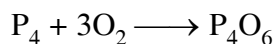
EUDIOMETRY

51. What volume (in ml) of $O_2(g)$ is needed for complete combustion of 40 ml ethane gas (C_2H_6) ? **MC0052**
52. 10 ml of CO is mixed with 25 ml air having 20% O_2 by volume. What would be the final volume (in ml) if none of CO and O_2 is left after the reaction? **MC0053**
53. Calculate the volume (in ml) of CO_2 evolved by the combustion of 50 ml of a mixture containing 40% C_2H_4 and 60% CH_4 (by volume) **MC0054**
54. 10 moles of a mixture of CO (g) and $CH_4(g)$ was mixed with 22 moles of O_2 gas and subjected to sparking. Find the moles of gas absorbed when the residual gases are passed through alc. KOH. **MC0055**
55. 60 ml of a mixture of nitrous oxide and nitric oxide was exploded with excess of hydrogen. If 38 ml of N_2 was formed, calculate % volume of NO gas in the mixture. **MC0056**
56. When 100 ml of a $O_2 - O_3$ mixture was passed through turpentine, there was reduction of volume by 20 ml. If 100 ml of such a mixture is heated, what will be the increase in volume (in ml)? **MC0057**
57. 10 ml of an oxide of nitrogen produce 20 ml NO_2 and 5 ml O_2 on complete decomposition. The oxide of nitrogen is- **MC0058**
58. A gaseous alkane is exploded with O_2 . The volume of O_2 required for complete combustion and the volume of CO_2 formed after combustion are in 7 : 4 ratio. What is the molecular formula of alkane ? **MC0059**
59. When a certain quantity of oxygen was ozonised in a suitable apparatus, the volume decreased by 4 ml. On addition of turpentine the volume further decreased by 8 ml. All volumes were measured at the same temperature and pressure. From these data, if formula of ozone is O_x then find x. **MC0060**
60. A 20 ml mixture of C_2H_4 and C_2H_2 undergoes sparking in gas eudiometer with just sufficient amount of O_2 and shows contraction of 37.5 ml. The volume (in ml) of C_2H_2 in the mixture is. **MC0061**

EXERCISE # S-II

MOLE

1. Two substance P_4 & O_2 are allowed to react completely to form mixture of P_4O_6 & P_4O_{10} leaving none of the reactants. Using this information calculate the moles of P_4O_6 and P_4O_{10} in the final mixture when the following amounts of P_4 & O_2 are taken.



- (i) If 1 mole P_4 & 4 mole of O_2
 (ii) If 3 mole P_4 & 11 mole of O_2
 (iii) If 3 mole P_4 & 13 mole of O_2 MC0062
2. By the reaction of carbon and oxygen, a mixture of CO and CO_2 is obtained. What is the mass percent of CO in of the mixture obtained when 20 grams of O_2 reacts with 12 grams of carbon ? MC0063

3. Nitrogen (N), phosphorus (P), and potassium (K) are the main nutrients in plant fertilizers. According to an industry convention, the numbers on the label refer to the mass % of N, P_2O_5 , and K_2O , in that order. If the N : P : K ratio of a 28 : 11.75 : 11.75 fertilizer in terms of moles of each elements, and express it as x : y : 1. Find value of y (P = 31, K = 39) MC0064

4. A 10 g sample of a mixture of calcium chloride and sodium chloride is treated with Na_2CO_3 to precipitate calcium as calcium carbonate. This $CaCO_3$ is heated to convert all the calcium to CaO and the final mass of CaO is 1.12gm. Calculate % by mass of NaCl in the original mixture. MC0065

5. A mixture of Ferric oxide (Fe_2O_3) and Al is used as a solid rocket fuel which reacts to give Al_2O_3 and Fe. No other reactants and products are involved. On complete reaction of 1 mole of Fe_2O_3 , 200 units of energy is released.

- (a) Write a balance reaction representing the above change.
 (b) If the ratio of masses of Fe_2O_3 and Al taken so that maximum energy per unit mass of fuel is released is y : 1 then value of y is

- (c) What would be energy released if 16 kg of Fe_2O_3 reacts with 2.7 kg of Al. MC0066

6. 5.33 mg of salt $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ is treated with excess of $AgNO_3(aq.)$, then mass (in mg) of AgCl precipitate obtained will be :

Given : [Cr = 52, Cl = 35.5, Ag = 108] MC0067

7. Mass % of oxygen in monovalent metal carbonate is 48%. If the number of atoms of metal present in 5mg of this metal carbonate sample is $y \times 10^{19}$ then value of y is ($N_A = 6.022 \times 10^{23}$). MC0068

8. The formula of compound composed of A & B which is given by A_xB_y , it is strongly heated in oxygen as per reaction-



If 2.5gm of A_xB_y on oxidation gives 3gm oxide of A, If ratio of x : y is z : 1 then value of z is.

[Atomic mass of A = 24 & B = 14] MC0069

9. Calculate the maximum mass of CaCl_2 produced when 2.4×10^{24} atoms of calcium is taken with 96 litre of Cl_2 gas at 380 mm pressure and at 27°C .
[$R = 0.08 \text{ atm L/mole-K}$ & $N_A = 6 \times 10^{23}$] **MC0070**
10. $\text{P}_4\text{S}_3 + 8\text{O}_2 \longrightarrow \text{P}_4\text{O}_{10} + 3\text{SO}_2$
Calculate mass in gm of P_4S_3 is required to produce at least 9.6 gm of each product. ($P = 31, S = 32$)
MC0071
11. Consider the given reaction
 $\text{H}_4\text{P}_2\text{O}_7 + 2\text{NaOH} \rightarrow \text{Na}_2\text{H}_2\text{P}_2\text{O}_7 + 2\text{H}_2\text{O}$
If 534 gm of $\text{H}_4\text{P}_2\text{O}_7$ is reacted with 3.0×10^{24} formula units of NaOH , then the total number of moles of H_2O produced is ($N_A = 6 \times 10^{23}$) ($P = 31$) **MC0072**

Comprehension based on “Law of Conservation of Mass” (12 & 13)

It states that matter can neither be created nor destroyed.

This law was put forth by Antoine Lavoisier in 1789. He performed careful experimental studies for combustion reactions for reaching to the above conclusion. This law formed the basis for several later developments in chemistry. Infact, this was the result of exact measurement of masses of reactants and products, and carefully planned experiments performed by Lavoisier.

12. What weight (in gm) of silver nitrate will react with 0.585 gm NaCl to produce 1.435 gm AgCl and 0.85 gm NaNO_3 **MC0073**
13. 6.3 gm sodium bicarbonate is added to 15 gm acetic acid solution. CO_2 formed is allowed to escape. The weight of the solution left is 18 gm. What is the mass (in gm) of CO_2 formed. **MC0073**

Comprehension based on “Law of Definite Proportions” (14 & 15)

This law was given by, a French chemist, Joseph Proust. He stated that a given compound always contains exactly the same proportion of elements by weight.

Proust worked with two samples of cupric carbonate—one of which was of natural origin and the other was synthetic one. He found that the composition of elements present in it was same for both the samples as shown below :

	% of copper	% of oxygen	% of carbon
Natural Sample	51.35	9.74	38.91
Synthetic Sample	51.35	9.74	38.91

Thus, irrespective of the source, a given compound always contains same elements in the same proportion. The validity of this law has been confirmed by various experiments. It is sometimes also referred to as **Law of constant composition**.

Limitation :

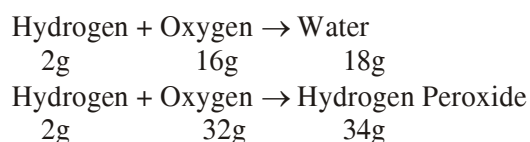
The law is not applicable if the compound is formed from different isotopes of an element. The two isotopes of carbon C-12 and C-14 form carbondioxide C^{12}O_2 and C^{14}O_2 . The ratio of C : O is 12 : 32 and 14 : 32 respectively. It is not a constant ratio.

14. 0.5 gm silver is dissolved in excess of nitric acid. This solution is treated with excess of NaCl solution when 0.66 gm AgCl is formed. One gram metallic silver wire is heated in dry Cl_2 , 1.32 gm AgCl is formed. Show that these data confirm the law of constant proportion. **MC0074**
15. 6.488 gm lead reacts with 1.002 gm oxygen to form an oxide. This oxide is also obtained by heating $\text{Pb}(\text{NO}_3)_2$. It is found that % of lead in this oxide is 86.62. Show that these data illustrate the law of definite proportions. **MC0075**

Comprehension based on "Law of Multiple Proportions" (16 to 17)

This law was proposed by Dalton in 1803. According to this law, if two elements combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element, are in the ratio of small whole numbers.

For example, hydrogen combines with oxygen to form two compounds, namely, water and hydrogen peroxide.



Here, the masses of oxygen (i.e. 16 gm and 32 gm) which combine with a fixed mass of hydrogen (2 gm) bear a simple ratio, i.e. 16:32 or 1: 2.

16. An element forms two oxides. In one oxide, one gram of the oxide contains 0.5 gm of the element. In another oxide, 4 gm of the oxide contains 0.8 gm of the element. Show that these data confirm the law of multiple proportion. **MC0076**
17. 0.11 gm of an oxide of nitrogen gives 56 mL N_2 at 273 K and 1 atm. 0.15 gm of another oxide of nitrogen gives 56 mL N_2 at 1 atm, 273K. Show that these data confirm the law of multiple proportion. **MC0077**
18. 10 ml of a mixture of CH_4 , C_2H_4 and CO_2 were exploded with excess of air. After explosion and further cooling, there was contraction of 17 ml and after treatment with KOH, there was further reduction of 14 ml. What is the composition of the mixture (in ml)? **MC0078**
19. 40 ml of a mixture of C_2H_2 and CO is mixed with 100 ml of O_2 gas and the mixture is exploded. The residual gases occupied 104 ml and when these are passed through KOH solution, the volume becomes 48 ml. All the volume are at same temperature and pressure. If ratio of volume of C_2H_2 & CO is $y : 1$, then value of y is **MC0079**
20. 10 mL of gaseous organic compound containing C, H and O only, was mixed with 100 mL of O_2 and exploded under identical conditions and then cooled. The volume left after cooling was 90 mL. On treatment with KOH a contraction of 20 mL was observed. If vapour density of compound is 23, if molecular formula of the compound is $\text{C}_x\text{H}_y\text{O}_z$, then find $(x + y + z)$. **MC0080**

EXERCISE # O-I

SINGLE CORRECT :

ATOMIC MASS, MOLECULAR MASS MOLE

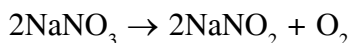
- One atomic mass unit in kilogram is
(A) $1/N_A$ (B) $12 / N_A$ (C) $1/1000 N_A$ (D) $1000 / N_A$ **MC0081**
- A gaseous mixture contains CO_2 (g) and N_2O (g) in a 2 : 5 ratio by mass. The ratio of the number of molecules of CO_2 (g) and N_2O (g) is
(A) 5 : 2 (B) 2 : 5 (C) 1 : 2 (D) 5 : 4 **MC0082**
- Which of the following contain largest number of carbon atoms?
(A) 15 gm ethane, C_2H_6 (B) 40.2 gm sodium oxalate, $\text{Na}_2\text{C}_2\text{O}_4$
(C) 72 gm glucose, $\text{C}_6\text{H}_{12}\text{O}_6$ (D) 35 gm pentene, C_5H_{10} **MC0083**
- The number of hydrogen atoms in 0.9 gm glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is same as
(A) 0.048 gm hydrazine, N_2H_4 (B) 0.17 gm ammonia, NH_3
(C) 0.30 gm ethane, C_2H_6 (D) 0.03 gm hydrogen, H_2 **MC0084**
- Ethanol, $\text{C}_2\text{H}_5\text{OH}$, is the substance commonly called alcohol. The density of liquid alcohol is 0.7893 g/ml at 293 K. If 1.2 mole of ethanol are needed for a particular experiment, what volume of ethanol should be measured out?
(A) 55 ml (B) 58 ml (C) 70 ml (D) 79 ml **MC0085**
- An iodized salt contains 0.5 % of NaI. A person consumes 3 gm of salt everyday. The number of iodide ions going into his body everyday is (I = 127)
(A) 10^{-4} (B) 6.02×10^{-4} (C) 6.02×10^{19} (D) 6.02×10^{23} **MC0086**
- The percentage by mole of NO_2 in a mixture of NO_2 (g) and NO (g) having average molecular mass 34 is :
(A) 25% (B) 20% (C) 40% (D) 75% **MC0087**
- The number of carbon atoms present in a signature, if a signature written by carbon pencil, weighing 1.2×10^{-3} g is
(A) 12.04×10^{20} (B) 6.02×10^{19} (C) 3.01×10^{19} (D) 6.02×10^{20} **MC0088**
- The average atomic mass of a mixture containing 79 mole % of ^{24}Mg and remaining 21 mole % of ^{25}Mg and ^{26}Mg , is 24.31. % mole of ^{26}Mg is
(A) 5 (B) 20 (C) 10 (D) 15 **MC0089**
- How many litres of oxygen at 1atm & 273K will be required to burn completely 2.2 g of propane (C_3H_8)
(A) 11.2 L (B) 22.4 L (C) 5.6 L (D) 44.8 L **MC0090**
- If $1/2$ moles of oxygen combine with aluminium to form Al_2O_3 then weight of Aluminium metal used in the reaction is (Al= 27)
(A) 27 g (B) 18 g (C) 54 g (D) 40.5 g **MC0091**
- Volume of CO_2 obtained at STP by the complete decomposition of 9.85 g BaCO_3 is (Ba = 137)
(A) 2.24 lit (B) 1.12 lit (C) 1.135 lit (D) 2.27 lit **MC0092**

13. The drain cleaner, **Drainex** contains small bits of aluminium ($Al = 27$) which reacts with caustic soda to produce dihydrogen. What is the volume (in ml) of dihydrogen at 27°C and 1.013 bar that is produced when 0.27 gm of aluminium reacts :



- (A) 0.369 (B) 369.0 (C) 246.0 (D) 540.0 **MC0093**

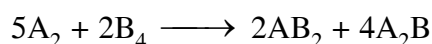
14. Volume of O_2 obtained at 2 atm & 546K, by the complete decomposition of 8.5 g $NaNO_3$ is



- (A) 2.24 lit (B) 1.12 lit (C) 0.84 lit (D) 0.56 lit **MC0094**

LIMITING REAGENT

15. The minimum mass of mixture of A_2 and B_4 required to produce at least 1 kg of each product is :
(Given At. mass of 'A' = 10 ; At. mass of 'B' = 120)

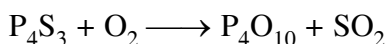


- (A) 2120 gm (B) 1060 gm (C) 560 gm (D) 1660 gm **MC0095**

16. The mass of CO_2 produced from 620 gm mixture of $C_2H_4O_2$ & O_2 , prepared to produce maximum energy is (Combustion reaction is exothermic)

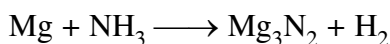
- (A) 413.33 gm (B) 593.04 gm (C) 440 gm (D) 320 gm **MC0096**

17. The mass of P_4O_{10} produced if 440 gm of P_4S_3 is mixed with 384 gm of O_2 is



- (A) 568 gm (B) 426 gm (C) 284 gm (D) 396 gm **MC0097**

18. The mass of Mg_3N_2 produced if 48 gm of Mg metal is reacted with 34 gm NH_3 gas is



- (A) $\frac{200}{3}$ gm (B) $\frac{100}{3}$ gm (C) $\frac{400}{3}$ gm (D) $\frac{150}{3}$ gm **MC0098**

PROBLEM RELATED WITH MIXTURE

19. An ideal gaseous mixture of ethane (C_2H_6) and ethene (C_2H_4) occupies 28 litre at 1atm, 0°C . The mixture reacts completely with 128 gm O_2 to produce CO_2 and H_2O . Mole fraction of C_2H_6 in the mixture is—

- (A) 0.6 (B) 0.4 (C) 0.5 (D) 0.8 **MC0099**

20. 280 g of a mixture containing CH_4 and C_2H_6 in 5 : 2 molar ratio is burnt in presence of excess of oxygen. Calculate total moles of CO_2 produced.

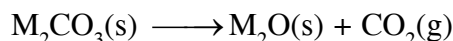
- (A) 9 (B) 18 (C) 7 (D) 12 **MC0100**

21. Mixture of $MgCO_3$ & $NaHCO_3$ on strong heating gives CO_2 & H_2O in 3 : 1 mole ratio. The weight % of $NaHCO_3$ present in the mixture is:

- (A) 30% (B) 80% (C) 40% (D) 50% **MC0101**

PERCENTAGE YIELD, DEGREE OF DISSOCIATION, PERCENTAGE PURITY

22. A metal carbonate decomposes according to following reaction



Percentage loss in mass on complete decomposition of $M_2CO_3(s)$

(Atomic mass of M = 102)

- (A) $\frac{100}{3}\%$ (B) $\frac{50}{3}\%$ (C) $\frac{25}{3}\%$ (D) 15% **MC0102**

23. 90 gm mixture of H_2 and O_2 is taken in stoichiometric ratio and gives H_2O with 50% yield. The produced mass of H_2O (in gm) is :

- (A) 45 gm (B) 36 gm (C) 20 gm (D) 90 gm **MC0103**

24. An impure sample of $CaCO_3$ contains 38% of Ca. The percentage of impurity present in the sample is :

- (A) 5% (B) 95% (C) 10% (D) 2.5% **MC0104**

25. The vapour density of sample of partially decomposed cyclobutane (C_4H_8) gas is 20. The degree of dissociation of C_4H_8 into C_2H_4 gas is -

- (A) 0.25 (B) 0.50 (C) 0.60 (D) 0.40 **MC0105**

26. A sample of NH_3 gas is 20% dissociated into N_2 and H_2 gases. The mass ratio of N_2 and NH_3 gases in the final sample is -

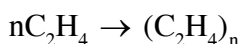
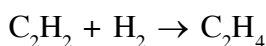
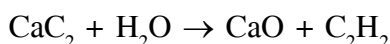
- (A) $\frac{7}{34}$ (B) $\frac{7}{17}$ (C) $\frac{14}{17}$ (D) $\frac{21}{17}$ **MC0106**

27. The density of a sample of SO_3 gas is 2.5 g/L at $0^\circ C$ and 1 atm. It's degree of dissociation into SO_2 and O_2 gases is -

- (A) $\frac{6}{7}$ (B) $\frac{1}{7}$ (C) $\frac{3}{7}$ (D) $\frac{5}{7}$ **MC0107**

SEQUENTIAL AND PARALLEL REACTIONS

28. Polyethene can be prepared by CaC_2 by the following sequence of reactions.



(Polyethene)

The mass in kg of polyethene that can be prepared by 20 kg CaC_2 .

- (A) 4.1 kg (B) 8.75 kg (C) 3.78 kg (D) 10 kg **MC0108**

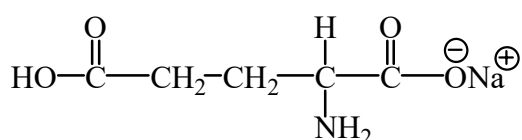
29. 25.4 gm of iodine and 14.2 gm of chlorine are made to react completely to yield a mixture of ICl and ICl_3 . Ratio of moles of ICl & ICl_3 formed is (Atomic mass : I = 127, Cl = 35.5)

- (A) 1 : 1 (B) 1 : 2 (C) 1 : 3 (D) 2 : 3 **MC0109**

30. One commercial system removes SO_2 emission from smoke at 95°C by the following set of reactions-
- $$\text{SO}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{SO}_2\text{Cl}_2(\text{g})$$
- $$\text{SO}_2\text{Cl}_2 + 2\text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + 2\text{HCl}$$
- $$\text{H}_2\text{SO}_4 + \text{Ca}(\text{OH})_2 \rightarrow \text{CaSO}_4 + 2\text{H}_2\text{O}$$
- Assuming the process to be overall 95% efficient. How many moles of CaSO_4 may be produced from 128 gm SO_2 . [Ca = 40, S- 32, O-16]
- (A) 1.9 moles (B) 2 mol (C) 3.8 mol (D) 0.95 mol **MC0110**
31. Equal masses of KClO_3 undergoes different reaction in two different container :
- (i) $2\text{KClO}_3 \longrightarrow 2\text{KCl} + 3\text{O}_2$
- (ii) $4\text{KClO}_3 \longrightarrow \text{KCl} + 3\text{KClO}_4$
- Mass ratio of the KCl produced in respective reaction is x : 1. Value of 'x' will be.
- (A) 4 (B) 2 (C) 0.25 (D) 3 **MC0111**

PERCENTAGE COMPOSITION, EMPIRICAL AND MOLECULAR FORMULA

32. A compound contains $10^{-2}\%$ of phosphorous. If atomic mass of phosphorus is 31, the molecular mass of the compound having one phosphorus atom per molecule is :-
- (A) 31 (B) 3.1×10^3 (C) 3.1×10^5 (D) 3.1×10^4 **MC0112**
33. 13.4 gm of a sample of unstable hydrated salt : $\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$ was strongly heated. Weight loss on heating is found to be equal to 6.3 gm. Calculate the value of x.
- (A) 6 (B) 5 (C) 7 (D) 8 **MC0113**
34. An organic compound contains 4% sulphur by mass. Its minimum molecular weight is :
- (A) 200 (B) 400 (C) 800 (D) 1600 **MC0114**
35. Monosodium glutamate (MSG) is salt of one of the most abundant naturally occurring non-essential amino acid which is commonly used in food products like in "MAGGI" having structural formula as



Mass % of Na in MSG is-

- (A) 14.8 (B) 15.1 (C) 13.6 (D) 16.5 **MC0115**
36. Which of the following series of compounds have same mass percentage of carbon ?
- (A) CO_2 , CO (B) CH_4 , C_2H_6 , C_2H_2
- (C) C_2H_2 , C_6H_6 , C_{10}H_8 (D) HCHO , CH_3COOH , $\text{C}_6\text{H}_{12}\text{O}_6$ **MC0116**
37. A compound contains 69.5% oxygen and 30.5% nitrogen and its molecular weight is 92. The formula of that compound is :-
- (A) N_2O (B) NO_2 (C) N_2O_4 (D) N_2O_5 **MC0117**
38. 1 lt. of a hydrocarbon weighs as much as one litre of CO_2 under identical conditions. The molecular formula of the hydrocarbon is -
- (A) C_3H_8 (B) C_2H_6 (C) C_2H_4 (D) C_3H_6 **MC0118**

39. Which of the following compounds has same empirical formula as that of glucose:-
 (A) CH_3CHO (B) CH_3COOH (C) CH_3OH (D) C_2H_6 **MC0119**
40. Two oxides of a metal contains 50% and 40% of a metal respectively. The formula of the first oxide is MO . Then the formula of the second oxide is -
 (A) MO_2 (B) M_2O_3 (C) M_2O (D) M_2O_5 **MC0120**
41. A compound of X and Y has equal mass of them. If their atomic weights are 30 and 20 respectively. The molecular formula of compound is -
 (A) X_2Y_2 (B) X_3Y_3 (C) X_2Y_3 (D) X_3Y_2 **MC0121**

EUDIOMETRY

42. 10 ml CH_4 gas is burnt completely in air ($\text{O}_2 = 20\%$, by volume). The minimum volume of air needed is -
 (A) 20 ml (B) 50 ml (C) 80 ml (D) 100 ml **MC0122**
43. $\text{C}_6\text{H}_5\text{OH}(\text{g}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
 Magnitude of volume change if 30 ml of $\text{C}_6\text{H}_5\text{OH}(\text{g})$ is burnt with excess amount of oxygen, is
 (A) 30 ml (B) 60 ml (C) 20 ml (D) 10 ml **MC0123**
44. A mixture of C_2H_2 and C_3H_8 occupied a certain volume at 80 mm Hg. The mixture was completely burnt to CO_2 and $\text{H}_2\text{O}(\text{l})$. The pressure of CO_2 was found to be 230 mm Hg at the same temperature and volume. The fraction of C_2H_2 in mixture is
 (A) 0.125 (B) 0.5 (C) 0.85 (D) 0.25 **MC0124**
45. 20 mL of a mixture of CO and H_2 were mixed with excess of O_2 and exploded & cooled. There was a volume contraction of 23 mL. All volume measurements corresponds to room temperature (27°C) and one atmospheric pressure. Determine the volume ratio $V_1 : V_2$ of CO and H_2 in the original mixture
 (A) 6.5 : 13.5 (B) 5 : 15 (C) 9 : 11 (D) 7 : 13 **MC0125**
46. Each volume of a gaseous organic compound containing C, H and S only produce 1 volume CO_2 , 2 volume H_2O vapours and 1 volume SO_2 gases on complete combustion. The molecular formula of compound is -
 (A) CH_2S (B) CH_4S (C) $\text{C}_2\text{H}_4\text{S}$ (D) $\text{C}_2\text{H}_6\text{S}$ **MC0126**
47. For a chemical reaction occurring at constant pressure and temperature.

$$2\text{A}(\text{g}) + 5\text{B}(\text{g}) \longrightarrow \text{C}(\text{g}) + 2\text{D}(\text{g})$$

 (A) contraction in volume is double the volume of A taken if B is taken in excess.
 (B) contraction in volume is more than the volume of B taken if A is in excess.
 (C) volume contracts by 20 mL if 10 mL A is reacted with 20 mL B.
 (D) no change in volume due to reaction **MC0127**

48. When one litre of CO_2 is passed over hot coke, the volume becomes 1.4 litres. The composition of final products will not be.

(A) $V_{\text{CO}_2} : V_{\text{CO}} = 3 : 4$ (B) $V_{\text{CO}} = 1.6 \text{ ltr.}$ (C) $n_{\text{CO}_2} : n_{\text{CO}} = 3 : 4$ (D) $\% V \text{ of CO} = \frac{400}{7}$

MC0128

49. 10 ml of a compound containing 'N' and 'O' is mixed with 30 ml of H_2 to produce H_2O (l) and 10 ml of N_2 (g). Molecular formula of compound if both reactants reacts completely, is

(A) N_2O (B) NO_2 (C) N_2O_3 (D) N_2O_5 MC0129

50. When a definite volume of a gaseous alkyne ($\text{C}_n\text{H}_{2n-2}$) is burnt completely in excess of air, a contraction in volume equal to twice the volume of alkyne burnt occurred. The value of 'n' is -

(A) 4 (B) 6 (C) 3 (D) 20 MC0130

EXERCISE # O-II

SINGLE CORRECT :

1. A sample of iron ore, weighing 0.700g, is dissolved in nitric acid. The solution is then diluted with water, followed with sufficient concentrated aqueous ammonia, to quantitative precipitation the iron as $\text{Fe}(\text{OH})_3$. The precipitate is filtered, ignited and weighed as Fe_2O_3 . If the mass of the ignited and dried precipitate is 0.541g, what is the mass percent of iron in the original iron ore sample ($\text{Fe} = 56$)
(A) 27.0 % (B) 48.1 % (C) 54.1 % (D) 81.1 % **MC0131**
2. A sample of pure Cu (4.00g) heated in a stream of oxygen for some time, gains in weight with the formation of black oxide of copper (CuO). The final mass is 4.90 g. What percent of copper remains unoxidized ($\text{Cu} = 64$)
(A) 90 % (B) 10 % (C) 20 % (D) 80 % **MC0132**
3. 1120 ml of ozonised oxygen ($\text{O}_2 + \text{O}_3$) at 1 atm & 273K weighs 1.76 gm. The reduction in volume on passing this through alkaline pyrogallol solution is -
(A) 896 ml (B) 224 ml (C) 448 ml (D) 672 ml **MC0133**

ASSERTION REASON:

4. **Statement-1 :** When a gaseous hydrocarbon is burnt in excess of oxygen and the products of combustion are cooled to the original temperature and pressure, a contraction in volume occurs.
Statement-2 : The contraction in volume is solely due to the liquifaction of water vapour.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true. **MC0134**
5. **Statement -1 :** $2\text{A} + 3\text{B} \longrightarrow \text{C}$
 $\frac{4}{3}$ moles of 'C' are always produced when 3 moles of 'A' & 4 moles of 'B' are added.
Statement -2 : 'B' is the limiting reactant for the given data.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is false, statement-2 is true.
(D) Statement-1 is true, statement-2 is false. **MC0135**
6. **Assertion :** During a chemical reaction, total moles remains constant.
Reason : During a chemical reaction, total mass remains constant.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true. **MC0136**

MULTIPLE CORRECT :

7. 40 gm of a carbonate of an **alkali metal** or **alkaline earth metal** containing some inert impurities was made to react with excess HCl solution. The liberated CO_2 occupied 12.315 lit. at 1 atm & 300 K. The correct option is

(A) Mass of impurity is 1 gm and metal is Be (B) Mass of impurity is 3 gm and metal is Li
(C) Mass of impurity is 5 gm and metal is Be (D) Mass of impurity is 2 gm and metal is Mg

MC0137

8. 1 mole of H_2SO_4 will exactly neutralise :

(A) 2 mole of ammonia (B) 1 mole of $\text{Ba}(\text{OH})_2$
(C) 0.5 mole of $\text{Ca}(\text{OH})_2$ (D) 2 mole of KOH

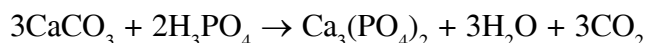
MC0138

9. 12 g of Mg was burnt in a closed vessel containing 32 g oxygen. Which of the following is /are correct.

(A) 2 gm of Mg will be left unburnt.
(B) 0.75 gm-molecule of O_2 will be left unreacted.
(C) 20 gm of MgO will be formed.
(D) The mixture at the end will weight 44 g.

MC0139

10. 50 gm of CaCO_3 is allowed to react with 68.6 gm of H_3PO_4 then select the correct option(s)-



(A) 51.67 gm salt is formed (B) Amount of unreacted reagent = 35.93 gm
(C) n_{CO_2} formed = 0.5 moles (D) $n_{\text{H}_2\text{O}}$ formed = 0.7 mole

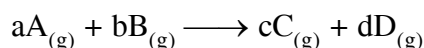
MC0140

11. Select the correct statement(s) for $(\text{NH}_4)_3\text{PO}_4$.

(A) Ratio of number of oxygen atoms to number of hydrogen atoms is 1 : 3
(B) Ratio of number of cations to number of anions is 3 : 1
(C) Ratio of number of gm-atoms of nitrogen to gm-atoms of oxygen is 3 : 2
(D) Total number of atoms in one mole of $(\text{NH}_4)_3\text{PO}_4$ is 20.

MC0141

12. Two gases A and B which react according to the equation



to give two gases C and D are taken (amount not known) in an Eudiometer tube (operating at a constant Pressure and temperature) to cause the above.

If on causing the reaction there is no volume change observed then which of the following statement is/are correct.

(A) $(a + b) = (c + d)$
(B) average molecular mass may increase or decrease if either of A or B is present in limited amount.
(C) Vapour Density of the mixture will remain same throughout the course of reaction.
(D) Total moles of all the component of mixture will change.

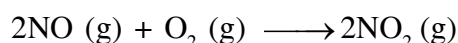
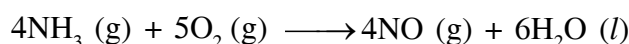
MC0142

13. 100 ml mixture of CO and CO₂ mixed with 30 mL of O₂ and sparked in eudiometer tube. The residual gas after treatment with aq. KOH has a volume of 10 mL which remains unchanged when treated with alkaline pyrogallol. If all the volumes are under the same conditions, point out **correct** option(s):
- (A) The volume of CO that reacts, is 60 mL
 (B) The volume of CO that remains unreacted, is 10 mL
 (C) The volume of O₂ that remains unreacted, is 10 mL
 (D) The volume of CO₂ that gets absorbed by aq.KOH, is 90 mL.

MC0143

Paragraph for Q.14 to Q.15

For the given series of reaction



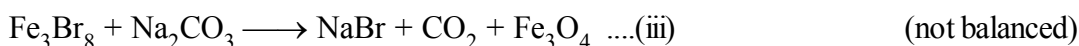
14. If 20 ml of NH₃ is mixed with 100 ml of O₂. Volume contraction at the completion of above reactions is
- (A) 20 ml (B) 85 ml (C) 35 ml (D) 100 ml
15. Total volume of O₂ used if 20 ml NH₃ is mixed with 100 ml O₂
- (A) 40 (B) 60 (C) 35 (D) None of these

MC0144

MC0144

Paragraph for Q.16 to Q.18

NaBr, used to produce AgBr for use in photography can be self prepared as follows :



(At. mass : Fe = 56, Br = 80)

16. Mass of iron required to produce 2.06×10^3 kg NaBr
- (A) 420 gm (B) 420 kg (C) 4.2×10^5 kg (D) 4.2×10^8 gm
17. If the yield of (ii) is 60% & (iii) reaction is 70% then mass of iron required to produce 2.06×10^3 kg NaBr
- (A) 10^5 kg (B) 10^5 gm (C) 10^3 kg (D) None
18. If yield of (iii) reaction is 90% then mole of CO₂ formed when 2.06×10^3 gm NaBr is formed
- (A) 20 (B) 10 (C) 9 (D) 440

MC0145

MC0146

MC0147

Comprehension Q.19 and Q.20 (2 questions)

Estimation of halogens (Carius method) : A known mass of compound is heated with conc. HNO_3 in the presence of AgNO_3 contained in a hard glass tube known as carius tube in a furnace. C and H are oxidised to CO_2 and H_2O . The halogen forms the corresponding AgX . It is filtered, dried, and weighed.

Estimation of sulphur : A known mass of compound is heated with fuming HNO_3 or sodium peroxide (Na_2O_2) in the presence of BaCl_2 solution in Carius tube. Sulphur is oxidised to H_2SO_4 and precipitated as BaSO_4 . It is filtered, dried and weighed.

- 19** 0.15gm of an organic compound gave 0.12 gm of silver bromide by the Carius method. Find the percentage of bromine in the compound. ($\text{Ag} = 108$, $\text{Br} = 80$)
 (A) 34.0 (B) 46.0 (C) 80.0 (D) 50.0 **MC0148**
- 20** 0.32 gm of an organic substance when treated by Carius method gave 0.466 gm of BaSO_4 . Calculate the percentage of sulphur in the compound. ($\text{Ba} = 137$)
 (A) 10.0 (B) 34.0 (C) 20.0 (D) 30.0 **MC0148**

Comprehension Q.21 and Q.22 (2 questions)

Estimation of phosphorous : A known mass of compound is heated with fuming HNO_3 or sodium peroxide (Na_2O_2) in Carius tube which converts phosphorous to H_3PO_4 . Magnesia mixture ($\text{MgCl}_2 + \text{NH}_4\text{Cl}$) is then added, which gives the precipitate of magnesium ammonium phosphate (MgNH_4PO_4) which on heating gives magnesium pyrophosphate ($\text{Mg}_2\text{P}_2\text{O}_7$), which is weighed.

- 21** 0.124 gm of an organic compound containing phosphorus gave 0.222 gm of $\text{Mg}_2\text{P}_2\text{O}_7$ by the usual analysis. Calculate the percentage of phosphorous in the compound. ($\text{Mg} = 24$, $\text{P} = 31$)
 (A) 25 (B) 75 (C) 62 (D) 50 **MC0149**
- 22** An organic compound has 6.2 % of phosphorus. On sequence of reaction, the phosphorous present in the 10 gm of organic compound is converted to $\text{Mg}_2\text{P}_2\text{O}_7$. Find the weight of $\text{Mg}_2\text{P}_2\text{O}_7$ formed.
 (A) 2.22 gm (B) 10.0 gm (C) 4.44 gm (D) 1.11 gm **MC0149**

TABLE TYPE :

	Column-I	Column-II	Column-III
(P)	60 gram sample of hydro carbon that contain 20% H and rest C	(1) %C = 40	(i) No. of atoms of C as well as O = $8N_A$
(Q)	240 gram urea	(2) %H = $\frac{20}{3}$	(ii) No. of C atoms = $4N_A$
(R)	120 gram acetic acid	(3) % O = $\frac{160}{3}$	(iii) No. of total atoms = $16N_A$
(S)	120 gram glucose	(4) % N = 46.7	(iv) No. of total atoms is 2 times of no. of H atom

- 23.** Out of below correct matching is -
 (A) P – 1 – i (B) P – 1 – ii (C) Q – 2 – iii (D) S – 2 – iv **MC0150**
- 24.** In which of following is incorrect -
 (A) Q – 2 – iv (B) R – 3 – iv (C) P – 4 – iii (D) R – 1 – ii **MC0150**
- 25.** Out of below correct matching is -
 (A) S – 4 – iv (B) R – 1 – ii (C) P – 4 – iii (D) P – 2 – ii **MC0150**

	Column-I (Gas taken)	Column-II (O₂ needed for complete combustion)	Column-III (Contraction in volume)
(1)	20 ml C ₂ H ₄	(i) 60 ml	(I) 50 ml
(2)	25 ml C ₃ H ₄	(ii) 100 ml	(II) 40 ml
(3)	30 ml C ₂ H ₆	(iii) 70 ml	(III) 75 ml
(4)	35 ml CH ₄	(iv) 105 ml	(IV) 70 ml

All volumes are measured at 25°C and 1 atm.

- 26.** Which of the following is correct match -
 (A) 1 – i – II (B) 1 – iii – IV (C) 2 – iv – II (D) 2 – ii – III **MC0151**
- 27.** Which of the following is correct match -
 (A) 3 – iii – III (B) 3 – iv – III (C) 4 – iii – III (D) 4 – iv – IV **MC0151**
- 28.** Which of the following is incorrect (**One or more than one correct**)
 (A) 2 – ii – I (B) 4 – iii – IV (C) 3 – iv – IV (D) 1 – iii – II **MC0151**

MATCH THE COLUMN :

- 29.** One type of artificial diamond (commonly called YAG for yttrium aluminium garnet) can be represented by the formula $\text{Y}_3\text{Al}_5\text{O}_{12}$. [**Y = 89, Al = 27**]

Column I	Column II	
Element	Weight percentage	
(A) Y	(P) 22.73%	
(B) Al	(Q) 32.32%	
(C) O	(R) 44.95%	MC0152

- 30.** The recommended daily dose is 17.6 milligrams of vitamin C (ascorbic acid) having formula $C_6H_8O_6$. Match the following. Given : $N_A = 6 \times 10^{23}$

Column I	Column II
(A) O-atoms present in daily dose	(P) 10^{-4} mole
(B) Moles of vitamin C in 1 gm of vitamin C	(Q) 5.68×10^{-3}
(C) Moles of vitamin C that should be consumed daily	(R) 3.6×10^{20}

MC0153

31. Gaseous alkane (C_nH_{2n+2}) exploded with oxygen. Ratio of the mol of O_2 for complete combustion to the mole of CO_2 formed is given in column-I & in column II formula is given.

Column-I

- (A) 7 : 4
(B) 2 : 1
(C) 5 : 3
(D) 13 : 8

Column-II

- (P) C_3H_8
(Q) C_4H_{10}
(R) C_2H_6
(S) CH_4

MC0154

MATCHING LIST TYPE :**32. Column-I**

- (P) $2H_2 + O_2 \rightarrow 2H_2O$
1g 1g
(Q) $3H_2 + N_2 \rightarrow 2NH_3$
1g 1g
(R) $H_2 + Cl_2 \rightarrow 2HCl$
1g 1g
(S) $2H_2 + C \rightarrow CH_4$
1g 1g

Column-II**(Mass of product)**

- (1) 1.028 g
(2) 1.333 g
(3) 1.125 g
(4) 1.214 g

Code:

	P	Q	R	S
(A)	3	4	1	2
(B)	2	4	1	3
(C)	4	3	1	2
(D)	2	3	1	4

MC0155

EXERCISE # JEE-MAINS

- A gaseous hydrocarbon gives upon combustion 0.72 g of water and 3.08 g of CO_2 . The empirical formula of the hydrocarbon is [JEE(Main)-2013]

(1) C_2H_4 (2) C_3H_4 (3) C_6H_5 (4) C_7H_8 MC0175
- The ratio of masses of oxygen and nitrogen in a particular gaseous mixture is 1 : 4. The ratio of number of their molecule is : [JEE(Main)-2014]

(1) 1 : 8 (2) 3 : 16 (3) 1 : 4 (4) 7 : 32 MC0174
- In Carius method of estimation of halogens, 250 mg of an organic compound gave 141 mg of AgBr. The percentage of bromine in the compound is : [JEE(Main)-2015]

(Atomic mass Ag = 108; Br = 80)

(1) 48 (2) 60 (3) 24 (4) 36 MC0173
- The most abundant elements by mass in the body of a healthy human adult are : Oxygen (61.4%) ; Carbon (22.9%), Hydrogen (10.0%) ; and Nitrogen (2.6%). The weight which a 75 kg person would gain if all ^1H atoms are replaced by ^2H atoms is [JEE(Main)-2017]

(1) 15 kg (2) 37.5 kg (3) 7.5 kg (4) 10 kg MC0171
- 1 gram of a carbonate (M_2CO_3) on treatment with excess HCl produces 0.01186 mole of CO_2 . the molar mass of M_2CO_3 in g mol^{-1} is :- [JEE(Main)-2017]

(1) 1186 (2) 84.3 (3) 118.6 (4) 11.86 MC0172
- The ratio of mass percent of C and H of an organic compound ($\text{C}_x\text{H}_y\text{O}_z$) is 6 : 1. If one molecule of the above compound ($\text{C}_x\text{H}_y\text{O}_z$) contains half as much oxygen as required to burn one molecule of compound C_xH_y completely to CO_2 and H_2O . The empirical formula of compound $\text{C}_x\text{H}_y\text{O}_z$ is [JEE(Main)-2018 (offline)]

(1) $\text{C}_2\text{H}_4\text{O}$ (2) $\text{C}_3\text{H}_4\text{O}_2$ (3) $\text{C}_2\text{H}_4\text{O}_3$ (4) $\text{C}_3\text{H}_6\text{O}_3$ MC0168
- For per gram of reactant, the maximum quantity of N_2 gas is produced in which of the following thermal decomposition reactions ? [JEE(Main)-2018 (online)]

(Given : Atomic wt. – Cr = 52u, Ba = 137u)

(1) $2\text{NH}_4\text{NO}_3(\text{s}) \rightarrow 2\text{N}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g}) + \text{O}_2(\text{g})$

(2) $\text{Ba}(\text{N}_3)_2(\text{s}) \rightarrow \text{Ba}(\text{s}) + 3\text{N}_2(\text{g})$

(3) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7(\text{s}) \rightarrow \text{N}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g})$

(4) $2\text{NH}_3(\text{g}) \rightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ MC0169
- An unknown chlorohydrocarbon has 3.55% of chlorine. If each molecule of the hydrocarbon has one chlorine atom only; chlorine atoms present in 1 g of chlorohydrocarbon are : [JEE(Main)-2018 (online)]

(Atomic wt. of Cl = 35.5 u; Avogadro constant = $6.023 \times 10^{23} \text{ mol}^{-1}$)

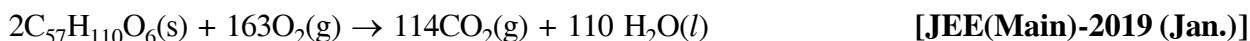
(1) 6.023×10^{21} (2) 6.023×10^{23} (3) 6.023×10^{20} (4) 6.023×10^9 MC0170

9. A 10 mg effervescent tablet containing sodium bicarbonate and oxalic acid releases 0.25 ml of CO_2 at $T = 298.15 \text{ K}$ and $p = 1 \text{ bar}$. If molar volume of CO_2 is 25.0 L under such condition, what is the percentage of sodium bicarbonate in each tablet ? [Molar mass of $\text{NaHCO}_3 = 84 \text{ g mol}^{-1}$]

[JEE(Main)-2019 (Jan.)]

- (1) 16.8 (2) 8.4 (3) 0.84 (4) 33.6 **MC0160**

- 10.** For the following reaction, the mass of water produced from 445 g of $\text{C}_{57}\text{H}_{110}\text{O}_6$ is :

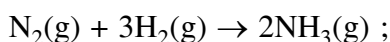


- (1) 495 g (2) 490 g (3) 890 g (4) 445 g **MC0161**

- 11.** The percentage composition of carbon by mole in methane is : **[JEE(Main)-2019 (-april)]**

- (1) 80% (2) 25% (3) 75% (4) 20%
- MC0162**

- 12.** For a reaction, **[JEE(Main)-2019 (-april)]**



identify dihydrogen (H_2) as a limiting reagent in the following reaction mixtures.

- (1) 14g of N_2 + 4g of H_2 (2) 28g of N_2 + 6g of H_2
(3) 56g of N_2 + 10g of H_2 (4) 35g of N_2 + 8g of H_2 **MC0163**

- 13.** 5 moles of AB_2 weigh 125×10^{-3} kg and 10 moles of A_2B_2 weigh 300×10^{-3} kg. The molar mass of A (M_A) and molar mass of B (M_B) in kg mol^{-1} are : **[JEE(Main)-2019 (April)]**

- (1) $M_A = 50 \times 10^{-3}$ and $M_B = 25 \times 10^{-3}$ (2) $M_A = 25 \times 10^{-3}$ and $M_B = 50 \times 10^{-3}$
 (3) $M_A = 5 \times 10^{-3}$ and $M_R = 10 \times 10^{-3}$ (4) $M_A = 10 \times 10^{-3}$ and $M_R = 5 \times 10^{-3}$

MC0164

- 14.** The minimum amount of $O_7(g)$ consumed per gram of reactant is for the reaction :

(Given atomic mass : Fe = 56, O = 16, Mg = 24, P = 31, C = 12, H = 1)

- (1) $\text{C}_3\text{H}_8(\text{g}) + 5 \text{O}_2(\text{g}) \rightarrow 3 \text{CO}_2(\text{g}) + 4 \text{H}_2\text{O}(\text{l})$ [JEE(Main)-2019 (April)]

- $$(2) \text{P}_4(\text{s}) + 5 \text{O}_2(\text{g}) \rightarrow \text{P}_4\text{O}_{10}(\text{s})$$

- $$(3) \quad 4 \text{ Fe(s)} + 3 \text{ O}_2\text{(g)} \rightarrow 2 \text{ Fe}_2\text{O}_3\text{(s)}$$

- (4) $2 \text{Mg(s)} + \text{O}_2\text{(g)} \rightarrow 2 \text{MgO(s)}$ **MC0165**

- 15.** At 300 K and 1 atmospheric pressure, 10 mL of a hydrocarbon required 55 mL of O_2 for complete combustion and 40 mL of CO_2 is formed. The formula of the hydrocarbon is :

[JEE(Main)-2019 (Jan)]

- (1) C_4H_8 (2) $\text{C}_4\text{H}_7\text{Cl}$ (3) C_4H_{10} (4) C_4H_6 **MC0166**

- 16.** 25 g of an unknown hydrocarbon upon burning produces 88 g of CO_2 and 9 g of H_2O . This unknown hydrocarbon contains. **[JEE(Main)-2019 (Jan)]**

- (1) 20g of carbon and 5 g of hydrogen (2) 24g of carbon and 1 g of hydrogen
(3) 18g of carbon and 7 g of hydrogen (4) 22g of carbon and 3 g of hydrogen

MC0167

17. NaClO_3 is used, even in spacecrafts, to produce O_2 . The daily consumption of pure O_2 by a person is 492L at 1 atm, 300K. How much amount of NaClO_3 , in grams, is required to produce O_2 for the daily consumption of a person at 1 atm, 300 K ? **[JEE(Main)-2020 (Jan)]**

$$\text{NaClO}_3(\text{s}) + \text{Fe}(\text{s}) \rightarrow \text{O}_2(\text{g}) + \text{NaCl}(\text{s}) + \text{FeO}(\text{s}) \quad R = 0.082 \text{ L atm mol}^{-1} \text{ K}^{-1} \quad \text{MC0156}$$

18. 5 g of zinc is treated separately with an excess of **[JEE(Main)-2020 (Jan)]**

- (a) dilute hydrochloric acid and
(b) aqueous sodium hydroxide.

The ratio of the volumes of H_2 evolved in these two reactions is :

(1) 1 : 4 (2) 1 : 2 (3) 2 : 1 (4) 1 : 1 **MC0157**

19. Amongst the following statements, that which was not proposed by Dalton was :

- (1) all the atoms of a given element have identical properties including identical mass. Atoms of different elements differ in mass.
 - (2) chemical reactions involve reorganisation of atoms. These are neither created nor destroyed in a chemical reaction.
 - (3) when gases combine or reproduced in a chemical reaction they do so in a simple ratio by volume provided all gases are at the same T & P.
 - (4) matter consists of indivisible atoms.
- [JEE(Main)-2020 (Jan)]**
MC0159

20. A 20.0 mL solution containing 0.2 g impure H_2O_2 reacts completely with 0.316 g of KMnO_4 in acid solution. The purity of H_2O_2 (in %) is _____ (mol. wt. of H_2O_2 = 34; mol. wt. of KMnO_4 = 158)

[JEE(Main)-2020]

MC0212

21. The formula of a gaseous hydrocarbon which requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times its own volume of CO_2 is C_xH_y . The value of y is _____.

[JEE(Main)-2021]

MC0213

22. Complete combustion of 1.80 g of an oxygen containing compound ($C_xH_yO_z$) gave 2.64 g of CO_2 and 1.08 g of H_2O . The percentage of oxygen in the organic compound is: **[JEE(Main)-2021]**

(1) 51.63 (2) 63.53 (3) 53.33 (4) 50.33 **MC0214**

23. The number of significant figures in 50000.020×10^{-3} is _____. **[JEE(Main)-2021]**

MC0215

24. Complete combustion of 750 g of an organic compound provides 420 g of CO_2 and 210 g of H_2O . The percentage composition of carbon and hydrogen in organic compound is 15.3 and _____ respectively. (Round off to the Nearest Integer) **[JEE(Main)-2021]**

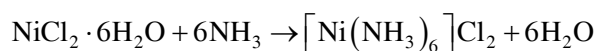
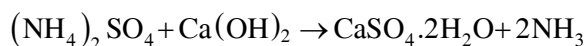
[JEE(Main)-2021]

MC0216

25. The number of chlorine atoms in 20 mL of chlorine gas at STP is _____ 10^{21} . (Round off to the Nearest Integer). [JEE(Main)-2021]
[Assume chlorine is an ideal gas at STP
 $R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$, $N_A = 6.023 \times 10^{23}$] MC0217
26. Complete combustion of 3 g of ethane gives $x \times 10^{22}$ molecules of water. The value of x is _____. (Round off to the Nearest Integer). [JEE(Main)-2021]
[Use : $N_A = 6.023 \times 10^{23}$; Atomic masses in u : C : 12.0 ; O : 16.0 ; H : 1.0] MC0218
27. 4g equimolar mixture of NaOH and Na_2CO_3 contains x g of NaOH and y g of Na_2CO_3 . The value of x is _____ g. (Nearest integer) [JEE(Main)-2021] MC0219
28. When 0.15 g of an organic compound was analyzed using Carius method for estimation of bromine, 0.2397 g of AgBr was obtained. The percentage of bromine in the organic compound is _____. (Nearest integer)
[Atomic mass : Silver = 108, Bromine = 80] [JEE(Main)-2021] MC0220
29. Consider the complete combustion of butane, the amount of butane utilized to produce 72.0 g of water is _____ $\times 10^{-1}$ g. (in nearest integer) [JEE(Main)-2021] MC0221
30. The number of significant figures in 0.00340 is _____. [JEE(Main)-2021] MC0222

EXERCISE # JEE-ADVANCED

1. A student performs a titration with different burettes and finds titre values of 25.2 mL, 25.25 mL, and 25.0 mL. The number of significant figures in the average titre value is :- **[JEE 2010]**
MC0223
2. Silver (atomic weight = 108 g mol⁻¹) has a density of 10.5 g cm⁻³. The number of silver atoms on a surface of area 10⁻¹² m² can be expressed in scientific notation as y × 10^x. The value of x is **[JEE 2010]**
MC0224
3. If the value of Avogadro number is 6.023 × 10²³ mol⁻¹ and the value of Boltzmann constant is 1.380 × 10⁻²³ JK⁻¹, then the number of significant digits in the calculated value of the universal gas constant is **[JEE 2014]**
MC0180
4. The ammonia prepared by treating ammonium sulphate with calcium hydroxide is completely used by NiCl₂·6H₂O to form a stable coordination compound. Assume that both the reactions are 100% complete. If 1584 g of ammonium sulphate and 952g of NiCl₂·6H₂O are used in the preparation, the combined weight (in kg) of gypsum and the nickel-ammonia coordination compound thus produced is _____. **[JEE 2018]**



(Atomic weights in g mol⁻¹: H = 1, N = 14, O = 16, S = 32, Cl = 35.5, Ca = 40, Ni = 59)

MC0178

5. Galena (an ore) is partially oxidized by passing air through it at high temperature. After some time, the passage of air is stopped, but the heating is continued in a closed furnace such that the contents undergo self-reduction. The weight (in kg) of Pb produced per kg of O₂ consumed is _____. **[JEE 2018]**

$$\text{PbS} + \text{O}_2 \longrightarrow \text{Pb} + \text{SO}_2$$

(Atomic weights in g mol⁻¹ : O = 16, S = 32, Pb = 207) **MC0179**

ANSWER KEY**EXERCISE # S-I**

- | | | | |
|--|---|---|--------------------------------------|
| 1. (i) 0.16 or 0.17
(iii) 23.38
(v) 12.04 | (ii) 3200.00
(iv) (18.06 or 18.07), (54.19 or 54.20) | | |
| 2. 1.99 | 3. 3.20 | 4. 180.00 | 5. 2.43 or 2.44 |
| 6. (9.48) | 7. (55.00) | 8. (34.05) | 9. (2.00) |
| 10. (102.50) | 11. 36.50 | 12. 27.60 | 13. 3.61 |
| 14. 0.63 | 15. 58.80 | 16. 6.41 or 6.42 | 17. 0.11 or 0.12 |
| 18. 0.25 | 19. 5.60 | 20. % Al = 60; % Mg = 40 | |
| 21. % CaCO ₃ = 28.40 or 28.41, % MgCO ₃ = 71.59 | | | |
| 22. % NaHCO ₃ = 16.80 ; % Na ₂ CO ₃ = 83.20 | | | |
| 23. 0.75 | 24. 1.70 | 25. 0.50 | 26. 39.18 |
| 27. 4.48 | 28. 61.50 | 29. 0.20 | 30. (50.00) |
| 31. (31.42 or 31.43) | 32. (0.64) | 33. (40.00) | 34. (0.72) |
| 35. (0.04) | 36. (3.04) | 37. 6.40 | 38. (a) 0.71 (b) 61.97 |
| 39. 1.20 | 40. (4.00) | 41. (33.33) | 42. 5.28 |
| 43. (4.00) | 44. (2.00) | 45. (0.07) | 46. (19.00) |
| 47. C ₆ H ₄ Cl ₂ | 48. (3.00) | 49. (C ₂ H ₄ O ₂) | 50. (0.50) |
| 51. 140.00 | 52. 30.00 | 53. 70.00 | 54. (10.00) |
| 55. 73.33 | 56. 10.00 | 57. (N ₂ O ₅) | 58. (C ₂ H ₆) |
| 59. (3.00) | 60. (5.00) | | |

EXERCISE # S-II

- | | | |
|--|--|------------|
| 1. (i) 0.5, 0.5 ; (ii) 2, 1 (iii) 1, 2 | 2. (65.62) | 3. 0.66 |
| 4. % NaCl = 77.80 | 5. (a) Fe ₂ O ₃ + 2 Al → Al ₂ O ₃ + 2Fe; (b) 2.96 ; (c) 10,000 units | |
| 6. 5.74 | 7. Ans. 6.02 | 8. (1.50) |
| 10. (11.00) | 11. (5.00) | 12. (1.70) |
| 13. (3.30) | | |
| 14. % Ag = $\frac{0.5}{0.66} \times 100 = 75.75$ | | |
| % Ag = $\frac{1}{1.32} \times 100 = 75.75$ | | |

In both AgCl, % Ag is same.

$$15. \quad \% \text{ Pb} = \frac{6.488}{(6.488+1.002)} \times 100 = 86.62\%$$

$$16. \quad \text{Oxide} = 1 \text{ gm} \Rightarrow \text{oxygen} = 0.5 \text{ gm} + \text{element} = 0.8 \text{ gm}$$

$$\text{Ratio} = 1 : 1$$

$$\text{Oxide} = 4 \text{ gm} \Rightarrow \text{oxygen} = 3.2 \text{ gm} + \text{element} = 0.8 \text{ gm}$$

$$\text{Ratio} = 4 : 1$$

$$17. \quad 0.11 \text{ gm oxide gives} \Rightarrow 0.07 \text{ gm N, Hence ; } 0.07 \text{ gm N} + 0.04 \text{ gm O}$$

$$0.15 \text{ gm oxide gives} \Rightarrow 0.07 \text{ gm N, Hence ; } 0.07 \text{ gm N} + 0.08 \text{ gm O}$$

$$18. \quad \text{CH}_4 = 4.50, \text{CO}_2 = 1.50, \text{C}_2\text{H}_4 = 4.00 \quad 19. (0.66 \text{ or } 0.67) \quad 20. 9.00$$

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	D	C	C	C	A	B	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	B	B	A	C	B	A	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	A	A	D	A	A	B	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	C	C	C	C	D	C	A	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	D	B	A	D	B	A	B	C	C

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	A	C	C	D	B	A,B,D	B,C,D	A,B,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B	A,C	A,B,D	C	C	B	C	B	A	C
Que.	21	22	23	24	25	26	27	28		
Ans.	D	A	D	C	B	A	B	C,D		
Que.	29			30			31			
Ans.	A-R	B-P	C-Q	A-R	B-Q	C-P	A-R	B-S	C-P	D-Q
Que.	32									
Ans.	A									

EXERCISE # JEE-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	3	3	2	3	4	3	2	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	3	3	4	2	2120 to 2140	4	3	85
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	8	3	8	3	1	18	1	68	464	3

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	
Ans.	3	7	4	2.99	6.46 or 6.47	

CONCENTRATION TERMS

1. SOLUTION

A solution is a homogenous mixture of two or more pure substances whose composition may be altered within certain limits. Though the solution is homogenous in nature, yet it retains the properties of its constituents.

Generally solution is composed of two components, **solute** and **solvent**. Such type of solution is known as binary solution.

Solvent is that component in solution whose physical state is the same as that of the resulting solution while other component is called as solute. If the physical state of both component is same, then the component in excess by moles is known as solvent and other one as solute. A solution may exist in any physical state such as liquid, solid and gaseous state.

Types of Solutions

Type of Solutions	Solute	Solvent	Common Example
Gaseous Solutions	Gas	Gas	Mixture of oxygen and nitrogen gases
	Liquid	Gas	Chloroform mixed with nitrogen gas
	Solid	Gas	Camphor in nitrogen gas
Liquid Solutions	Gas	Liquid	Oxygen dissolved in water
	Liquid	Liquid	Ethanol dissolved in water
	Solid	Liquid	Glucose dissolved in water
Solid Solutions	Gas	Solid	Solution of hydrogen in palladium
	Liquid	Solid	Amalgam of mercury with sodium
	Solid	Solid	Copper dissolved in gold

2. CONCENTRATION OR STRENGTH OF SOLUTION :

The concentration of a solution is the amount of solute dissolved in a known amount of the solvent or solution. Solution can be described as dilute or concentrated solution as per their concentration. A dilute solution has a very small quantity of solute while concentrated solution has a large quantity of solute in solution. Various concentration terms are as follows:

2.1 Mass percentage :

It may be defined as the number of parts of mass of solute per hundred parts by mass of solution.

$$\% \text{ by mass} \left(\frac{w}{W} \right) = \frac{\text{wt. of solute}}{\text{wt. of solution}} \times 100$$

[X % by mass means 100 gm solution contains X gm solute and hence (100 – X) gm solvent]

2.2 Mass-volume percentage (W/V %) :

It may be defined as the mass of solute (in gm) present in 100 cm³ of solution. For example, If 100 cm³ of solution contains 5 g of sodium hydroxide, then the mass-volume percentage will be 5% solution.

$$\% \left(\frac{w}{V} \right) = \frac{\text{wt. of solute}}{\text{volume of solution}} \times 100$$

$$[X \% \left(\frac{w}{V} \right) \text{ means } 100 \text{ ml solution contains } X \text{ gm solute}]$$

2.3 Volume Percent :

It can be represented as % v/v or % volume and normally used for the solutions in which both components are in liquids state. It is the number of parts of by volume solute per hundred parts by volume of solution. Therefore,

$$\% \left(\frac{v}{V} \right) = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100$$

$$\mathbf{2.4 \quad Mole \% = \frac{\text{Moles of solute}}{\text{Total moles}} \times 100}$$

- For gases, **% by volume** is same as **mole %**

2.5 Mole Fraction (X) :

Mole fraction may be defined as the ratio of number of moles of one component to the total number of moles of all the components (solute and solvent) present in solution. It is denoted by letter X and the sum of all mole fractions in a solution is always equals to one.

$$\text{Mole fraction (X)} = \frac{\text{Moles of solute}}{\text{Total moles}}$$

Mole fraction does not depend upon temperature and can be extended to solutions having more than two components.

2.6 Molarity (M) :

Molarity is most common unit for concentration of solution. It is defined as the number of moles of solute present in one litre or one dm³ of the solution or millimol of solute present in one mL of solution.

$$\text{Molarity (M)} = \frac{\text{Mole of solute}}{\text{volume of solution in litre}}$$

- 2.7 Molality (m) :** The number of mole of the solute present in 1000 g of the solvent is known as molality of solution. It represented by letter 'm'.

$$\text{Molality (m)} = \frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$$

The unit of molality is mol/kg and is not affected by temperature.

- 2.8 Parts per million (ppm) :** The very low concentration of solute in solution can be expressed in ppm. It is the number of parts by mass of solute per million parts by mass of the solution.

$$\text{Parts per million (ppm)} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$$

- ◆ Get yourselves very much comfortable in their inter conversion. It is very handy.

Concentration Type	Mathematical Formula	Concept
Percentage by mass	$\% \left(\frac{w}{w} \right) = \frac{\text{Mass of solute} \times 100}{\text{Mass of solution}}$	Mass (in gm) of solute present in 100 gm of solution.
Volume percentage	$\% \left(\frac{v}{v} \right) = \frac{\text{Volume of solute} \times 100}{\text{Volume of solution}}$	Volume (cm ³) of solute present in 100 cm ³ of solution.
Mass-volume percentage	$\% \left(\frac{w}{v} \right) = \frac{\text{Mass of solute} \times 100}{\text{Volume of solution}}$	Mass (in gm) of solute present in 100 cm ³ of solution.
Parts per million	$\text{ppm} = \frac{\text{Mass of solute} \times 10^6}{\text{Mass of solution}}$	Parts by mass of solute per million parts by mass of the solution
Mole fraction	$X_A = \frac{\text{Mole of A}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + \dots}$ $X_B = \frac{\text{Mole of B}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + \dots}$	Ratio of number of moles of one component to the total number of moles.
Molarity	$M = \frac{\text{Mole of solute}}{\text{Volume of solution (in L)}}$	Moles of solute in one litre of solution.
Molality	$m = \frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$	Moles of solute in one kg of solvent

Ex.1 Calculate the mole fractions of the components of the solution composed by 92 g glycerol and 90 g water ? ($M(\text{water}) = 18$; $M(\text{glycerol}) = 92$)

Sol. Moles of water = $90 \text{ g} / 18 \text{ g} = 5 \text{ mol}$ water
 Moles of glycerol = $92 \text{ g} / 92 \text{ g} = 1 \text{ mol}$ glycerol
 Total moles in solution = $5 + 1 = 6 \text{ mol}$
 Mole fraction of water = $5 \text{ mol} / 6 \text{ mol} = 0.833$
 Mole fraction of glycerol = $1 \text{ mol} / 6 \text{ mol} = 0.167$

Ex.2 What will be the molarity of solution when water is added to 16.4g $\text{Ca}(\text{NO}_3)_2$ to make 100 mL of solution?

Sol. Mol of $\text{Ca}(\text{NO}_3)_2 = 16.4 / 164 = 0.1$
 Molarity = Mole of solute / Volume of solution (L) = $0.10 \text{ mol} / 0.10 \text{ L}$
 Therefore, Molarity of given solution = 1.0 M

Ex.3 Calculate the molality of a solution containing 20 g of sodium hydroxide (NaOH) in 250 g of water?

Sol. Moles of sodium hydroxide = $20 / 40 = 0.5 \text{ mol}$ NaOH
 $250 \text{ gm} = 0.25 \text{ kg}$ of water
 Hence molality of solution = Mole of solute / Mass of solvent (kg) = $0.5 \text{ mol} / 0.25 \text{ kg}$
 or Molality(m) = 2.0 m

Ex.4 Calculate the gram of copper sulphate (CuSO_4) needed to prepare 250.0 mL of 1.00 M CuSO_4 ?

Sol. Moles of $\text{CuSO}_4 = M \times V = 1 \times \frac{250}{1000}$
 Molar mass of copper sulphate = 159.6 g/mol
 Hence Mass of copper sulphate (gm) = Moles of $\text{CuSO}_4 \times$ Molar mass of copper sulphate.

$$= 1 \times \frac{250}{1000} \times 159.6 \text{ g/mol} = 39.9 \text{ gm of Copper sulphate}$$

Ex.5 How many gram of H_2SO_4 are present in 500 ml of 0.2M H_2SO_4 solution ?

Sol. $M = \frac{\text{moles}}{\text{vol.}} \Rightarrow \text{moles of } \text{H}_2\text{SO}_4 = M \times V = 0.2 \times \frac{500}{1000} \text{ L} = 0.1$
 Mass of $\text{H}_2\text{SO}_4 = 0.1 \times 98 = 9.8 \text{ g}$

(a) $\text{Weight percent} = \frac{\text{wt. of solute}}{\text{wt. of solution}} \times 100 = \frac{4.450}{86.65} \times 100 = 5.14$

(b) Mole fraction :

$$\text{Mole of solute} = \frac{\text{wt. of solute}}{\text{mol wt. of solute}} = \frac{4.45}{98} = 0.0454$$

$$\text{Mole of solvent} = \frac{82.20}{18} = 4.566$$

Total moles in solution = $0.0454 + 4.566 = 4.6114$

$$\text{Mole fraction of solute} = \frac{0.0454}{4.6114} = 0.0098$$

(c) $\text{Mole percent} = \text{mole fraction of solute} \times 100 = 0.0098 \times 100 = 0.98$

$$\begin{aligned} \text{(d) Molality} &= \frac{\text{moles of solute}}{\text{mass of solvent (in gm)}} \times 1000 \\ &= \frac{0.0454 \times 1000}{82.2} = 0.552 \end{aligned}$$

(e) $\text{Molarity} = \frac{\text{moles of solute}}{\text{litre of solution}}$

$$\text{Volume of solution} = \frac{\text{Mass}}{\text{Density}} = \frac{86.65}{1.029} \text{ ml} = \frac{86.65}{1.029 \times 1000} \text{ litre}$$

$$\text{Molarity} = \frac{\frac{0.0454}{86.54}}{\frac{1.029 \times 1000}{86.65}} = 0.539$$

Ex.11 Find number of Na^+ & PO_4^{3-} ions in 250 ml of 0.2M Na_3PO_4 solution.



50 millimoles (m.m.) 150 mm 50 mm

$$\text{No. of Na}^+ \text{ ions} = 150 \times 10^{-3} \times N_A ; \text{No. of PO}_4^{-3} \text{ ions} = 50 \times 10^{-3} \times N_A$$

Ex.12 80g NaOH was added to 2L water. Find molality of solution if density of water = 1g/mL

Sol. $m = \frac{\text{moles of NaOH}}{\text{mass of H}_2\text{O}} \times 1000 = \frac{80/40}{2 \times 1000} \times 1000 = 1 \text{ molal}$

Ex.13 The average concentration of Na^+ ion in human body is 3.0 to 3.9 gm per litre. The molarity of Na^+ ion is about.

0.15 M

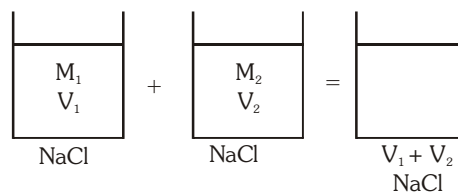
Sol. $M_{Na^+} = \frac{n_{\text{solute}}}{\text{volume of solution in Lt}} = \frac{\frac{3+3.9}{2}}{23} = \frac{6.9}{46} = 0.15M$

3. MIXING OF AND DILUTION SOLUTIONS :

It is based on law of conservation of moles of solute

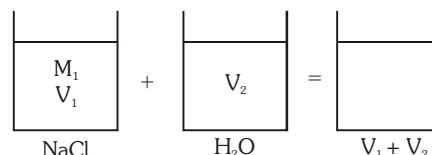
(i) Two solutions having same solute

$$\text{Final molarity} = \frac{\text{Total moles}}{\text{Total volume}} = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$$



(ii) Dilution Effect : When a solution is diluted, the moles of solute do not change but molarity changes.

$$\text{Final molarity} = \frac{M_1 V_1}{V_1 + V_2}$$



Ex.14 50 ml 0.2 M H_2SO_4 is mixed with 50 ml 0.3M H_2SO_4 . Find molarity of final solution.

Sol. $M_f = \frac{\text{Total moles of } H_2SO_4}{\text{Total volume}} = \frac{50 \times 0.2 \times 10^{-3} + 50 \times 10^{-3} \times 0.3}{(50 + 50) \times 10^{-3}} = 0.25 \text{ M.}$

Ex.15 Find final molarity in each case :

(i) 500 ml 0.1 M HCl + 500 ml 0.2M HCl

(ii) 50 ml, 0.1M HCl + 150 ml, 0.3M HCl + 300 ml H_2O

(iii) 4.9g H_2SO_4 + 250 ml H_2O + 250 ml 0.1 M H_2SO_4

Sol.

(i) $M_f = \frac{500 \times 0.1 + 500 \times 0.2}{500 + 500} = 0.15 \text{ M.}$

(ii) $M_f = \frac{50 \times 0.1 + 150 \times 0.3}{50 + 150 + 300} = \frac{50}{500} = 0.1 \text{ M}$

(iii) $M_f = \frac{\frac{4.9}{98} + \frac{250}{1000} \times 0.1}{\left(\frac{250 + 250}{1000} \right)} = \frac{50 + 25}{500} = 0.15 \text{ M}$

Ex.16 How much water should be added to 2M HCl solution to form 1 litre of 0.5 M HCl ?

Sol. Let V be initial volume

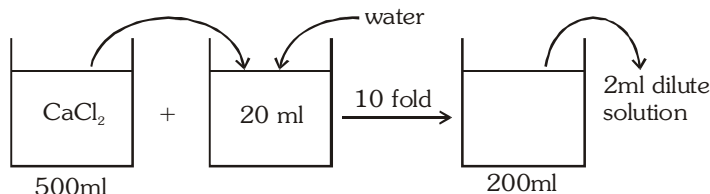
Then mol of HCl = constant

$$2 \times V = 1 \times 0.5 \Rightarrow V = 0.25 \text{ L}$$

$$\text{Volume of water added} = 1 - 0.25 = 0.75 \text{ L}$$

Ex.17 1.11g CaCl_2 is added to water forming 500 ml of solution. 20 ml of this solution is taken and diluted 10 folds. Find moles of Cl^- ions in 2 ml of diluted solution.

Sol. Moles of $\text{CaCl}_2 = \frac{1.11}{111} = 0.01 \text{ mol}$



$$\text{Moles of } \text{CaCl}_2 \text{ in } 20\text{ml solution} = \frac{0.01}{500} \times 20 = \frac{0.01}{25}$$

$$\text{In } 200 \text{ ml solution, moles of } \text{CaCl}_2 = \frac{0.01}{25} \quad [\text{Dilution does not change moles of solute}]$$

$$\text{In } 2 \text{ ml of dilute solution moles of } \text{CaCl}_2 = \frac{0.01/25}{200} \times 2 = \frac{0.01}{2500} = 4 \times 10^{-6}$$

$$\therefore \text{ moles of } \text{Cl}^- = 2 \times 4 \times 10^{-6} = 8 \times 10^{-6}$$

Ex.18 What volumes of 1M & 2M H_2SO_4 solution are required to produce 2L of 1.75M H_2SO_4 solution?

Sol. Let XL be vol. of 1M solution.

$$\therefore (2 - X)\text{L is vol. of } 2\text{M solution.}$$

$$\text{Moles of } \text{H}_2\text{SO}_4 : 2 \times 1.75 = 1(X) + (2 - X)2 \Rightarrow X = 0.5 \text{ L}$$

i.e. 0.5L of 1M & 1.5 L of 2M solution are required.

Ex.19 A solution is made by mixing 300 ml 1.5M $\text{Al}_2(\text{SO}_4)_3$ + 300 ml 2M CaSO_4 + 400 ml 3.5M CaCl_2

Find final molarity of (1) SO_4^{2-} , (2) Ca^{2+} , (3) Cl^- . [Assume complete dissociation of these compounds].

Sol. (1) $[\text{SO}_4^{2-}]_f = \frac{\text{Total moles}}{\text{Total volume}} = \frac{300 \times 1.5 \times 10^{-3} \times 3 + 300 \times 2 \times 10^{-3}}{(300 + 300 + 400) \times 10^{-3}} = 1.95\text{M}$

(2) $[\text{Ca}^{2+}]_f = \frac{300 \times 2 + 400 \times 3.5}{1000} = 2\text{M}$

(3) $[\text{Cl}^-]_f = \frac{400 \times 3.5 \times 2}{1000} = 2.8\text{M}$

Ex.20 A solution of KCl has a density of 1.69 g mL^{-1} and is 67% by weight. Find the density of the solution if it is diluted so that the percentage by weight of KCl in the diluted solution is 30%.

Sol. Let the volume of the KCl solution be 100 mL,

$$\text{Weight of KCl solution} = 100 \times 1.69 = 169 \text{ g}$$

$$100 \text{ g of solution contains} = 67 \text{ g of KCl}$$

$$169 \text{ g of solution} = \frac{67}{100} \times 169 = 113.23\text{g}$$

Let x mL of H_2O be added.

New volume of solution = $(100 + x)$ mL

New weight of solution = $(169 + x)$ g

(Since x mL of $H_2O = x$ g of H_2O , $d_{H_2O} = 1$)

New percentage of the solution = 30%

$$\% \text{ by weight} = \frac{\text{weight of solute} \times 100}{\text{weight of solution}}$$

$$30 = \frac{113.23}{(169 + x)} \times 100$$

$$x = 208.43 \text{ mL} = 208.43 \text{ g}$$

$$\text{New density} = \frac{\text{New weight of solution}}{\text{New volume of solution}} = \frac{(169 + x)}{(100 + x)}$$

$$\therefore d = 1.224 \text{ gm/ml}$$

Ex.21 Calculate the amount of the water ("in mL") which must be added per mL to a given solution of concentration of 40 mg silver nitrate per mL, to yield a solution of concentration of 16 mg silver nitrate per mL ?

Sol. Before dilution After dilution

$$(n_{\text{solute}}) = (n_{\text{solute}})$$

$$M_i V_i = M_f V_f$$

$$\frac{40}{\frac{170}{1}} \times 1 = \frac{16}{\frac{170}{1}} (1 + V)$$

$$\Rightarrow V = 1.5 \text{ mL}$$

Ex.22. 100 mL, 3%(w/v) NaOH solution is mixed with 100 mL, 9%(w/v) NaOH solution. The molarity of final solution is-

Sol. Total NaOH in 100 mL (1st solution) = 3 gm

Total NaOH in 100 mL (2nd solution) = 9 gm

$$\therefore \text{Molarity} = \left(\frac{12/40}{200/1000} \right) = 1.5M$$

Ex.23. 1120 gm of 2 'm' urea solution is mixed with 2480 gm of 4 'm' urea solution. Calculate the molality of the resulting solution?

Sol. Let 2 m, 1120 g solution have mass of solute = w gm

$$\therefore \text{solvent} = (1120 - w) \text{ gm}$$

& Let 4 m, 2480 g solution have mass of solute = w' gm

$$\therefore \text{solvent} = (2480 - w') \text{ gm}$$

$$\text{molality} = \frac{n_{\text{solute}}}{\text{wt. of solvent in kg}}$$

$$2 = \frac{w/60}{1120 - w} \times 1000$$

$$w = 120 \text{ gm}$$

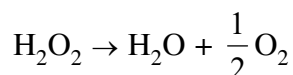
$$\& \quad 4 = \frac{w'/60}{2480 - w'} \times 1000$$

$$w' = 480 \text{ gm}$$

$$\text{resulting molality } m = \frac{120 + 480}{\frac{60}{1120 - 120 + 2480 - 480}} \times 1000 = 3.33 \text{ m}$$

4. VOLUME STRENGTH OF H_2O_2 SOLUTION :

It represents the volume of O_2 (in litre) at STP that can be obtained from 1 litre of H_2O_2 solution, when H_2O_2 decomposes as :



Volume strength is normally represented as 'x vol' or 'x V'.

$$\text{Volume Strength of } \text{H}_2\text{O}_2 \text{ Solution} = 11.35 \times \text{molarity}$$

Ex.24 Find the % w/v of "10 V" H_2O_2 solution-

$$\text{Sol. Molarity (M) of solution} = \frac{\text{volume strength}}{11.35} = \frac{10}{11.35}$$

$$\% \left(\frac{w}{v} \right) = \frac{M \times \text{mol. wt. of solute}}{10} = \frac{10}{11.35} \times \frac{34}{10} = 3\%$$

Ex.25 $2\text{H}_2\text{O}_2(\text{aq}) \longrightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$

Under conditions where 1 mole of gas occupies 24 dm^3 , $X \text{ L of } \frac{1}{24} \text{ M solution of } \text{H}_2\text{O}_2 \text{ produces } 3 \text{ dm}^3 \text{ of } \text{O}_2$. Thus X is :-

$$\text{Sol. moles of } \text{H}_2\text{O}_2 = \frac{1}{24} \times X$$

$$\text{moles of } \text{O}_2 = \frac{3}{24} = \frac{1}{8}$$

$$\text{moles of } \text{H}_2\text{O}_2 = \frac{1}{4} = \frac{X}{24}$$

$$X = 6$$

PREVIOUS YEARS SOLVED EXAMPLES

Q.1 One gm of charcoal absorbs 100 ml 0.5 M CH_3COOH to form a monolayer, and thereby the molarity of CH_3COOH reduces to 0.49. Calculate the surface area of the charcoal adsorbed by each molecule of acetic acid. Surface area of charcoal = $3.01 \times 10^2 \text{ m}^2/\text{gm}$. [JEE'2003]

Ans. $5 \times 10^{-19} \text{ m}^2$

Sol. Final molarity = $0.5 - 0.49 = 0.01 \text{ M}$

$$\text{mole} = M \times v = 0.01 \times \frac{100}{1000} = 10^{-3}$$

$$\text{no of molecule} = \text{moles} \times N_A = 10^{-3} \times N_A = 6.02 \times 10^{20}$$

$$1 \text{ gm contain charcoal} = 3.01 \times 10^2 \text{ m}^2$$

$$6.02 \times 10^{20} \text{ molecule of acetic acid absorbed charcoal} = 3.01 \times 10^2$$

$$1 \text{ molecule of acetic acid adsorbed charcoal} = 3.01 \times 10^2 \text{ m}^2$$

$$6.02 \times 10^{20} \text{ molecule of acetic acid absorbed charcoal} = 3.01 \times 10^2$$

$$1 \text{ molecule of acetic acid adsorbed charcoal} = \frac{3.01 \times 10^2}{6.02 \times 10^{20}} = 5 \times 10^{-19} \text{ m}^2$$

Q.2 Calculate the molarity of pure water using its density to be 1000 kg m^{-3} . [JEE'2003]

Ans. 55.5 mol L^{-1}

Sol. $M = \frac{1000}{\text{MW}} = \frac{1000}{18} = 55.5$

Q.3 6.02×10^{21} molecules of urea are present in 100 ml of its solution. The concentration of urea solution is - [AIEEE-2004]

(A) 0.001 M

(B) 0.01 M

(C) 0.02 M

(D) 0.1 M

Ans. (D)

Sol. Moles of urea = $\frac{6.02 \times 10^{21}}{6.02 \times 10^{23}} = \frac{1}{100}$

$$\text{Molarity} = \frac{1000}{100 \times 100} = 0.1 \text{ M}$$

Important Notes

EXERCISE # S-I

CONCENTRATION TERMS

- (a) Calculate number of milimoles of 4.9 g H_2SO_4 CT0115

(b) 50 mL, 0.2 M NaOH is mixed with 20 mL, 0.1 M NaOH. Calculate number of milimoles of Na^+ in final solution. CT0115

(c) 3.65 g HCl, 9.80 g H_2SO_4 and 4.9 g H_3PO_4 are mixed together. Calculate the total milimoles of H^+ in the final solution. CT0115

(d) 10 mL, 0.1 M NaOH and 10 mL, 0.2 M $\text{Ca}(\text{OH})_2$ are added to 80 mL water. Calculate final milimoles & molarity of $[\text{OH}^-]$ in final solution. CT0115
- Calculate the molarity of the following solutions :

(a) 4g of caustic (NaOH) soda is dissolved in 200 mL of the solution. CT0001

(b) 5.3 g of anhydrous sodium (Na_2CO_3) carbonate is dissolved in 100 mL of solution. CT0001

(c) 0.365 g of pure HCl gas is dissolved in 50 mL of solution. CT0001
- Density of a solution containing 13% by mass of sulphuric acid is 0.98 g/mL. Then molarity of solution will be CT0002
- 15 g of methyl alcohol is present in 100 mL of solution. If density of solution is 0.90 g mL^{-1} , calculate the mass percentage of methyl alcohol in solution CT0003
- Units of parts per million (ppm) or per billion (ppb) are often used to describe the concentrations of solutes in very dilute solutions. The units are defined as the number of grams of solute per million or per billion grams of solvent. Bay of Bengal has 2.1 ppm of lithium ions. If the molality of Li^+ is $x \times 10^{-4} \text{ m}$, then find x. ($\text{Li} = 7$) CT0004
- A 7.0 M solution of KOH in water contains 28% by mass of KOH. What is density of solution in gm/ml ? ($\text{K} = 39$) CT0005
- The concentration of a solution is 8% (w/w) and 10% (w/v). Calculate density (in gm/mL) of solution? CT0006
- The mole fraction of solute in aqueous urea solution is 0.2. Calculate the mass percent of solute ? CT0007
- The concentration of $\text{Ca}(\text{HCO}_3)_2$ in a sample of hard water is 405 ppm. The density of water sample is 1.0 gm/ml. If the molarity of solution is $x \times 10^{-3} \text{ M}$ then find x ? CT0008
- How much BaCl_2 (in gm) would be needed to make 250 ml of a solution having the same concentration of Cl^- as one containing 1.825 gm HCl per 100 ml ? ($\text{Ba} = 137$) CT0009
- Calculate **molality (m)** of each ion present in the aqueous solution of $2\text{M NH}_4\text{Cl}$ assuming 100% dissociation according to reaction.



Given : Density of solution = 3.107 gm / ml.

CT0010

PROBLEMS RELATED WITH MIXING & DILUTION

12. Find out the volume (L) of 98% w/w H_2SO_4 (density = 1.8 gm/ ml), must be diluted to prepare 12.6 litres of 2.0 M sulphuric acid solution. **CT0011**
13. 500 ml of 2 M NaCl solution was mixed with 200 ml of 1/4 M NaCl solution. Calculate the molarity of NaCl in final solution. **CT0012**
14. A mixture containing equimolar amounts of $\text{Ca}(\text{OH})_2$ and $\text{Al}(\text{OH})_3$ requires 0.5 L of 4.0 M HCl to react with it completely. Total moles of the mixture is : **CT0013**
15. 500 gm of urea solution of mole fraction 0.2 is diluted to 1500 gm. Calculate the mole fraction of solute in the diluted solution ? **CT0014**
16. When V ml of 2.2 M H_2SO_4 solution is mixed with 10 V ml of water, the volume contraction of 2% take place. Calculate the molarity of diluted solution ? **CT0015**
17. What volume (in ml) of 0.8 M AlCl_3 solution should be mixed with 50 ml of 0.2M CaCl_2 solution to get solution of chloride ion concentration equal to 0.6 M ? **CT0016**
18. A solution containing 200 ml 0.5 M KCl is mixed with 50 ml 19% w/v MgCl_2 and resulting solution is diluted 8 times. Molarity of chloride ion in final solution is : **CT0017**

VOLUME STRENGTH

19. 500 ml of a H_2O_2 solution on complete decomposition produces 2 moles of H_2O . Calculate the volume strength of H_2O_2 solution? **CT0020**
20. The volume strength of 100 ml H_2O_2 solution which produce 5.6 litre of oxygen gas at 1 bar & 0°C. **CT0021**

EXERCISE # S-II

1. What volume of 0.2 M NaOH (in L) solution should be mixed to 500 ml of 0.5 M NaOH solution so that 300 ml of final solution is completely neutralised by 20 ml of 2 M H_3PO_4 solution. CT0022

2. How much minimum volume (in L) of $\left(\frac{5}{51}\right)$ M aluminium sulphate solution should be added to excess calcium nitrate to obtain atleast 1 gm of each salt in the reaction.

$$\text{Al}_2(\text{SO}_4)_3 + 3\text{Ca}(\text{NO}_3)_2 \longrightarrow 2\text{Al}(\text{NO}_3)_3 + 3\text{CaSO}_4$$
 CT0023

3. One litre of milk weighs 1.035 kg. The butter fat is 4% (v/v) of milk and has density of 875 kg/m^3 . If the density of fat free skimmed milk is 'x' gm/ml, the value of (x) is ? CT0024

4. 100 ml of 0.1 M solution of AB ($d = 1.5 \text{ gm/ml}$) is mixed with 100 ml of 0.2 M solution of CB_2 ($d = 2.5 \text{ gm/ml}$). Calculate the molarity of B^- in final solution if the density of final solution is 4 gm/ml . Assuming AB and CB_2 are non reacting & dissociates completely into A^+ , B^- , C^{+2} . CT0025

5. 60 ml of a "x" % w/w alcohol by weight ($d = 0.6 \text{ g/cm}^3$) must be used to prepare 200 cm^3 of 12% alcohol by weight ($d = 0.90 \text{ g/cm}^3$). Calculate the mass of alcohol (in gm) in original sample. CT0026

6. If 0.5 M methanol undergo self dissociation like $\text{CH}_3\text{OH} \rightleftharpoons \text{CH}_3\text{O}^- + \text{H}^+$ & if concentration of H^+ is $2.5 \times 10^{-4} \text{ M}$ then calculate % dissociation of methanol. CT0027

7. Determine the volume (in L) of diluted nitric acid ($d = 1.11 \text{ g mL}^{-1}$, 20% w/v HNO_3) that can be prepared by diluting 50 mL of conc. HNO_3 with water ($d = 1.42 \text{ g mL}^{-1}$, 70% w/v). CT0028

8. 50 ml of '20V' H_2O_2 is mixed with 200 ml, '10V' H_2O_2 . The volume strength of resulting solution is CT0029

9. 500 ml of 0.90M CH_3COOH solution is mixed with 600 ml 12% w/v CH_3COOH solution then calculate the final molarity of solution. CT0030

10. 45.4 V H_2O_2 solution (500 ml) when exposed to atmosphere loses 11.2 litre of O_2 at 1 atm, & 273 K. New molarity of H_2O_2 solution (Assume no change in volume) CT0031

EXERCISE # O-I

CONCENTRATION TERMS

- 125 ml of 8% w/w NaOH solution (sp. gravity 1) is added to 125 ml of 10% w/v HCl solution. The nature of resultant solution would be _____.
(A) Acidic (B) Basic (C) Neutral (D) Can not be predicted **CT0032**
- The molarity of pure water is :
(A) 100 M (B) 55.6 M (C) 50 M (D) 18M **CT0033**
- Mole fraction of $C_3H_5(OH)_3$ (glycerine) in a solution of 36 g of water and 46 g of glycerine is :
(A) 0.46 (B) 0.36 (C) 0.20 (D) 0.40 **CT0034**
- A molal solution is one that contains one mole of a solute in
(A) 1000 g of the solvent (B) one litre of the solution
(C) one litre of the solvent (D) 22.4 litres of the solution **CT0035**
- The molarity of a solution of sodium chloride in water containing 5.85 gm of sodium chloride in 500 ml of solution is :-
(A) 0.25 M (B) 2.0 M (C) 1.0 M (D) 0.2 M **CT0036**
- The molarity of 98% by wt. H_2SO_4 ($d = 1.8$ g/ml) is
(A) 6 M (B) 18 M (C) 10 M (D) 4 M **CT0037**
- Which one of the following modes of expressing concentration of solution is independent of temperature -
(A) Molarity (B) Molality (C) % w/v (D) Grams per litre **CT0038**
- Equal weight of NaCl and KCl are dissolved separately in equal volumes of solutions. Molarity of the solutions will be -
(A) Equal (B) Greater for NaCl
(C) Greater for KCl (D) Uncomparable. **CT0039**
- Molality of 20% (w/w) aq. glucose solution is :
(A) $\frac{25}{18}$ m (B) $\frac{10}{9}$ m (C) $\frac{25}{9}$ m (D) $\frac{5}{18}$ m **CT0042**
- Molarity of liquid HCl, if density is 1.17 g/cc. :
(A) 36.5 M (B) 18.25 M (C) 32.05 M (D) 42.10 M **CT0043**
- Molarity and Molality of a solute (M. wt = 50) in aqueous solution is 9 and 18 respectively. What is the density of solution.
(A) 1 g/cc (B) 0.95 g/cc (C) 1.05 g/cc (D) 2 g/cc **CT0093**
- The relationship between mole fraction (X_A) of the solute & molality 'm' of its solution in ammonia would be
(A) $\frac{55.56(X_A)}{1-X_A} = m$ (B) $\frac{58.82(X_A)}{1-X_A} = m$ (C) $\frac{58.82(1-X_A)}{X_A} = m$ (D) $\frac{55.56(1-X_A)}{X_A} = m$ **CT0052**

13. 3.0 molal NaOH solution has a density of 1.12 g/mL. The molarity of the solution is-
 (A) 2.97 M (B) 3 M (C) 3.05 M (D) 3.5 M **CT0053**
14. The concentrated sulphuric acid that is peddled commercially is 95% H_2SO_4 by weight. If the density of this commercial acid is 1.834 g cm^{-3} , the molarity of this solution is :-
 (A) 17.8 M (B) 15.7 M (C) 10.5 M (D) 12.0 M **CT0087**
15. The density of a solution prepared by dissolving 120 g of urea (mol. mass = 60 u) in 1000 g of water is 1.15 g/mL. The molarity of this solution is
 (A) 2.05 M (B) 0.50 M (C) 1.78 M (D) 1.02 M **CT0088**
16. A 5.2 molal aqueous solution of methyl alcohol, CH_3OH , is supplied. What is the mole fraction of methyl alcohol in the solution ?
 (A) 0.086 (B) 0.050 (C) 0.100 (D) 0.190 **CT0089**

PROBLEMS RELATED WITH MIXING AND DILUTION

17. How much water should be added to 200 cc of semimolar solution of NaOH to make it exactly decimolar?
 (A) 1000 cc (B) 400 cc (C) 800 cc (D) 600 cc **CT0040**
18. 100 ml of 0.3 M HCl solution is mixed with 200 ml of 0.3 M H_2SO_4 solution. What is the molarity of H^+ in resultant solution ?
 (A) 0.9 M (B) 0.6 M (C) 0.4 M (D) 0.5 M **CT0041**
19. The molarity of a solution made by mixing 50 ml of conc. H_2SO_4 (18 M) with 50 ml of water, is:
 (A) 36 M (B) 18 M (C) 9 M (D) 6M **CT0044**
20. Equal volumes of 10% (w/v) of HCl is mixed with 10% (w/v) NaOH solution. The resultant solution be.
 (A) basic (B) neutral (C) acidic (D) can't be predicted. **CT0045**
21. What volume of 0.2M NaOH solution is needed for complete neutralisation of 0.49 gm orthophosphoric acid -
 (A) 75 ml (B) 300 ml (C) 0.075 ml (D) 50 ml **CT0046**
22. Assuming complete precipitation of AgCl, calculate the sum of the molar concentration of all the ions if 2 lit of 2M Ag_2SO_4 is mixed with 4 lit of 1 M NaCl solution is :
 (A) 4M (B) 2M (C) 3 M (D) 2.5 M **CT0050**

VOLUME STRENGTH

23. 20 ml of '20 vol' H_2O_2 solution is diluted to 80 ml. The final volume strength of solution is
 (A) '80 vol' (B) '25 vol' (C) '5 vol' (D) '8 vol' **CT0096**

EXERCISE # O-II

ASSERTION/REASON :

1. **Statement -1** : Molality of pure ethanol is lesser than pure water.
Statement -2 : Density of ethanol is lesser than density of water.
[Given : $d_{\text{ethanol}} = 0.789 \text{ gm/ml}$; $d_{\text{water}} = 1 \text{ gm/ml}$]
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is false, statement-2 is true.
 (D) Statement-1 is true, statement-2 is false. **CT0054**
2. **Statement-1** : Molarity and molality have almost same value for a very dilute aqueous solution.
Statement-2 : In all very dilute solution, the mass of solvent (in gm) is equal to the volume of solution (in ml).
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true. **CT0055**
3. **Statement-1** : The mass fraction of solute in a solution is always greater than its mole fraction.
Statement-2 : Mole fraction of solvent in an aqueous solution of ethanol must be greater than that of solute.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true. **CT0056**

ONE OR MORE THAN ONE CORRECT :

4. The molar concentration of HCl (aq.) is 10^{-5} M . Which of the following statements are correct.
($d_{\text{solution}} = 1 \text{ gm/cc}$)
 (A) The mole fraction of HCl $\cong 1.8 \times 10^{-7}$
 (B) The concentration of HCl in ppm is 3.65 ppm
 (C) The molality of HCl solution is approximately 10^{-5} m
 (D) The (w/v)% of solution is $3.65 \times 10^{-5} \%$ **CT0057**

- PARAGRAPH BASED :**

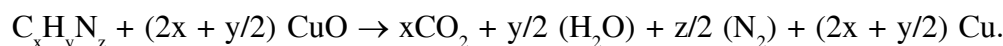
2 litre of 9.8 % w/w H_2SO_4 (d = 1.5 gm/ml) solution is mixed with 3 litre of 1 M KOH solution.

- mode06\B08C-BD\Kota\JEE[Advanced]\Nurture\Chem\Module\2-Mole, Conc, Equilib-Redox & Eq\Eng\2. Conc. Term\02_Exercise.p65

Comprehension Q.11 and Q.14 (4 questions)

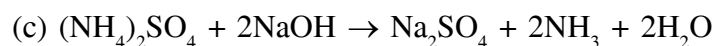
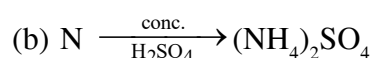
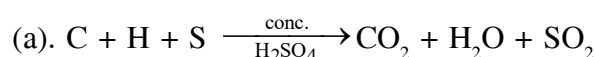
Estimation of nitrogen : There are two methods for the estimation of nitrogen (i) Dumas method and (ii) Kjeldahl's method.

- i. **Dumas method :** A known mass of compound is heated with copper oxide (CuO) in an atmosphere of CO₂, which gives free nitrogen along with CO₂ and H₂O.



The gaseous mixture is passed over a heated copper gauze which converts traces of nitrogen oxides formed to N₂. The gaseous mixture is collected over an aqueous solution of KOH which absorbs CO₂, and nitrogen is collected in the upper part of the graduated tube.

- ii. **Kjeldahl's method :** A known mass of organic compound (0.5 gm) is mixed with K₂SO₄ (10 gm) and CuSO₄ (1.0 gm) or a drop of mercury (Hg) and conc. H₂SO₄ (25 ml), and heated in Kjeldahl's flask. CuSO₄ or Hg acts as a catalyst, while K₂SO₄ raises the boiling point of H₂SO₄. The nitrogen in the organic compound is quantitatively converted to ammonium sulphate. The resulting mixture is then distilled with excess of NaOH solution and the NH₃ evolved is passed into a known but excess volume of standard HCl or H₂SO₄. The acid left unused is estimated by titration with some standard alkali. The amount of acid used against NH₃ can thus be known and from this the percentage of nitrogen is calculated.



iii. This method is not applicable to compounds containing N in nitro and azo groups, and N present in the ring (e.g., pyridine) as N of these compounds does not change to (NH₄)₂SO₄ (ammonium sulphate) under these reaction conditions.

11. 0.30 gm of an organic compound gave 82.1 ml of nitrogen collected at 300K and 775 mm pressure in Dumas method. Calculate the percentage of nitrogen in the compound. (Vapour pressure of water or aqueous tension of water at 300K is 15 mm.)
(A) 31.11 (B) 15.56 (C) 28.0 (D) 31.72 **CT0064**
12. 0.50 gm of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 ml of 0.5M H₂SO₄. The residual acid required 60 ml of M/2 NaOH solution. Find the percentage of nitrogen in the compound.
(A) 50 (B) 56 (C) 66 (D) 40 **CT0065**
13. 0.4 gm of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 ml of 0.5M H₃PO₃. The residual acid required 30 ml of 0.5M Ca(OH)₂. Find the percentage of N₂ in the compound.
(A) 20 (B) 50 (C) 70 (D) 45 **CT0066**
14. 0.002 gm of an organic compound was treated according to Kjeldahl's method. 0.2 × 10⁻⁴ mol of H₂SO₄ was required to neutralise NH₃. Calculate the percentage of N₂.
(A) 50 (B) 28 (C) 70 (D) 18 **CT0067**

TABLE TYPE QUESTION

	Column-I	Column-II	Column-III
(A)	2 M - aqueous NaOH solution (density = 1.25 gm/ml)	(i) 2 mole solute/litre solution	(I) 6 % (w/v) solution
(B)	1.5 m - aqueous NaOH solution (density = 1.06 gm/ml)	(ii) 1.5 mole solute/litre solution	(II) 8 % (w/v) solution
(C)	0.5 M aqueous Glucose solution (density = 1.09 gm/ml)	(iii) 0.5 mole solute/litre solution	(III) 9 % (w/v) solution
(D)	1.5 M aqueous Urea solution (density = 1.15 gm/ml)	(iv) 1.5 mole solute/kg solvent	(IV) 9 gm solute per 100 gm solvent

15. Which of the following is incorrect match ?

- (A) A – i – II (B) B – ii – I (C) C – iii – IV (D) D – iv – III **CT0068**

16. Which of the following is correct match ?

- (A) A – i – IV (B) B – iv – I (C) C – iii – I (D) D – ii – I **CT0068**

17. Which of the following is correct match ?

- (A) A – ii – III (B) B – ii – III (C) C – ii – III (D) D – ii – III **CT0068**

MATCH THE COLUMN :

18. Match the column-

Column-I
(Concentration of aqueous solution)

(A) 2M NaOH solution

(B) $8\% \left(\frac{w}{V} \right)$ KOH solution

(C) $25\% \left(\frac{w}{W} \right)$ CaCO₃ solution

(D) $X_{C_3H_7OH} = \frac{1}{11}$

Column-II

(Density of given solutions is 1.2 g/ml)

(P) 16gm solute in 240gm solution

(Q) 60gm solute in 240 gm solution

(R) 8gm solute in 100 ml solution

(S) 30 gm solute in 100 ml solution

(T) 1 mole solute in 400 gm solution **CT0069**

19. Match the column:

Column I

- (A) 20 V H_2O_2
 (B) 24.5 % w/v H_2SO_4
 (C) Pure water
 (D) 5% w/w NaOH ($d_{\text{solution}} = 1.2 \text{ gm/ml}$)

Column II

- (P) 2.5 M
 (Q) 1.76 M
 (R) 1.5 M
 (S) 55.5 M

CT0070

20.

Column-I

- (A) 120 g CH_3COOH in 1 L solution
 ($d_{\text{sol}} = 1.2 \text{ g/mL}$)
 (B) 120 g glucose dissolved in 1 L solution
 ($d_{\text{sol}} = 1.2 \text{ g/mL}$)
 (C) $X_{\text{NH}_2\text{CONH}_2} = 1/31$ (aqueous solution)
 (D) 19.6% (w/v) H_2SO_4 solution $\rightarrow$
 ($d_{\text{solution}} = 1.2 \text{ g/mL}$)

Column-II

- (P) $M = 2$
 (Q) 10% w/w solution
 (R) 12% w/v solution
 (S) $m = 1.85$
 (T) $m = 0.617$

CT0071

EXERCISE # J-MAINS

- 10 mL of 2(M) NaOH solution is added to 200 mL of 0.5 (M) of NaOH solution. What is the final concentration ?
[JEE(Main-online)-2013]
(A) 0.57 M (B) 5.7 M (C) 11.4 M (D) 1.14 M **CT0085**
- The density of 3M solution of sodium chloride is 1.252 g mL^{-1} . The molality of the solution will be (molar mass, NaCl = 58.5 g mol^{-1})
[JEE(Main-online)-2013]
(A) 2.18 m (B) 3.00 m (C) 2.60 m (D) 2.79 m **CT0086**
- The amount of BaSO_4 formed upon mixing 100 mL of 20.8% BaCl_2 solution with 50 mL of 9.8% H_2SO_4 solution will be :
[JEE(Main-online)-2014]
(Ba = 137, Cl = 35.5, S=32, H = 1 and O = 16)
(A) 33.2 g (B) 11.65 g (C) 23.3 g (D) 30.6 g **CT0083**
- For the estimation of nitrogen, 1.4 g of an organic compound was digested by Kjeldahl method and the evolved ammonia was absorbed in 60 mL of $\frac{\text{M}}{10}$ sulphuric acid. The unreacted acid required 20 mL of $\frac{\text{M}}{10}$ sodium hydroxide for complete neutralization. The percentage of nitrogen in the compound is :
[JEE(Main-online)-2014]
(A) 3% (B) 5% (C) 6% (D) 10% **CT0084**
- The amount of sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) required to prepare 2 L of its 0.1 M aqueous solution is :
[JEE-Main(Jan.)-2019]
(A) 68.4 g (B) 17.1 g (C) 34.2 g (D) 136.8 g **CT0076**
- A solution of sodium sulfate contains 92 g of Na^+ ions per kilogram of water. The molality of Na^+ ions in that solution in mol kg^{-1} is:
[JEE-Main(Jan.)-2019]
(A) 16 (B) 8 (C) 4 (D) 12 **CT0077**
- 8g of NaOH is dissolved in 18g of H_2O . Mole fraction of NaOH in solution and molality (in mol kg^{-1}) of the solutions respectively are :
[JEE-Main(Jan.)-2019]
(A) 0.167, 11.11 (B) 0.2, 22.20 (C) 0.2, 11.11 (D) 0.167, 22.20 **CT0078**
- The volume strength of 1M H_2O_2 is: (Molar mass of $\text{H}_2\text{O}_2 = 34 \text{ g mol}^{-1}$)
[JEE-Main(Jan.)-2019]
(A) 16.8 (B) 11.35 (C) 22.4 (D) 5.6 **CT0079**
- The strength of 11.2 volume solution of H_2O_2 is :
[Given that molar mass of H = 1 g mol^{-1} and O = 16 g mol^{-1}]
[JEE-Main(april)-2019]
(A) 13.6% (B) 3.4% (C) 34% (D) 1.7% **CT0080**
- What would be the molality of 20% (mass/mass) aqueous solution of KI?
(molar mass of KI = 166 g mol^{-1})
[JEE-Main(april)-2019]
(A) 1.08 (B) 1.48 (C) 1.51 (D) 1.35 **CT0081**
- The mole fraction of a solvent in aqueous solution of a solute is 0.8. The molality (in mol kg^{-1}) of the aqueous solution is
[JEE-Main(april)-2019]
(A) 13.88×10^{-1} (B) 13.88×10^{-2} (C) 13.88 (D) 13.88×10^{-3} **CT0082**

12. The molarity of HNO_3 in a sample which has density 1.4 g/mL and mass percentage of 63% is (Molecular Weight of $\text{HNO}_3 = 63$) **[JEE-Main(Jan)-2020]**
CT0072
13. 10.30 mg of O_2 is dissolved into a liter of sea water of density 1.03 g/mL. The concentration of O_2 in ppm is **[JEE-Main(Jan)-2020]**
CT0073
14. The ammonia (NH_3) released on quantitative reaction of 0.6 g urea (NH_2CONH_2) with sodium hydroxide (NaOH) can be neutralized by : **[JEE-Main(Jan)-2020]**
(A) 100 ml of 0.1 M HCl (B) 200 ml of 0.4 M HCl
(C) 100 ml of 0.2 M HCl (D) 200 ml of 0.2 M HCl **CT0074**
15. Ferrous sulphate heptahydrate is used to fortify foods with iron. The amount (in grams) of the salt required to achieve 10 ppm of iron in 100 kg of wheat is _____. **[JEE-Main(Jan)-2020]**
Atomic weight : Fe = 55.85 ; S = 32.0 ; O = 16.00 **CT0075**
16. The volume strength of 8.9 M H_2O_2 solution calculated at 273 K and 1 atm is _____. ($R=0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$) (rounded off to the nearest integer) **[JEE(Main-online)-2020]**
CT0097
17. 6.023×10^{22} molecules are present in 10 g of a substance 'x'. The molarity of a solution containing 5 g of substance 'x' in 2 L solution is _____ $\times 10^{-3}$. **[JEE(Main-online)-2020]**
CT0098
18. The strengths of 5.6 volume hydrogen peroxide (of density 1 g/mL) in terms of mass percentage and molarity (M), respectively, are: **[JEE(Main-online)-2020]**
(Take molar mass of hydrogen peroxide as 34 g/mol)
(1) 1.7 and 0.25 (2) 1.7 and 0.5 (3) 0.85 and 0.5 (4) 0.85 and 0.25 **CT0099**
19. 4.5 g of compound A (MW = 90) was used to make 250 mL of its aqueous solution. The molarity of the solution in M is $x \times 10^{-1}$. The value of x is _____. (Rounded off to the nearest integer) **[JEE(Main-online)-2021]**
CT0100
20. The NaNO_3 weighed out to make 50 mL of an aqueous solution containing 70.0 mg Na^+ per mL is _____ g. (Rounded off to the nearest integer) **[JEE(Main-online)-2021]**
[Given : Atomic weight in g mol^{-1} – Na : 23 ; N : 14 ; O : 16] **CT0101**
21. A 6.50 molal solution of KOH (aq.) has a density of 1.89 g cm^{-3} . The molarity of the solution is _____ mol dm^{-3} . (Round off to the Nearest Integer). **[JEE(Main-online)-2021]**
[Atomic masses: K :39.0 u; O :16.0 u; H :1.0 u] **CT0102**
22. The mole fraction of a solute in a 100 molal aqueous solution _____ $\times 10^{-2}$. (Round off to the Nearest Integer). **[JEE(Main-online)-2021]**
[Given : Atomic masses : H : 1.0 u, O : 16.0 u] **CT0103**
23. 0.8 g of an organic compound was analysed by Kjeldahl's method for the estimation of nitrogen. If the percentage of nitrogen in the compound was found to be 42%, then _____ mL of 1 M H_2SO_4 would have been neutralized by the ammonia evolved during the analysis. **CT0105**

24. The density of NaOH solution is 1.2 g cm^{-3} . The molality of this solution is _____ m.
(Round off to the Nearest Integer) [JEE(Main-online)-2021]
[Use : Atomic masses : Na : 23.0 u O : 16.0 u, H : 1.0 u
Density of H_2O : 1.0 g cm^{-3}] CT0106
25. An aqueous KCl solution of density 1.20 g mL^{-1} has a molality of 3.30 mol kg^{-1} . The molarity of the solution in mol L^{-1} is _____ (Nearest integer) [JEE(Main-online)-2021]
[Molar mass of KCl = 74.5] CT0107
26. 100 mL of Na_3PO_4 solution contains 3.45 g of sodium. The molarity of the solution is _____ $\times 10^{-2} \text{ mol L}^{-1}$. (Nearest integer) [JEE(Main-online)-2021]
[Atomic Masses - Na : 23.0 u, O : 16.0 u, P : 31.0 u] CT0108
27. The molarity of the solution prepared by dissolving 6.3 g of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) in 250 mL of water in mol L^{-1} is $x \times 10^{-2}$. The value of x is _____. (Nearest integer) [JEE(Main-online)-2021]
[Atomic mass : H : 1.0, C : 12.0, O : 16.0] CT0109
28. Sodium oxide reacts with water to produce sodium hydroxide. 20.0 g of sodium oxide is dissolved in 500 mL of water. Neglecting the change in volume, the concentration of the resulting NaOH solution is _____ $\times 10^{-1} \text{ M}$. (Nearest integer) [JEE(Main-online)-2021]
[Atomic mass : Na = 23.0, O = 16.0, H = 1.0] CT0110
29. If 80 g of copper sulphate $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is dissolved in deionised water to make 5 L of solution. The concentration of the copper sulphate solution is $x \times 10^{-3} \text{ mol L}^{-1}$. The value of x is _____. [JEE(Main-online)-2021]
[Atomic masses Cu : 63.54 u, S : 32 u, O : 16 u, H : 1 u] CT0111
30. Chlorophyll extracted from the crushed green leaves was dissolved in water to make 2 L solution of Mg of concentration 48 ppm. The number of atoms of Mg in this solution is $x \times 10^{20}$ atoms. The value of x is _____. (Nearest Integer)
(Given : Atomic mass of Mg is 24 g mol^{-1} , $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$) [JEE(Main-online) July-2022] CT0116
31. When 800 mL of 0.5 M nitric acid is heated in a beaker, its volume is reduced to half and 11.5 g of nitric acid is evaporated. The molarity of the remaining nitric acid solution is $x \times 10^{-2} \text{ M}$. (Nearest Integer) [JEE(Main-online) June-2022] CT0117
(Molar mass of nitric acid is 63 g mol^{-1})

EXERCISE # JEE-ADVANCED

1. Dissolving 120 g of urea (mol. wt. 60) in 1000 g of water gave a solution of density 1.15 g/mL. The molarity of the solution is [JEE-2011]
(A) 1.78 M (B) 2.00 M (C) 2.05 M (D) 2.22 M CT0092
2. 29.2 % (w/w) HCl stock solution has a density of 1.25 g mL⁻¹. The molecular weight of HCl is 36.5 g mol⁻¹. The volume (mL) of stock solution required to prepare a 200 mL solution of 0.4 M HCl is. [JEE-2012]
CT0112
3. A compound H₂X with molar weight of 80 g is dissolved in a solvent having density of 0.4 g /ml, Assuming no change in volume upon dissolution, the **molality** of a 3.2 molar solution is. [JEE 2014]
CT0091
4. The mole fraction of a solute in a solution is 0.1. At 298 K, molarity of this solution is the same as its molality. Density of this solution at 298 K is 2.0 g cm⁻³. The ratio of the molecular weights of the solute and solvent, $\left(\frac{MW_{\text{solute}}}{MW_{\text{solvent}}} \right)$, is [JEE-2016]
CT0113
5. The mole fraction of urea in an aqueous urea solution containing 900 g of water is 0.05. If the density of the solution is 1.2 g cm⁻³, the molarity of urea solution is [JEE 2019]
(Given data : Molar masses of urea and water are 60 g mol⁻¹ and 18 g mol⁻¹, respectively)
CT0090

ANSWER KEY**EXERCISE # S-I**

1. (a) 50 milimoles, (b) 12 milimoles, (c) 450 milimoles, (d) 5 milimoles, 0.05 M
2. (a) 0.50 M, (b) 0.50 M, (c) 0.20 M
3. 1.30 M 4. 16.66 or 16.67 5. 3.00 6. 1.40
7. 1.25 8. 45.45 9. 2.50 10. 13.00 gm
11. 0.66 or 0.67 12. 1.40 13. 1.50 M 14. (0.80)
15. 0.05 16. 0.20 M 17. 5.55 or 5.56 18. (0.15 M)
19. 45.40 V 20. (56.00 V)

EXERCISE # S-II

1. 0.25 2. 0.02 3. 1.04 4. 0.50 M
5. 21.60 6. 0.05 7. 0.17 or 0.18 8. (12.00 V)
9. (1.50 M) 10. (2.00 M)

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	C	A	D	B	B	B	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	B	A	A	A	C	D	C	C
Que.	21	22	23							
Ans.	A	B	C							

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	D	A,C,D	A,C	B,D	B,D	A,C	C	C
Que.	11	12	13	14	15	16	17			
Ans.	A	B	C	B	D	B	D			
Que.	18				19					
Ans.	A-P,R	B-P,R	C-Q,S,T	D-S,Q	A-Q	B-P	C-S	D-R		
Que.	20									
Ans.	A-P,Q,R,S	B-Q,R,T	C-Q,S	D-P						

EXERCISE # J-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	B	D	A	C	A	B	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	14.00	10	C	4.95 to 4.97	100	25	B	2	13
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	9	64	12	5	3	50	20	13	64	24
Que.	31									
Ans.	54									

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	
Ans.	C	8	8.00	9	2.98 or 2.99	

REDOX REACTIONS & EQUIVALENT CONCEPTS

1. OXIDATION & REDUCTION

Let us do a comparative study of oxidation and Reduction ;

Oxidation	Reduction
(1) Addition of oxygen $\text{Mg} \rightarrow \text{MgO}$	(1) Removal of oxygen $\text{CuO} \rightarrow \text{Cu}$
(2) Removal of Hydrogen $\text{H}_2\text{S} \rightarrow \text{S}$	(2) Addition of Hydrogen $\text{S} \rightarrow \text{H}_2\text{S}$
(3) Addition of electronegative atom or group $\text{Mg} \rightarrow \text{MgCl}_2$	(3) Removal of Electronegative atom or group $\text{FeCl}_3 \rightarrow \text{FeCl}_2$
(4) Removal of electropositive atom or group $\text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow \text{K}_3[\text{Fe}(\text{CN})_6]$	(4) Addition of electropositive atom or group $\text{Cl}_2 \rightarrow \text{MgCl}_2$
(5) Increase in positive charge $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$	(5) Decrease in positive charge $\text{Fe}^{3+} \rightarrow \text{Fe}^{2+}$
(6) Increase in oxidation number $+2 \quad +4$ $\text{SnCl}_2 \rightarrow \text{SnCl}_4$	(6) Decrease in oxidation number $+7$ $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$
(7) Loss of electron $\text{Sn}^{2+} \rightarrow \text{Sn}^{4+} + 2\text{e}^-$	(7) Gain of electron $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$

2. OXIDATION NUMBER OR OXIDATION STATE :

- ◆ Oxidation number denotes the oxidation state of an element in a compound ascertained according to a set of rules formulated on the basis that electron pair in a covalent bond belongs entirely to more electronegative element.

For example , in CO_2 , the oxidation state of carbon is +4 and it is also its oxidation number. Similarly, the oxidation state as well as the oxidation number of oxygen is -2. This implies that the oxidation number denotes the oxidation state of an element in a compound.

Now, understand the determination of oxidation number with the help of these examples :

(a) HCl

Structural formula : H – Cl

As chlorine is more electronegative than hydrogen, counting shared electron pair for chlorine will result $\text{H}^+ \text{Cl}^-$ and hence,

oxidation number of H = +1

oxidation number of Cl = -1

(b) CO_2 Structural formula : $\text{O} = \text{C} = \text{O}$

As oxygen is more electronegative than carbon, counting shared electron pair for oxygen with result $\text{O}^{2-} \text{C}^{2+} \text{O}^{2-}$ and hence,

Oxidation number of C = +4

Oxidation number of O = -2

2.1 Rules to determine oxidation number :

The following rules are helpful in calculating oxidation number of the elements in their different compounds. It must be remembered that the basis of these rule is the electronegativity of the element.

(i) Oxidation number of an element in free state or in allotropic forms is always zero.

e.g. $\overset{0}{\text{O}_2}$, $\overset{0}{\text{S}_8}$, $\overset{0}{\text{P}_4}$, $\overset{0}{\text{O}_3}$, $\overset{0}{\text{Na}}$

(ii) Sum of the oxidation number of all the atoms in a molecule is zero.

(iii) Sum of the oxidation number of all the atoms in an ions is equal to the charge on the ion.

(iv) If the group number of an element in periodic table is 'n' (I, II, III, etc) then its oxidation number may vary from (+n) to (n - 8).

e.g. N-atom belongs to V group in the periodic table therefore as per rule its oxidation number

may vary from -3 to +5 ($\overset{-3}{\text{NH}_3}$, $\overset{+2}{\text{NO}}$, $\overset{+3}{\text{N}_2\text{O}_3}$, $\overset{+4}{\text{NO}_2}$, $\overset{+5}{\text{N}_2\text{O}_5}$)

Exceptions :

Cu(II) : +1, +2 ;

Au (I) : +1, +3 , etc

(v) **Metals :**

(a) Alkali metal (Li, Na, K, Rb, Cs, Fr) always have oxidation number +1 in all of their compounds.

(b) Alkaline earth metal (Be, Mg, Ca , Sr, Ba, Ra) always have oxidation number +2 in all of their compounds.

(c) Aluminium always have +3 oxidation number in all of its compounds.

(vi) **Hydrogen atom :**

H atom has oxidation number equal to +1 in all of its compound, but in metallic hydrides (e.g. NaH, KH), it is -1.

(vii) **Oxygen atom :**

Oxygen atom has oxidation number equal to -2 in all of its compounds except

- (a) peroxides (e.g. H_2O_2 , Na_2O_2), where it is -1
- (b) super oxides (e.g. KO_2) where it is $-\frac{1}{2}$
- (c) ozonides (KO_3), where it is $-\frac{1}{3}$
- (d) oxygen fluorides (O_nF_2), where it is $\left(+\frac{2}{n}\right)$.

(viii) Halogen atom :

Fluorine is most electronegative atom. It always has oxidation no. equal to -1 in all of its compound.

All other halogen atoms (Cl, Br, I) has oxidation number equal to -1 in all of their compounds. But if halogen atom is attached with an more electronegative atom, than halogen atom then it will show positive oxidation numbers.

Eg. KClO_3^{+5} , HIO_3^{+5} , HClO_4^{+7} , KBrO_3^{+5} , etc

- (ix) If two or more than two atoms of an element are present in the molecule or ion such as $\text{Na}_2\text{S}_2\text{O}_3$, $\text{Cr}_2\text{O}_7^{2-}$, etc, the oxidation number of the atom of that element will then be the average of the oxidation number of all the atoms of that element.

2.2 SOME SPECIAL CASES :

In order to determine the exact or individual oxidation number we need to take help from the structures of the molecules. Some special cases are discussed as follows :

(i) CrO_5

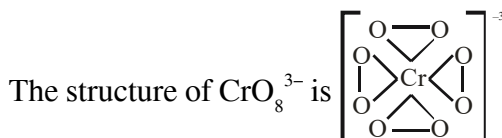


From the structure it is evident that in CrO_5 there are two peroxide linkages. The contribution of each peroxide linkage is -2 . Let the O.N. of Cr is x .

$$\therefore x + (-2)2 + (-2) = 0$$

$$\therefore \text{O.N. of Cr} = x = +6$$

(ii) K_3CrO_8



From the structure it is evident that in CrO_8^{3-} there are four peroxide linkages. The contribution of each peroxide linkage is -2 . Let the O.N. of Cr is x .

(iii) H_2SO_5

The structure of H_2SO_5 is $\text{H}-\text{O}-\text{O}-\text{S}(\text{O})_2-\text{O}-\text{H}$

From the structure, it is evident that in H_2SO_5 , there are one peroxide linkage, Let the O.N. of S = x

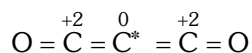
$$\therefore +1 - 2 + x + (-2) 2 + (-2) + 1 = 0$$

$$\therefore \text{O.N. of S} = x = 6$$

2.3 PARADOX OF FRACTIONAL OXIDATION NUMBER

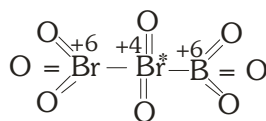
Fractional oxidation state is the average oxidation state of the element under examination and the structural parameters reveal that the element for whom fractional oxidation state is realised is actually present in different oxidation states. Structure of the species C_3O_2 , Br_3O_8 and $\text{S}_4\text{O}_6^{2-}$ reveal the following bonding situations. The element marked with asterisk in each species is exhibiting the different oxidation state (oxidation number) from rest of the atoms of the same element in each of the species.

- ◆ This reveals that in C_3O_2 , two carbon atoms are present in +2 oxidation state each whereas the third one is present in zero oxidation state and the average is $4/3$. However the realistic picture is +2 for two terminal carbons and zero for the middle carbon.



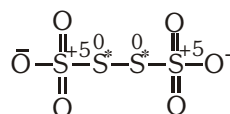
Structure of C_3O_2
(Carbon suboxide)

- ◆ Likewise in Br_3O_8 , each of the two terminal bromine atoms are present in +6 oxidation state and the middle bromine is present in +4 oxidation state. Once again the average, that is different from reality is $16/3$.



Structure of Br_3O_8 (tribromooctaoxide)

- ◆ In the same fashion, in the species $\text{S}_4\text{O}_6^{2-}$, is 2.5, whereas the reality being +5, 0, 0 and +5 oxidation number respectively for each sulphur.



Structure of $\text{S}_4\text{O}_6^{2-}$ (tetrathionate ion)

Ex.1 Calculate oxidation number of underlined element in the following:

- (i) $\text{Na}_2\underline{\text{S}}_2\text{O}_3$ (ii) $\text{Na}_2\underline{\text{S}}_4\text{O}_6$ (iii) $\underline{\text{Cl}}\text{O}^-$, (iv) $\underline{\text{N}}\text{O}_2^-$,
 (v) $\underline{\text{N}}\text{O}_3^-$ (vi) $\underline{\text{C}}\text{Cl}_4$ (vii) $\text{K}_2\underline{\text{Cr}}\text{O}_4$ (viii) $\text{KMn}\underline{\text{O}}_4$

Sol. (i) Let oxidation number of S-atom is x.

$$(+1) \times 2 + (x) \times 2 + (-2) \times 3 = 0$$

$$x = +2$$

(ii) Let oxidation number of S-atom is x

$$(+1) \times 2 + (x) \times 2 + (-2) \times 6 = 0$$

$$x = +2.5$$

(iii) $x - 2 = -1$ (where 'x' is O.N. of Cl)

$$\therefore x = +1$$

(iv) $2 \times (-2) + x = -1$

(where 'x' is O.N. of N)

$$\therefore x = +3$$

(v) $x + 3 \times (-2) = -1$

(where 'x' is O.N. of N)

$$x = +5$$

(vi) $x + 4 \times (-1) = 0$

(where 'x' is O.N. of C)

$$\therefore x = +4$$

(vii) $2 \times +1 + x + 4 \times (-2) = 0$

(where 'x' is O.N. of Cr)

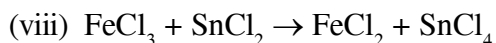
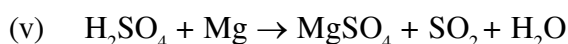
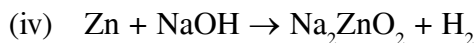
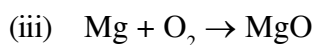
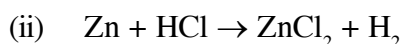
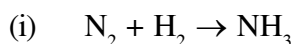
$$\therefore x = +6$$

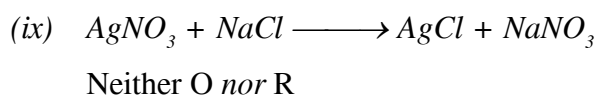
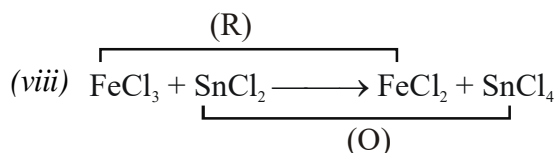
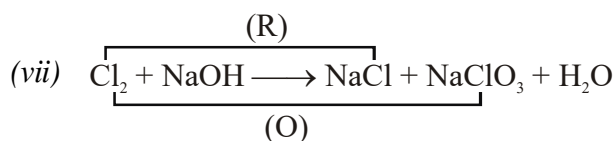
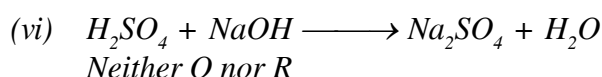
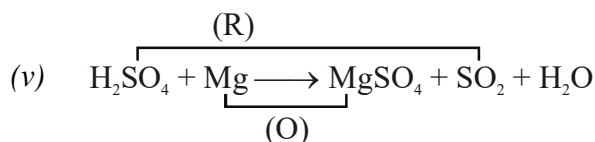
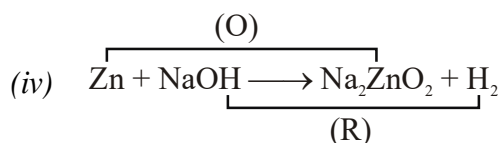
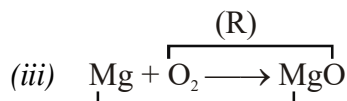
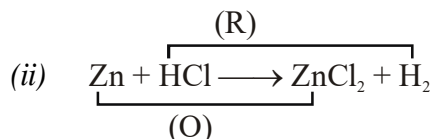
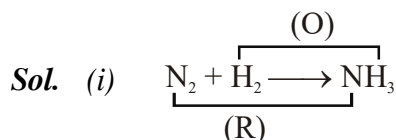
(vi) $+1 + x + 4 \times (-2) = 0$

(where x is O.N. of Mn).

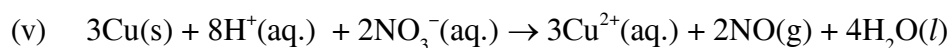
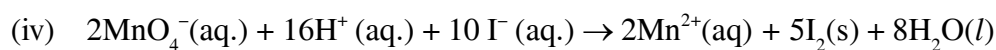
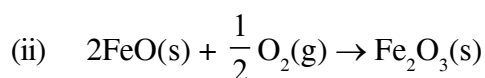
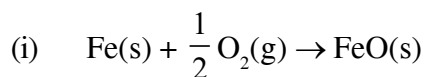
$$\therefore x = +7$$

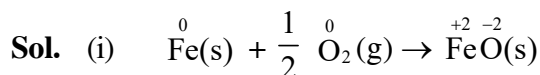
Ex.2. Identify the species undergoing oxidation and reduction in the following reactions :



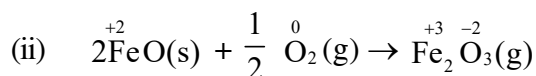


Ex.3. Identify the oxidising and reducing agents in the following reactions :

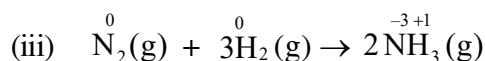




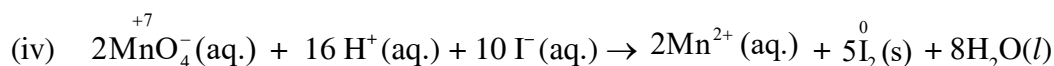
R.A. O.A.



R.A. O.A.

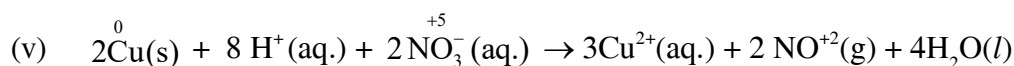


O.A. R.A.



O.A.

R.A.



R.A.

O.A.

3 OXIDISING AND REDUCING AGENTS :

3.1 Oxidising agent or Oxidant :

Oxidising agents are those compound which can oxidise others and reduced itself during the chemical reaction. Those reagents whose O.N. decrease or which gain electron(s) in a redox reaction are termed as oxidants.

e.g. KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, HNO_3 , conc. H_2SO_4 etc, are some powerful oxidising agents.

3.2 Reducing agent or Reductant :

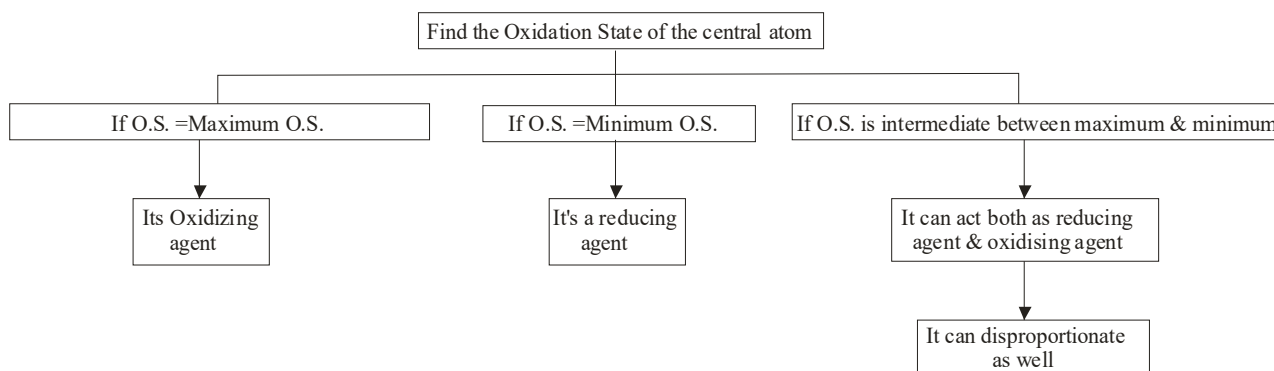
Reducing agents are those compound which can reduce others and oxidise itself during the chemical reaction. Those reagents whose O.N. increase or which loses electron in a redox reaction are termed as reductants.

e.g. KI , $\text{Na}_2\text{S}_2\text{O}_3$, etc are some powerful reducing agents.

Note : There are some compounds also which can work both oxidising agent and reducing agent.

e.g. H_2O_2 , NO_2^-

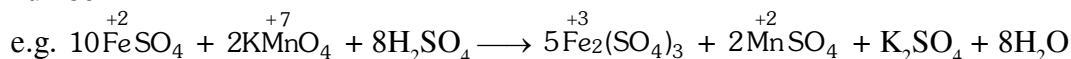
3.3 How to identify whether a particular substance is an oxidising or reducing agent :



4 REDOX REACTIONS

A reaction in which oxidation and reduction simultaneously take place.

In all redox reactions, the total increase in oxidation number must equal the total decrease in oxidation number.

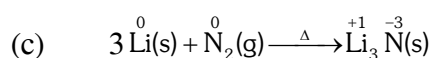
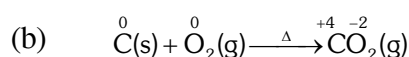
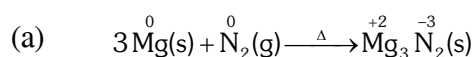


TYPE OF REDOX REACTION

4.1 Combination Reactions :

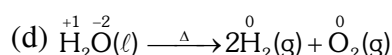
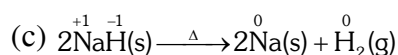
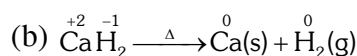
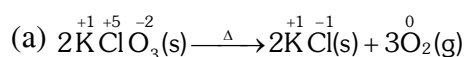
When two or more element combine to form compound, then such combination reactions are redox reaction.

For example : $\text{A} + \text{B} \longrightarrow \text{C}$



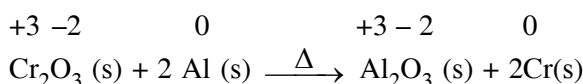
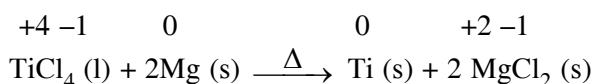
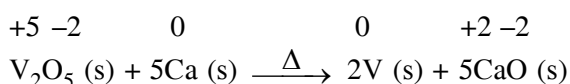
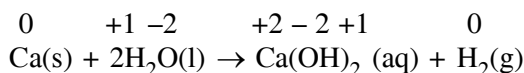
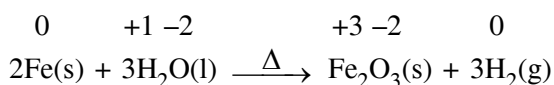
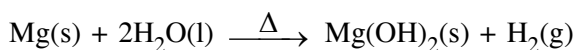
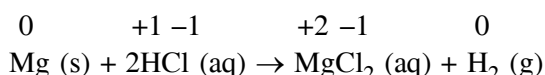
4.2 Decomposition Reactions :

Decomposition reactions are opposite of combination. Decomposition reactions are those in which when a compound breakdown to form two or more components. If at least one of the product is in the elemental state, such decomposition reactions are definitely redox. For example :

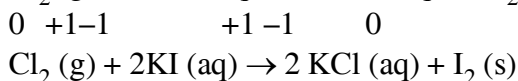
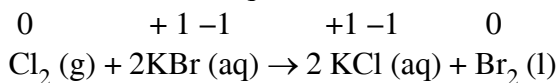


$$\overset{+2}{\text{Ca}} \overset{+4-2}{\text{CO}_3}(\text{s}) \xrightarrow{\Delta} \overset{+2}{\text{Ca}} \overset{-2}{\text{O}}(\text{s}) + \overset{+4}{\text{C}} \overset{-2}{\text{O}_2}(\text{g})$$

A reaction in which an atom or ion in a compound is replaced by an atom or ion of another element is called a displacement reaction. It may be denoted as :


$$\overset{+2}{\text{Cu}}\overset{+6}{\text{S}}\overset{-2}{\text{O}_4}(\text{aq}) + \text{Zn}(\text{s}) \rightarrow \text{Cu}(\text{s}) + \overset{+2}{\text{Zn}}\overset{+6}{\text{S}}\overset{-2}{\text{O}_4}(\text{aq})$$

$$2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$$

$$0 \quad +1 \quad -2 \quad +2 \quad -2 \quad +1 \quad 0$$

$$\begin{array}{ccccccc} 0 & +1 & -1 & & +2 & -1 & 0 \\ \text{Zn(s)} & + & 2\text{HCl(aq)} & \rightarrow & \text{ZnCl}_2\text{(aq)} & + & \text{H}_2\text{(g)} \end{array}$$

$$\begin{array}{ccccccc} +1 & -2 & & 0 & & +1 & -1 & & 0 \\ 2\text{H}_2\text{O} & (\text{l}) & + & 2\text{F}_2 & (\text{g}) & \rightarrow & 4\text{HF} & (\text{aq}) & + & \text{O}_2 & (\text{g}) \end{array}$$

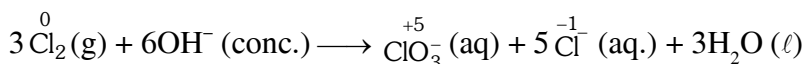
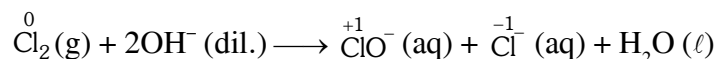
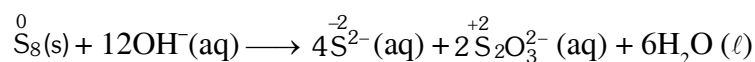
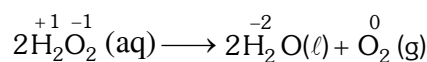
It is for this reason that the displacement reactions of chlorine, bromine and iodine using fluorine are not generally carried out in aqueous solution. On the other hand, chlorine can displace bromide and iodide ions in an aqueous solution as shown below:



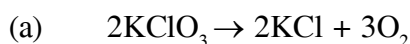
4.4 Disproportionations reactions :

A redox reaction in which the same element present in a particular compound (molecule or ion) in a definite oxidation state is oxidized as well as reduced simultaneously is a disproportionation reactions.

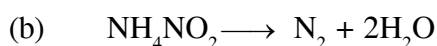
Disproportionation reactions are a special type of redox reactions. One of the reactants in a disproportionation reaction always contains an element that can exist in at least three oxidation states. The element in the form of reacting substance is in the intermediate oxidation state and both higher and lower oxidation states of that element are formed in the reaction.



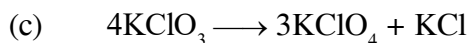
◆ **Consider the following reactions :**



KClO_3 plays a role of oxidant and reductant both. Because same element is not oxidised and reduced. Here, Cl present in KClO_3 is reduced and O present in KClO_3 is oxidized. So **it is not a disproportion reaction** although it looks like one.

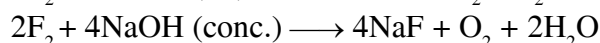
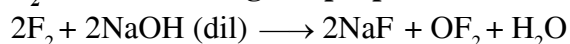


Nitrogen in this compound has -3 and $+3$ oxidation number but belongs to different ions so it is not a disproportion reaction.



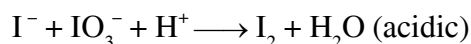
It is case of disproportionation reaction in which Cl is disproportionating.

Note : F_2 does not undergo disproportionation as it is the most electronegative element.



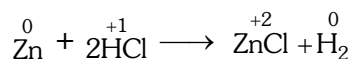
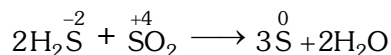
4.5 Comproportionation reaction :

Reverse of disproportionation is called comproportionation. It is a class of redox reaction in which an element from two different oxidation state gets converted into a single oxidation state. In some of the disproportionation reactions by changing the medium (from acidic to basic or reverse) the reaction goes in backward direction and can be taken as an example of comproportionation.

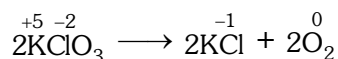


4.6 Intermolecular and intramolecular redox reactions :

In **intermolecular redox reaction**, oxidation and reduction of different species occur. For example :



In **Intramolecular redox reaction**, the same species get oxidised as well as reduced. For example:



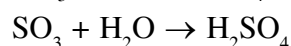
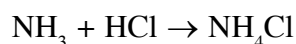
Oxidation and reduction occurred for the same compound, KClO_3 .

Disproportionation reaction is a type of intramolecular redox reaction, in which oxidation and reduction occur due to the same element, from same molecule or ion.

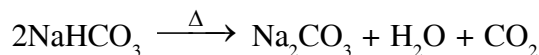
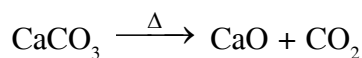
5. NON-REDOX REACTIONS :

These are the reactions in which neither oxidation nor reduction take place. Some example of non-redox reactions are :

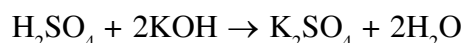
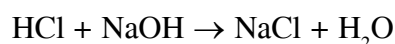
(i) Combination reactions like :



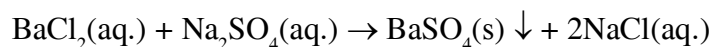
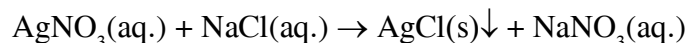
(ii) Decomposition reactions like :



(iii) Neutralisation reactions like :



(iv) Precipitation reactions like :

**6. BALANCING OF REDOX REACTION**

All balanced equations must satisfy two criteria :

(i) **Atom balance (mass balance) :**

There must be the same number of atoms of each element in reactant and product side.

(ii) **Charge balance :**

The sum of actual total charges on both side of the equation must be equal.

There are two methods for balancing the redox reactions

(a) Oxidation - number method

(b) Ion electron method or half cell method

6.1 Oxidation number method :

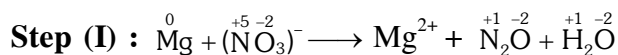
This method was given by Jonson. In a balanced redox reaction, **total increase in oxidation number must be equal to total decreases in oxidation number**. This equivalence provides the basis for balancing redox reactions. The general procedure involves the following steps :

- Step-I :** Write the correct formula for each reactant and product.
- Step-II :** Identify atoms which undergoes change in oxidation number in the reaction by assigning the oxidation number to all elements in the reaction.
- Step-III :** Balance the number of atoms of the elements by simple counting, for which changes in oxidation numbers are taking place.
- Step-IV :** Calculate the increase or decrease in the oxidation number per atom and for the entire molecule/ion in which it occurs. If these are not equal, then multiply by suitable number so that they become equal.
- Step-V :** Balance the atoms of all other elements (for which change in oxidation number was not occurring), by simple counting (adjusting the stoichiometric coefficients of other compounds with the help of stoichiometric coefficients of the components fixed in step-III).
- Step-VI :** If the given reaction is in ionic form, ascertain that the total charges in reactants and products are equal. Addition of H^+ or OH^- ions and / or H_2O molecule may be helpful in balancing charge and even, atoms in step- IV. For reaction in acidic medium, H^+ ions and for reaction in basic medium, OH^- ions are used, as required.

Ex.4 Balance the following chemical reaction by oxidation number method and write their skeleton equation :

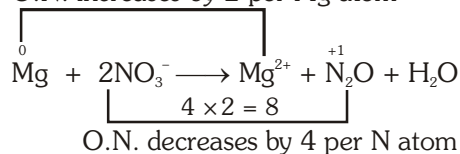
The nitrate ions in acidic medium oxidize magnesium to Mg^{2+} ions but itself gets reduced to nitrous oxide.

Sol. The skeleton equation for the given process is

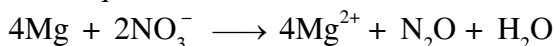


Multiply NO_3^- by 2 to equalize N atoms

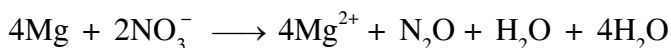
Step (II) : O.N. increases by 2 per Mg atom



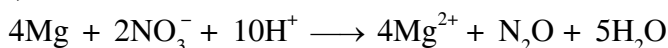
Step (III) : Equalize increase/decrease in O.N. by multiplying Mg by 4 and $2NO_3^-$ by 1.



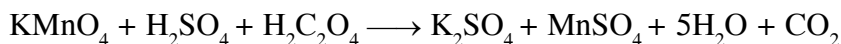
Step (IV) : Balance O atoms



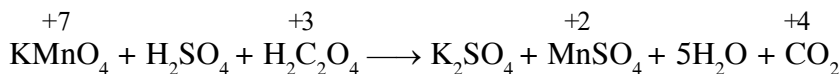
Step (V) : Balance H atoms as is done in acidic medium.



Ex5. Balance the following redox reaction oxidation number method.



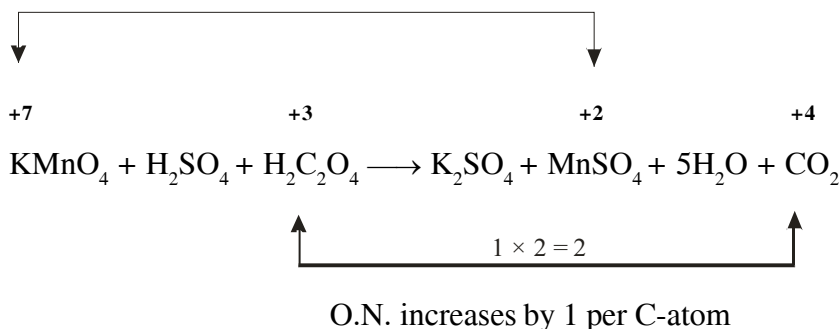
Sol. Step-I :



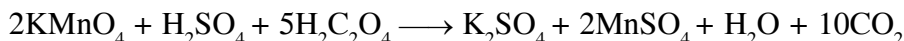
Multiply CO_2 by 2 to equalise the number of C-atoms

Step-II :

O.N. decreases by 5 per Mn atom

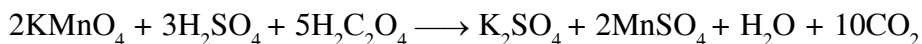


Step-III : Equalise increase and decrease in oxidation number by multiplying KMnO_4 by 2 and $\text{H}_2\text{C}_2\text{O}_4$ by 5.

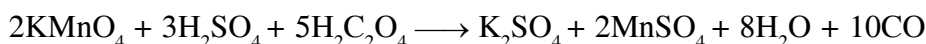


Step-IV : Balance the atoms of other elements as

- K-atoms are already balanced
- S-atoms are three in right side and hence, it should also be three in left side.



H-atoms are 16 in left side and hence it should also be 16 in right side.



- (d) Now check. There are 40 - O-atoms in both sides. Reaction is balanced.

6.2 Ion electron method or half cell method :

In this method, the two half equations are balanced separately and then added together to give balanced equation. The general procedure involves the following steps :

Step-I : Produce unbalanced equation for the reactions in ionic form.

Step-II : Separate the equation into two half reactions : oxidation half and reduction half.

Step-III : Balance the number of atoms of the elements by simple counting, for which changes in oxidation numbers are taking place.

Step-IV : For reactions occurring in acidic medium, add H_2O to balance O-atoms and H^+ to balance H-atoms.

Step-V : Add electrons (s) to one side of half reactions to balance the charges. Make the numbers of electrons equal in both half reactions by multiplying them with suitable numbers.

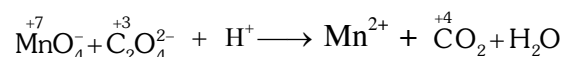
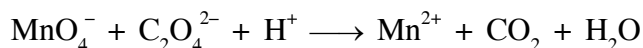
Step-VI : Add both half reactions to achieve the overall reactions and cancel the electrons in both sides. Check the number of atoms of each element and total charges in each side, to reveal that the reaction is fully balanced in ionic form.

For the reaction in a basic medium, first balanced the atoms as is done in acidic medium. Then for each H^+ ion, add an equal number of OH^- ions to both sides of the equation. Where H^+ and OH^- appear on the same side of the equation, combine these to give H_2O .

Ex.6 Write the skeleton equation for each of the following processes and balance them by ion electron method :

- (i) *Permanganate ion oxidizes oxalate ions in acidic medium to carbon dioxide and gets reduced itself to Mn^{2+} ions.*
- (ii) *Chlorine reacts with base to form chlorate ion and chloride ion.*

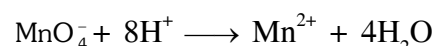
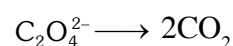
Sol. (i) The skeleton equation for the process :



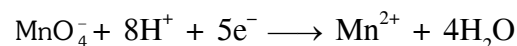
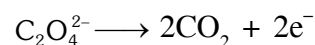
Step (I) : Writing oxidation and reduction half reaction :



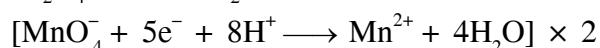
Step (II) : Balancing 'O' and 'H'-atom by adding H_2O molecules and H^+ ions :



Step (III) : Balancing charge by electrons :

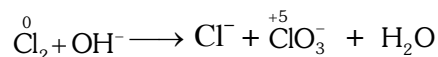


Step (IV) : Multiply the oxidation half reaction by 2 and reduction half reaction by 5 to equalize the electrons lost and gained and add the two half reactions.



This is balance equation

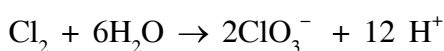
Sol. (ii) The skeleton equation for the given process :



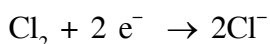
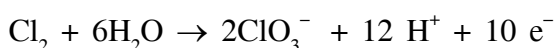
Step (I) : Write the oxidation and reduction half reactions



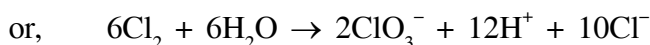
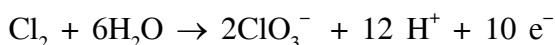
Step (II) : Balancing O- and H-atom by H_2O molecules and H^+ ions.



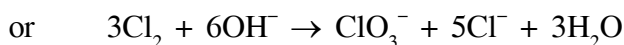
Step (III) : Balancing charge by electron(s)



Step (IV) : Multiply the reduction half reaction by 5 to make the number of electrons lost and gained equal and add both half reactions.



Step (V) : As the reaction given is in basic medium, add 6 OH^- ions in each side.



This is balanced chemical equation.

7. Equivalent weight (E) :

7.1 Equivalent weight, $E = \frac{\text{Molecular weight}}{\text{valency factor (v.f)}}$

Valency factor may also be named as n-factor.

- ◆ Equivalent mass is a pure number. When expressed in gram, it is called gram equivalent mass.
- ◆ The equivalent mass of substance may have different values under different conditions, because n-factor depends on the nature of that substance in that reaction.

7.2 n-factor calculation :

7.2.1 For Acids :

n-factor = number of replaceable H^+ ions = basicity of the acid

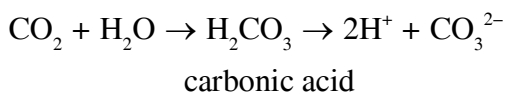
Here, basicity should be taken as moles of H^+ ions furnished per mole of the acid in water.

For example, for the acids HCl , H_2SO_4 and H_3PO_4 , n-factors are 1, 2 and 3 respectively and their

equivalent weights are $\left(\frac{36.5}{1} = 36\right)$, $\left(\frac{98}{2} = 49\right)$ and $\left(\frac{98}{3} = 31.67\right)$, respectively

(a) Non-metal oxides, which may also behave as acid in reactions :

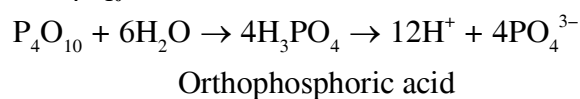
(i) CO_2 :



Hence, n-factor of $CO_2 = 2$

and Eq. wt. of $CO_2 = \frac{44}{2} = 22$

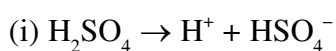
(ii) P_4O_{10} :



Hence, n-factor of $P_4O_{10} = 12$

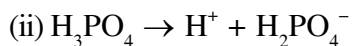
and Eq. wt. of $P_4O_{10} = \frac{284}{12} = 23.67$

(b) Partial neutralisation :



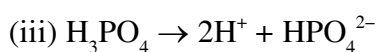
n-factor = 1

$$E = \frac{98}{1} = 98$$



n-factor = 1

$$E = \frac{98}{1} = 98$$



n-factor = 2

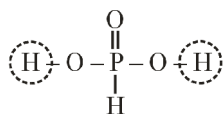
$$E = \frac{98}{2} = 49$$

(c) Special attention :

(i) H_3PO_3

n-factor = 2

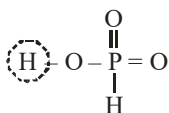
$$E = \frac{82}{2} = 41$$



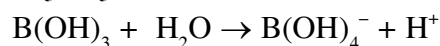
(ii) H_3PO_2

n-factor = 1

$$E = \frac{66}{1} = 66$$



(iii) H_3BO_3



Boric Acid

n-factor = 1

$$E = \frac{61.8}{1} = 61.8$$

7.2.2 For Base :

n-factor = number of replaceable OH^- ions = acidity of the base.

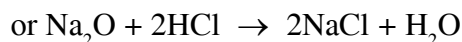
Here, acidity should be taken as moles of OH^- ions furnished per mole of base, in water, or moles of H^+ ions needed to neutralise each mole of the base.

For example, for the bases NaOH , Ca(OH)_2 and Al(OH)_3 , the n-factors are 1, 2 and 3, respectively

and the equivalent weights are $\left(\frac{40}{1} = 40\right)$, $\left(\frac{74}{2} = 37\right)$ and $\left(\frac{78}{3} = 26\right)$ respectively.

(a) Metal oxides, which may also behave as base in reactions :

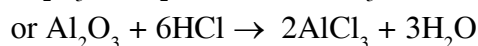
(i) Na_2O :



n-factor = 2

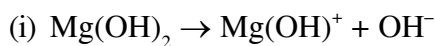
$$E = \frac{62}{2} = 31$$

(ii) Al_2O_3



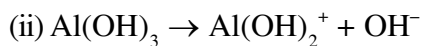
n-factor = 6

$$E = \frac{102}{6} = 17$$

(b) Partial neutralisation :

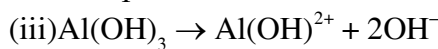
n-factor = 1

$$E = \frac{102}{6} = 58$$



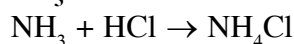
n-factor = 1

$$E = \frac{78}{1} = 78$$



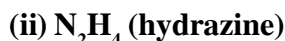
n-factor = 2

$$E = \frac{78}{2} = 39$$

(c) N-containing base :

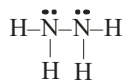
n-factor = 1

$$E = \frac{17}{1} = 17$$

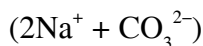


n-factor = 2

$$E = \frac{32}{2} = 16$$

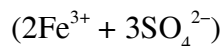
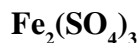
**7.2.3 For Salts :**

n-factor = Total number of positive charge or negative charge present in the salt.

For Example :

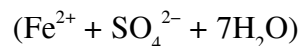
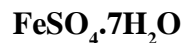
n-factor = 2

$$E = \frac{M}{2}$$



$2 \times 3 = 6$

$$\frac{M}{6}$$



2

$$\frac{M}{2}$$

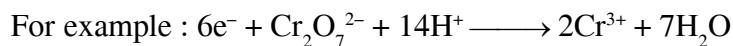
7.2.4 Eq. wt. of oxidising / reducing agents in redox reaction :

The equivalent weight of an oxidising agent is its weight which accepts or loose one mole electron in the chemical reaction.

The n-factor of oxidising or reducing is the number of electrons gained or lost per particle or change in oxidation number per particle

(a) **Eq. wt. of an oxidant or oxidising agent**

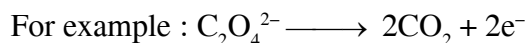
$$= \frac{\text{Mol. wt.}}{\text{Mole of electrons gained per mole}} = \frac{\text{Mol. wt.}}{\text{decrease in oxidation no.}}$$



$$\text{Eq. wt. of } \text{K}_2\text{Cr}_2\text{O}_7 = \frac{\text{Mol. wt.}}{6}$$

(b) **Similarly eq. wt. of a reductant or reducing agent**

$$= \frac{\text{Mol. wt.}}{\text{moles of electrons lost per mole}} = \frac{\text{Mol. wt.}}{\text{increase in oxidation no.}}$$

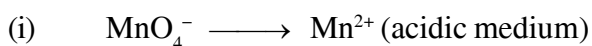


$$\text{Eq. wt. of } \text{H}_2\text{C}_2\text{O}_4 = \frac{\text{Mol. wt.}}{2}$$

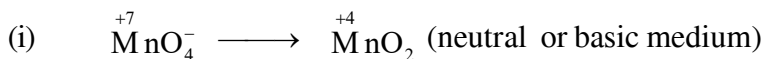
(c) *In different condition a compound may have different equivalent weights, because, it depends upon the number of electrons gained or lost by that compound in that reaction.*

For example :

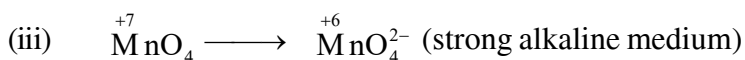
+7



$$\text{Eq. wt. of } \text{KMnO}_4 = \frac{\text{Mol. wt. of } \text{KMnO}_4}{5} = \frac{158}{5} = 31.6$$

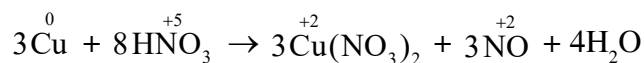


$$\text{Eq. wt. } \text{KMnO}_4 = \frac{\text{Mol. wt. of } \text{KMnO}_4}{3} = \frac{158}{3} = 52.7$$



$$\text{Eq. wt. of } \text{KMnO}_4 = \frac{\text{Mol. wt. of } \text{KMnO}_4}{1} = 158$$

- (d) In any redox reaction, n-factor of all component is determined in terms of electrons n-actor is never determined as acid, base or salt, in redox reactions. For example:



The reaction is balanced by the loss or gain of 6 electrons per reaction. Hence,

$$\text{n-factor or Cu} = \frac{6}{3} = 2 \text{ and } E_{\text{Cu}} = \frac{M}{2}$$

$$\text{n-factor or HNO}_3 = \frac{6}{8} = \frac{3}{4} \text{ and } E_{\text{HNO}_3} = \frac{M}{3/4} = \frac{4M}{3}$$

$$\text{n-factor or Cu}(\text{NO}_3)_2 = \frac{6}{3} = 2 \text{ and } E_{\text{Cu}(\text{NO}_3)_2} = \frac{M}{2}$$

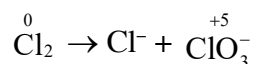
$$\text{n-factor or NO} = \frac{6}{2} = 3 \text{ and } E_{(\text{NO})} = \frac{M}{3}$$

$$\text{and, n-factor or H}_2\text{O} = \frac{6}{4} = \frac{3}{2} \text{ and } E_{\text{H}_2\text{O}} = \frac{M}{3/2} = \frac{2M}{3}$$

- (e) The n-factor of substance undergoing disproportionation (or comproportionation) is given

$$\text{as : n-factor} = \frac{n_1 \cdot n_2}{n_1 + n_2}$$

where n_1 and n_2 are n-factors of the substance only for oxidation and reductions half reactions. For example,



n_1 for Cl_2 undergoing oxidation only = $5 \times 2 = 10$

n_2 for Cl_2 undergoing reduction only = $1 \times 2 = 2$

$$\therefore \text{n-factor of Cl}_2 = \frac{n_1 \cdot n_2}{n_1 + n_2} = \frac{10 \times 2}{10 + 2} = \frac{5}{3}$$

$$\text{and } E_{\text{Cl}_2} = \frac{M}{5/3} = \frac{3M}{5}$$

7.3 SOME OXIDIZING AGENTS WITH THEIR EQUIVALENT WEIGHTS :

Species	Changed to	Reaction	Electrons exchanged or change in O.N.	Eq. wt.
MnO_4^-	Mn^{+2} in acidic medium	$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	5	$E = \frac{M}{5}$
MnO_4^-	MnO_2 in neutral medium	$\text{MnO}_4^- + 3\text{e}^- + 2\text{H}_2\text{O} \longrightarrow \text{MnO}_2 + 4\text{OH}^-$	3	$E = \frac{M}{3}$
MnO_4^-	MnO_4^{2-} in basic medium	$\text{MnO}_4^- + \text{e}^- \longrightarrow \text{MnO}_4^{2-}$	1	$E = \frac{M}{1}$
$\text{Cr}_2\text{O}_7^{2-}$	Cr^{3+} in acidic medium	$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	6	$E = \frac{M}{6}$
MnO_2	Mn^{2+} in acidic medium	$\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^- \longrightarrow \text{Mn}^{2+} + 2\text{H}_2\text{O}$	2	$E = \frac{M}{2}$
Cl_2	Cl^-	$\text{Cl}_2 + 2\text{e}^- \longrightarrow 2\text{Cl}^-$	2	$E = \frac{M}{2}$
H_2O_2	H_2O	$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow 2\text{H}_2\text{O}$	2	$E = \frac{M}{2}$

7.4 SOME REDUCING AGENTS WITH THEIR EQUIVALENT WEIGHTS :

$\text{S}_2\text{O}_3^{2-}$	$\text{S}_4\text{O}_6^{2-}$	$2\text{S}_2\text{O}_3^{2-} \longrightarrow \text{S}_4\text{O}_6^{2-} + 2\text{e}^-$	2	$E = \frac{2M}{2}$
H_2O_2	O_2	$\text{H}_2\text{O}_2 \longrightarrow \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$	2	$E = \frac{M}{2}$
Fe^{2+}	Fe^{3+}	$\text{Fe}^{2+} \longrightarrow \text{Fe}^{3+} + \text{e}^-$	1	$E = \frac{M}{1}$

8. g-EQUIVALENT OR EQUIVALENTS ($n_{\text{eq.}}$)

$$\text{No. of g-equivalents, } n_{\text{eq}} = \frac{\text{wt. (in gm)}}{\text{gmeq. wt.}} = \frac{W}{E}$$

$$\text{Now, } E = \frac{M}{n - \text{factor}}$$

$$\therefore n_{\text{eq.}} = \frac{W}{M / n - \text{factor}} = \frac{W}{M} \times n - \text{factor} = n \times n - \text{factor}$$

$$\text{or, no. of g-equivalents} = \text{no. of moles} \times n - \text{factors}$$

Ex.7 Calculate the number of g-equivalents in the following

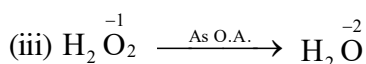
- (i) 4 gm NaOH
 (ii) 2.45 gm H₂SO₄
 (iii) 0.2 mole H₂O₂ (as O.A.)
 (iv) 0.05 mole CH₄ (in complete combustion)

Sol. (i) n-factor of NaOH = 1

$$\therefore n_{\text{eq}} = n \times \text{n-factor} = \frac{4}{40} \times 1 = 0.1$$

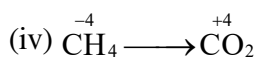
(ii) n-factor of H₂SO₄ = 2

$$\therefore n_{\text{eq}} = n \times \text{n-factor} = \frac{2.45}{98} \times 2 = 0.05$$



$$\text{n-factor of H}_2\text{O}_2 = 1 \times 2 = 2$$

$$\therefore n_{\text{eq}} = n \times \text{n-factor} = 0.2 \times 2 = 0.4$$

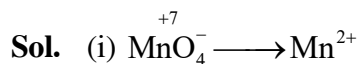


$$\text{n-factor of CH}_4 = 8$$

$$\therefore n_{\text{eq}} = n \times \text{n-factor} = 0.05 \times 8 = 0.4$$

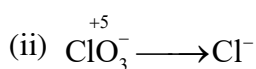
Ex.8 Calculate the number of g-equivalents of solute in the following solution

- (i) 100 ml 0.2 M - KMnO₄ (as O.A. in acidic medium)
 (ii) 4L of 0.05 M - HClO₃ (in the change ClO₃⁻ → Cl⁻)



$$\text{n-factor} = 5$$

$$\therefore n_{\text{eq}} = n \times \text{n-factor} = \left(\frac{100 \times 0.2}{1000} \right) \times 5 = 0.1$$



$$\text{n-factor} = 6$$

$$\therefore n_{\text{eq}} = n \times \text{n-factor} = (4 \times 0.05) \times 6 = 1.2$$

9. NORMALITY

Normality of solution is defined as the number of equivalent of solute present in one litre (1000 mL) solutions. Let a solution is prepared by dissolving W g of solute of eq. wt. E in V mL water.

$$\diamond \quad \text{No. of equivalent of solute} = \frac{W}{E}$$

$$\therefore \quad V \text{ mL of solution have } \frac{W}{E} \text{ equivalent of solute}$$

$$\therefore \quad 1000 \text{ mL solution have } \frac{W \times 1000}{E \times V} \text{ equivalent of solute}$$

$$\therefore \quad \text{Normality (N)} = \frac{W \times 1000}{E \times V}$$

$$\diamond \quad \text{Normality (N)} = \text{Molarity (M)} \times n\text{-factor}$$

$$\text{No. of g-equivalents of solute, } n_{eq} = \frac{W}{E} = \frac{V.N}{1000}$$

$$0.1 \text{ M } - \text{H}_2\text{SO}_4 \text{ solution} = 0.2 \text{ N } - \text{H}_2\text{SO}_4 \text{ solution}$$

Ex.9 4.9 gm H_3PO_4 is dissolved in sufficient water to get 300 ml solution. Calculate the normality of solution.

$$\text{Sol. Eq. wt. of } \text{H}_3\text{PO}_4 = \frac{\text{Mol. wt.}}{n\text{-factor}} \Rightarrow E = \frac{98}{3}$$

$$\text{Now, normality} = \frac{W \times 1000}{E \times V} = \frac{4.9 \times 1000}{\frac{98}{3} \times 300} = 0.5 \text{ N}$$

Ex10. Calculate the no. of g-equivalents of the solute in 250 ml of 0.4 N- KMnO_4 solution.

$$\text{Sol. } n_{eq} = \frac{V.N}{1000} = \frac{250 \times 0.4}{1000} = 0.1$$

Ex11. The normality of KMnO_4 solution is 0.4 N w.r.t. reaction in acidic medium what should be the normality of same solution w.r.t. neutral medium.

Sol. Molarity of any solution is fixed but normality of same solution may change, if n-factor of solute changes in different reactions, Now ,

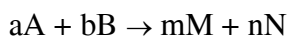
$$M = N \times n\text{-factor} = \text{constant}$$

$$\text{or, } 0.4 \times 5 = N \times 3$$

$$\therefore \text{ Normality in neutral medium, } N = \frac{2}{3}$$

10. LAW OF EQUIVALENCE

The law states that one equivalent of a substance combine with one equivalent of the other substance and one equivalent of each product is formed. In a chemical reaction, equivalent (or mill equivalent) of reactants react in equal amount to give same no. of equivalent (or milli equivalents) of products separately.



$$\text{eq. of A} = \text{eq. of B} = \text{eq. of M} = \text{Eq. of N}$$

$$\text{or } m.\text{eq of A} = m.\text{eq of B} = m.\text{eq of M} = m.\text{eq of N}$$

Ex.12 The number of moles of oxalate ions oxidized by one mole of MnO_4^- ion in acidic medium.

(A) $\frac{5}{2}$

(B) $\frac{2}{5}$

(C) $\frac{3}{5}$

(D) $\frac{5}{3}$

Sol : Equivalents of $\text{C}_2\text{O}_4^{2-} = \text{equivalents of } \text{MnO}_4^-$

$$x (\text{mole}) \times 2 = 1 \times 5$$

$$x = \frac{5}{2}$$

Ex.13 What weight of Na_2CO_3 of 85% purity would be required to react completely with 45.6 mL of 0.235N H_2SO_4 ?

Sol. Meq. of $\text{Na}_2\text{CO}_3 = \text{Meq. of } \text{H}_2\text{SO}_4$

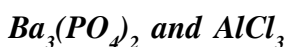
$$\text{or, } \frac{W_{\text{Na}_2\text{CO}_3}}{106/2} \times 1000 = 45.6 \times 0.235$$

$$\therefore W_{\text{Na}_2\text{CO}_3} = 0.5679 \text{ g}$$

For 85 g of pure Na_2CO_3 , weighed sample = 100 g

$$\therefore \text{For } 0.5679 \text{ g of pure } \text{Na}_2\text{CO}_3, \text{ weighed sample} = \frac{100}{85} \times 0.5679 = 0.6681 \text{ g}$$

Ex.14 Calculate the molar ratio in which the following two substances would react ?

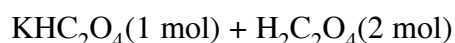


Sol. $n_{\text{eq}} \text{Ba}_3(\text{PO}_4)_2 = n_{\text{eq}} \text{AlCl}_3$

$$\text{or } n_1 \times 6 = n_2 \times 3$$

$$\therefore n_1 : n_2 = 1 : 2$$

Ex.15 Calculate volume of 0.4 M KMnO_4 required to react with following in acidic medium

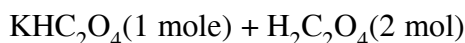


Sol. $n_{\text{eq}} \text{KMnO}_4 = n_{\text{eq}} \text{KHC}_2\text{O}_4 + n_{\text{eq}} \text{H}_2\text{C}_2\text{O}_4$

$$\text{or } (V \times 0.4) \times 5 = 1 \times 2 + 2 \times 2$$

$$\therefore V = 3.0 \text{ L}$$

Ex.16. Calculate volume of 0.4 M NaOH required to react with following mixture.



Sol. $n_{\text{eq}} \text{NaOH} = n_{\text{eq}} \text{KHC}_2\text{O}_4 + n_{\text{eq}} \text{H}_2\text{C}_2\text{O}_4$

or, $(V \times 0.4) \times 1 = 1 \times 1 + 2 \times 2$

$\therefore V = 12.5 \text{ L}$

11. TITRATION OR TITRIMETRIC (VOLUMETRIC) ANALYSIS :

Volumetric analysis involves determination of the volume of a solution of accurately known concentration, which is required to react quantitatively with the measured volume of the solution of a substance, concentration of which is to be determined. The solution of accurately known concentration is called **standard solution**. The reagent of known concentration is called **titrant** and the substance being titrated is called **titrand**.

To carry out volumetric analysis, standard solution is usually added from the long graduated tube called **burette**.

The process of adding the standard solution to the solution of unknown concentration until the reactions in just complete is called **titration**.

11.1 REQUIREMENT FOR THE REACTION IN THE VOLUMETRIC ANALYSIS -

- (i) The substance of which amount is to be determined by volumetric analysis must react completely and rapidly with the other reagent in stoichiometric proportion.
- (ii) The reaction should be fast and there must be alternation in physical or chemical property of the solution at the equivalence point, which can be detected by an indicator, or by measuring the potential difference or current, etc.

11.2 STANDARD SOLUTION :

A solution of exactly known concentration is called standard solution. Any substance, which is stable at room temperature and does not react with solvent in which it is dissolved, can be directly weighed to prepare its standard solution. Description and preparation of the solutions is given below :

11.2.1 PRIMARY STANDARDS :

A primary standard is a compound of sufficient purity in which total amount of impurities does not exceed 0.01 - 0.02 %. The standard solution can be prepared by direct weighing of a sample of primary standard followed by its dissolution in water (or solvent) to obtain a definite volume of solution. The substance to be used as a primary standard should also satisfy the following requirements:

- (i) It must be easily available in pure and dry form.
- (ii) It should not undergo change in air. i.e. it should not be hygroscopic, oxidised by air or affected by gases such as carbon dioxide present in the atmosphere or lose water of crystallization, so that it can be stored safely.
- (iii) It should be easy to detect the impurities present in it.
- (iv) It should have high relative molecular mass so that weighing errors are negligible.
- (v) Its reaction with another substance should be instantaneous and stoichiometric.
- (vi) The substance should be readily soluble in water.

It is difficult to obtain an ideal primary standard. Therefore, substances having characteristics nearest to the primary standards are usually employed.

Unstable hydrated salts, as a rule, should not be used as primary standards. However, sodium carbonate, sodium tetraborate, potassium hydrogenphthalate, oxalic acid, ferrous ammonium sulphate etc. can be used as primary standards because of their sufficient stabilities.

(a) **Primary standard acids :** Oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$), benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$), potassium hydrogenphthalate ($\text{KC}_8\text{H}_5\text{O}_4$), etc.

(b) **Primary standard bases :** Washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$), borax ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$), etc.

(c) **Primary standard oxidising agent :** $\text{K}_2\text{Cr}_2\text{O}_7$.

* KMnO_4 is not primary standard oxidising agent because it undergoes oxidation in water.



(d) **Primary standard reducing agent :** Mohr's salt ($\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$), Sodium oxalate ($\text{Na}_2\text{C}_2\text{O}_4$), etc.

11.2.2 SECONDARY STANDARDS :

A solution of secondary standard is the one which may be used for standardization after finding out its exact concentration by titration against a standard solution of primary standard.

A secondary standard cannot be used for preparing standard solution by direct weighing. Sodium hydroxide and potassium permanganate are examples of secondary standards.

11.3 EQUIVALENCE POINT :

The point at which reaction is completed in titration is called equivalence point or the theoretical or stoichiometric end point.

11.4 END POINT :

The point at which reaction of titration is stopped is called end point. The end point is detected either by some physical change produced in the reaction mixture itself or by addition of an auxiliary reagent, known as indicator; alternatively some other physical measurements may be used. At the completion of the reaction, the indicator shows a visible change (i.e., colour change or turbidity) in the solution being titrated.

In an ideal titration, the visible end point coincides with the equivalence point.

The role of indicator in colour change will be discussed in detail in the chapter, ionic equilibrium.

11.5 Types of titration :

(a) ACID - BASE TITRATION :

(i) **Acidimetry :** Standard solution of acid is used to determine concentration of solution of base.

(ii) **Alkalimetry :** standard basic solution is used to determine the concentration of solution of acid.

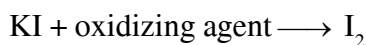
(b) REDOX TITRATION :

(i) **Dichromometry :** Standard $\text{K}_2\text{Cr}_2\text{O}_7$ solution is used to determine the concentration of solution of reducing agent.

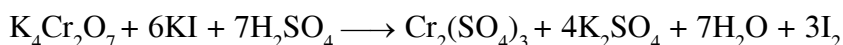
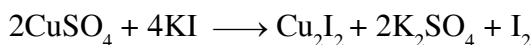
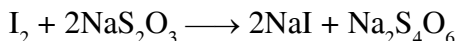
(ii) **Permanganometry :** standard KMnO_4 solution is used to determine the concentration of solution of reducing agent.

(iii) Iodometry :

In iodometric titrations, an oxidizing agent is allowed to react in neutral medium or in acidic medium with excess of potassium iodide to liberate free iodine.



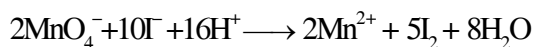
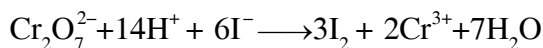
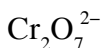
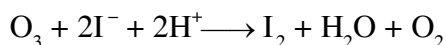
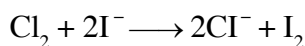
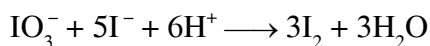
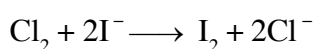
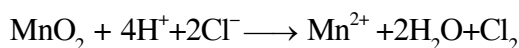
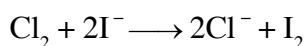
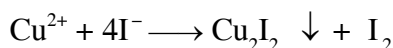
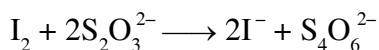
Free iodine is titrated against a standard reducing agent usually with sodium thiosulphate. Halogen, dichromates, cupric ion, peroxides etc., can be estimated by this method.



□ SOME IODOMETRIC TITRATIONS (TITRATING SOLUTIONS IS $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$)

Estimation of

Reaction



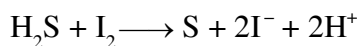
(iv) Iodimetry :

Standard I_2 solution is used to determine the concentration of solution of reducing agent.

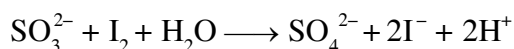
□ SOME IODIMETRIC TITRATION (TITRATING SOLUTIONS IS I_2 IN KI)

Estimation of

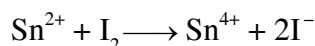
Reaction



(in acidic medium)



(in acidic medium)



(in acidic medium)

In iodimetric and iodometric titration, starch solution is used as an indicator. Starch solution gives blue or violet colour with free iodine. At the end point, the blue or violet colour disappears when iodine is completely changed to iodide.

11.6 Calculation of concentration :

Let there be a solution of substance A of unknown concentration. We are given solution of another substance B whose concentration is known (N_1). We take a certain known volume of A in a flask (V_2) and then we add B to A slowly till all the A is consumed by B (this can be known with the help of indicators). Let us, assume that the volume of B consumed is V_1 . According to the law of equivalence, the number of g equivalents of B at the end point.

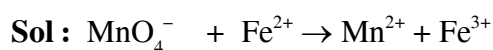
$$\therefore N_1 V_1 = N_2 V_2,$$

where N_2 is the conc. of A. From this we can calculate the value of N_2 .

11.7 BACK TITRATION

Back titration is a titration done in reverse ; instead of titrating the original sample, a known excess of standard reagent is added to the solution, and the excess is titrated. A back titration is useful if the endpoint of the reverse titration is easier to identify than the endpoint of the normal titration, as with the precipitation reactions.

Ex.17 How many milliliters of 0.02000 M KMnO_4 solution would be required to exactly titrate 25.00 mL of 0.2000 M $\text{Fe}(\text{NO}_3)_2$ solution.



$$\text{n-factor} = 5 \quad \text{n-factor} = 1$$

$$\text{Meq of } \text{KMnO}_4 = \text{Meq. of } \text{Fe}(\text{NO}_3)_2$$

$$\text{or, } V \times (0.02 \times 5) = 25 \times (0.2 \times 1)$$

$$\therefore V = 50 \text{ ml}$$

Ex.18 3.55 g sample of bleaching powder suspended in H_2O was treated with enough acetic acid and KI solution. Iodine thus liberated required 80 mL of 0.2 M hypo for titration. Calculate the % of available chlorine.



$$\text{or, } \frac{w}{35.5} = \left(\frac{80 \times 0.2}{1000} \right) \times 1 \quad [\text{hypo is } \text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}]$$

$$\therefore w = 16 \times 35.5 \times 10^{-3} \text{ gm}$$

$$\text{Hence, \% of available chlorine} = \frac{16 \times 35.5 \times 10^{-3}}{3.55} \times 100 = 16\%$$

Ex.19 50 gm of a sample of Ca(OH)_2 is dissolved in 50 ml of 0.5 N HCl solution. The excess of HCl was titrated with 0.3N – NaOH. The volume of NaOH used was 20cc. Calculate % purity of Ca(OH)_2

Sol. $n_{\text{eq.}} \text{Ca(OH)}_2 + n_{\text{eq.}} \text{NaOH} = n_{\text{eq.}} \text{HCl}$

$$\text{or, } \left(\frac{w}{74} \right) \times 2 + \left(\frac{20 \times 0.3}{1000} \right) = \frac{50 \times 0.5}{1000}$$

$\therefore$ wt. of Ca(OH)_2 taken, $w = 0.703$ gm

$$\text{Hence, \% purity} = \frac{0.703}{50} \times 100 = 1.406 \%$$

Ex.20 To 50 L of 0.2 N NaOH, 2.5 L of 2N HCl and 15 L of 0.1 N FeCl_3 solutions are added. What weight of Fe_2O_3 can be obtained from the precipitate? Also report the normality of NaOH left in resultant solution :

Sol. $n_{\text{eq.}} \text{NaOH taken} = 50 \times 0.2 = 10$

$n_{\text{eq.}} \text{HCl taken} = 2.5 \times 2 = 5$

$n_{\text{eq.}} \text{FeCl}_3 \text{ taken} = 15 \times 0.1 = 1.5$

NaOH will react with both, HCl and FeCl_3 , and from the amounts taken, it is clear that NaOH is in excess.

Now, $n_{\text{eq.}} \text{FeCl}_3 = n_{\text{eq.}} \text{Fe}_2\text{O}_3 \text{ formed}$

$$\text{or, } 1.5 = \left(\frac{W}{160} \right) \times 6$$

$\therefore$ wt of Fe_2O_3 formed, $w = 40$ gm.

Now, $n_{\text{eq.}} \text{NaOH remained} = 10 - (5 + 1.5) = 3.5$ and the final volume of solution

$$= 50 + 2.5 + 15 = 67.5 \text{ L}$$

$$\therefore \text{Normality of NaOH left} = \frac{3.5}{67.5} = 0.052 \text{ N}$$

Ex.21 1.20 g sample of Na_2CO_3 and K_2CO_3 was dissolved in water to form 100 mL of a Solution. 20 mL of this solution required 40 mL of 0.1 N HCl for complete neutralization. Calculate the weight of Na_2CO_3 in the mixture. If another 20 mL of this solution is treated with excess of BaCl_2 what will be the weight of the precipitate ?

Sol. Let, weight of $\text{Na}_2\text{CO}_3 = x$ g

Weight of $\text{K}_2\text{CO}_3 = y$ g

$$\therefore x + y = 1.20 \text{ g} \quad \dots\dots\dots(i)$$

For neutralization reaction of 100 mL

Meq. of $\text{Na}_2\text{CO}_3 + \text{Meq. of } \text{K}_2\text{CO}_3 = \text{Meq. of HCl}$

$$\Rightarrow \frac{x}{106} \times 2 \times 1000 + \frac{y}{138} \times 2 \times 1000 = \frac{40 \times 0.1 \times 100}{20}$$

$$\therefore 69x + 53y = 73.14 \quad \dots\dots\dots(ii)$$

From Eqs. (i) and (ii), we get

$$x = 0.5962 \text{ g}$$

$$y = 0.604 \text{ g}$$

Solution of Na_2CO_3 and K_2CO_3 gives ppt. of BaCO_3 with BaCl_2

(Meq. of $\text{Na}_2\text{CO}_3 + \text{Meq. of } \text{K}_2\text{CO}_3$) in 20 mL = Meq. of BaCO_3

$\Rightarrow$ Meq. of HCl for 20 mL mixture = Meq. of BaCO_3

$\Rightarrow$ Meq. of $\text{BaCO}_3 = 40 \times 0.1 = 4$

$$\frac{W_{\text{BaCO}_3}}{M_{\text{BaCO}_3}} \times 1000 = 40 \times 0.1 = 4$$

$$\frac{W_{\text{BaCO}_3}}{197} \times 2 \times 1000 = 4$$

$$\therefore W_{\text{BaCO}_3} = 0.394 \text{ g}$$

Ex.22 One gm of impure sodium carbonate is dissolved in water and the solution is made up to 250 mL. To 50 mL of this made up solution, 50 mL of 0.1N – HCl is added and the mix after shaking well required 10 mL of 0.16 N – NaOH solution for complete titration. Calculate the % purity of the sample.

Sol. $n_{\text{eq.}} \text{Na}_2\text{CO}_3$ (in diluted solution) + $n_{\text{eq.}} \text{NaOH.} = n_{\text{eq.}} \text{HCl}$

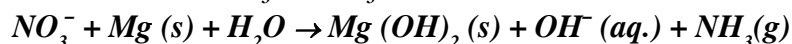
$$\text{or } \left(\frac{50 \times M}{1000} \right) \times 2 + \frac{10 \times 0.16}{1000} = \frac{50 \times 0.1}{1000}$$

$\therefore$ Molarity of diluted solution, $M = 0.034$

$$\text{Hence, moles of } \text{Na}_2\text{CO}_3 \text{ taken} = \frac{250 \times 0.034}{1000} = 8.5 \times 10^{-3}$$

$$\therefore \% \text{ purity of sample} = \frac{8.5 \times 10^{-3} \times 106}{1} \times 100 = 90.1\%$$

Ex.23 Mg can reduce NO_3^- to NH_3 in basic medium.



A 25.0 mL sample of NO_3^- solution was treated with Mg. The NH_3 (g) was passed into 100 mL of 0.15 N HCl. The excess of HCl required 32.10 mL of 0.10 N NaOH for neutralization. What was the molarity of NO_3^- ions in the original sample ?

Sol. From conservation of N-atoms,

$$\text{moles of } \text{NO}_3^- = \text{moles of } \text{NH}_3 = n$$

$$\text{Now, } n_{\text{eq. HCl}} = n_{\text{eq. NH}_3} + n_{\text{eq. NaOH}}$$

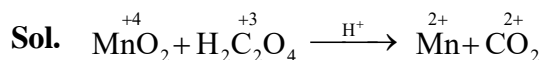
$$\text{or, } \frac{100 \times 0.15}{1000} = n \times 1 + \frac{32.10 \times 0.10}{1000}$$

$$\therefore \text{ moles of } \text{NO}_3^-, n = 11.79 \times 10^{-3}$$

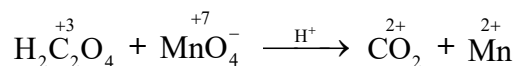
$$\text{Now, } \frac{25 \times M_{\text{NO}_3^-}}{1000} = 11.79 \times 10^{-3}$$

$$\therefore \text{ molarity of original } \text{NO}_3^- \text{ solution, } M_{\text{NO}_3^-} = 0.47\text{M}$$

Ex.24 3.2 g of pyrolusite (MnO_2) was treated with 50 mL of 0.5 M oxalic acid and some sulphuric acid. The oxalic acid left undecomposed was raised to 250 mL in a flask. 25 mL of this solution when treated with 0.02 M KMnO_4 required 32 mL of the solution : Find the % of MnO_2 in the sample.



$$(\text{n-factor} = 2) \quad (\text{n-factor} = 2)$$



$$(\text{n-factor} = 2) \quad (\text{n-factor} = 5)$$

$\text{H}_2\text{C}_2\text{O}_4$ reacts with MnO_2 and the remaining $\text{H}_2\text{C}_2\text{O}_4$ reacts with KMnO_4 .

$$\text{Moles of } \text{H}_2\text{C}_2\text{O}_4 \text{ taken} = \frac{50 \times 0.5}{1000} = 25 \times 10^{-3}$$

$$\text{Now, } n_{\text{eq. H}_2\text{C}_2\text{O}_4} = n_{\text{eq. KMnO}_4}$$

$$\text{or, } n \times 2 = \left(\frac{32 \times 0.02}{1000} \right) \times 5$$

$$\therefore \text{ moles of } \text{H}_2\text{C}_2\text{O}_4 \text{ remained in 25 ml solution, } n = 1.6 \times 10^{-3}$$

$$\text{Hence, moles of } \text{H}_2\text{C}_2\text{O}_4 \text{ with } \text{MnO}_2 = 25 \times 10^{-3} - 1.6 \times 10^{-3} \times \frac{250}{25} = 9 \times 10^{-3}$$

$$\text{Now, } n_{\text{eq. MnO}_2} = n_{\text{eq. H}_2\text{C}_2\text{O}_4}$$

$$\text{or, } \left(\frac{w}{87} \right) \times 2 = 9 \times 10^{-3} \times 2$$

$$\text{or, } w = 9 \times 87 \times 10^{-3} \text{ gm}$$

$$\therefore \% \text{ of pure } \text{MnO}_2 \text{ in pyrolusite} = \frac{9 \times 87 \times 10^{-3}}{3.2} \times 100 = 24.47 \%$$

Ex.25 An aqueous solution containing 0.10 g KIO_3 (formula wt. 214.0) was treated with an excess of KI solution. The solution was acidified with HCl . The liberated I_2 consumed 45.0 mL of thiosulphate solution to decolourise the blue starch – iodine complex. Calculate the molarity of the sodium thiosulphate solution. [JEE 1998]



$$\frac{0.1}{214} \text{ mole} \qquad 3 \times \frac{0.1}{214} \text{ mole}$$



$$\frac{0.3}{214} \text{ mole} \quad 2 \times \frac{0.3}{214} \text{ mole} = \frac{45 \times M}{1000}$$

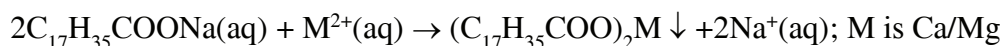
$$\therefore \text{molarity of } Na_2S_2O_3 \text{ solution} = 0.062 \text{ M}$$

12. HARDNESS OF WATER :

Presence of calcium and magnesium salts in water makes water, hard. **Hard water** does not give lather with soap.

Water free from soluble salts of calcium and magnesium is called **Soft water**. It gives lather with soap easily.

Hard water forms scum/precipitate with soap. Soap containing sodium stearate ($C_{17}H_{35}COONa$) reacts with hard water to precipitate out Ca/Mg stearate.



Hardness of water is of two types

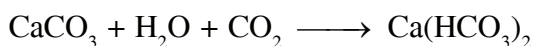
(a) Temporary Hardness

(b) Permanent Hardness

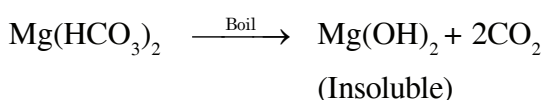
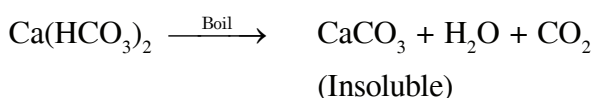
12.1 TEMPORARY HARDNESS :-

This is due to presence of bicarbonate of calcium and magnesium. Rain water dissolves a small quantities of CO_2 from the atmosphere forming a very dilute solution of carbonic acid.

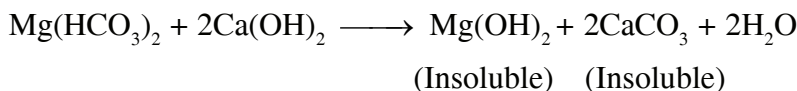
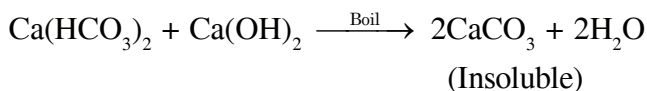
This water attacks Ca and Mg carbonate in any rock



Temporary hardness in water is easily removed by boiling, as the bicarbonates decomposes readily and the insoluble carbonates are precipitated.



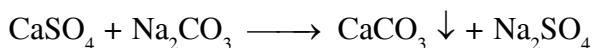
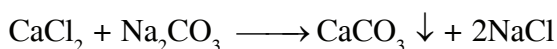
Temporarily hardness can also be removed by **Clark's process** which involves the addition of slaked lime Ca(OH)_2



12.2 PERMANENT HARDNESS :

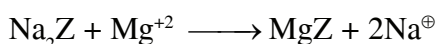
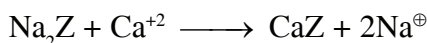
Permanent hardness is due to presence of sulphates and chlorides of both calcium and magnesium. This type of hardness cannot be removed by boiling or by Ca(OH)_2 ; Substance used to remove the hardness of water are known as water softener. This various water softeners are -

- (i) **Washing soda :-** It removes both the temporary and permanent hardness by converting soluble Ca and Mg compounds into insoluble carbonates.

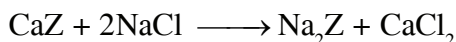


- (ii) **Permutit:** Permutit is technical name given to certain hydrated silicates of aluminium and sodium. or sodium aluminium orthosilicate or sodium zeolite. This method is also called ion - exchange method or zeolite method.

Example : $\text{Na}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot x\text{H}_2\text{O}$ (symbolised as Na_2Z)

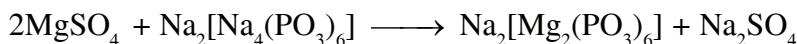
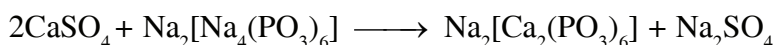


These ions can be re-exchanged by treating it with brine (NaCl) solution.



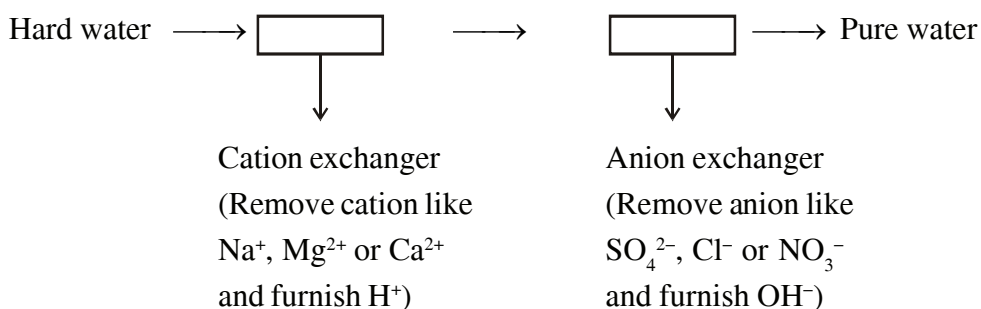
This method is useful for the removal of both temporary and permanent hardness of water.

- (iii) **Calgon :** The complex salt of metaphosphoric acid, sodium hexametaphosphate ($\text{Na}_4(\text{PO}_3)_6$) is called Calgon. It is represented as $\text{Na}_2[\text{Na}_4(\text{PO}_3)_6]$.



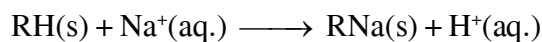
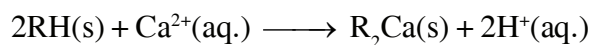
The complex formed keep Ca and Mg ions in water but unreactive towards soaps.

- (iv) **Ion exchange Resins:-** Ion exchange resins are the most popular water softener these days. This resins are synthetic substance. The cation exchanger consists of granular insoluble organic acid resins having giant molecules with $-\text{SO}_3\text{H}$ or $-\text{COOH}$ groups while the anion exchanges contains giant organic molecules with basic groups derived from amine. Ion exchange resins removes all soluble mineral from water.

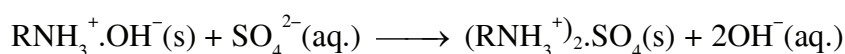
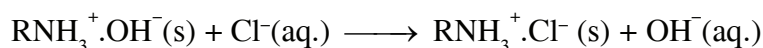
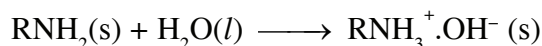


The water coming from cation exchanger is acidic due to H^+ ions. This water is then passed through another bed containing anion exchanger. This exchanger removes anion like Cl^- , SO_4^{2-} , NO_3^- by exchanging with OH^- ions.

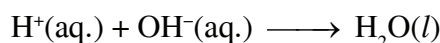
Reaction at Cation exchanger :



Reaction at Anion exchanger :



OH^- produced neutralises H^+ ions in water from cation exchange.



The exhausted cation and anion exchange resin beds are regenerated by treatment with dilute acid and alkali solutions, respectively.

12.3 DEGREE OF HARDNESS :

Degree of Hardness of water is measured in terms of ppm of $CaCO_3$ which is defined as number of parts of $CaCO_3$ by mass present in one million parts by mass of water, equivalent to the of various calcium & magnesium salts.

For example, consider the sample of hard water which is found to contain 36 mg of $MgSO_4$ per kg of water.

gm eq. of $CaCO_3$ = gm eq. of $MgSO_4$

$$\frac{W_{CaCO_3}}{100} \times 2 = \frac{36 \times 10^{-3}}{120} \times 2$$

$$W_{CaCO_3} = 3 \times 10^{-2} \text{ gm}$$

$$\text{degree of hardness} = \frac{3 \times 10^{-2}}{1000} \times 10^6 = 30 \text{ ppm } CaCO_3$$

Ex.26 50 L sample of tap water contains 2.40 gm $MgSO_4$ and 1.17 gm $NaCl$. Calculate the hardness of water.

Sol. Hardness is due to $MgSO_4$ only

$$n_{eq.} CaCO_3 = n_{eq.} MgSO_4$$

$$\text{or, } \left(\frac{w}{100} \right) \times 2 = \left(\frac{2.40}{120} \right) \times 2$$

$$\therefore w_{CaCO_3} = 2 \text{ gm}$$

$$\text{Hence, hardness} = \frac{2}{50 \times 10^3} \times 10^6 = 40 \text{ ppm } CaCO_3$$

Ex.27 200 ml of a water sample contain 8.2 mg $\text{Ca}(\text{NO}_3)_2$, 5.7 mg MgCl_2 , 6.48 mg $\text{Ca}(\text{HCO}_3)_2$ and 2.13 mg Na_2SO_4 . Calculate the temporary, permanent and total hardness of water sample.

Sol. Temporary hardness due to $\text{Ca}(\text{HCO}_3)_2$ only

$$n_{\text{eq.}} \text{CaCO}_3 = n_{\text{eq.}} \text{Ca}(\text{HCO}_3)_2$$

$$\text{or, } \left(\frac{w}{100} \right) \times 2 = \left(\frac{6.48}{162} \right) \times 2$$

$$\therefore w_{\text{CaCO}_3} = 4 \text{ mg}$$

$$\text{Hence, temporary hardness} = \frac{4 \times 10^{-3}}{200} \times 10^6 = 20 \text{ ppm CaCO}_3$$

Permanent hardness is due to $\text{Ca}(\text{NO}_3)_2$ and MgCl_2 only

$$n_{\text{eq.}} \text{CaCO}_3 = n_{\text{eq.}} \text{Ca}(\text{NO}_3)_2 + n_{\text{eq.}} \text{MgCl}_2$$

$$\text{or, } \left(\frac{w}{100} \right) \times 2 = \left(\frac{8.2}{164} \right) \times 2 + \left(\frac{5.7}{95} \right) \times 2$$

$$\therefore w_{\text{CaCO}_3} = 11 \text{ mg}$$

$$\text{Hence, permanent hardness} = \frac{11 \times 10^{-3}}{200} \times 10^6 = 55 \text{ ppm CaCO}_3$$

and, the total hardness = 20 + 55 = 75 ppm. CaCO_3

Ex.28 50 ml of tap water exactly require 20 ml (N/50) HCl solution for titration of all the bicarbonates present in water. Calculate the temporary hardness in water.

Sol. $n_{\text{eq.}} \text{HCl} = n_{\text{eq.}} \text{HCO}_3^- = n_{\text{eq.}} \text{CaCO}_3$

$$\text{or, } \frac{20 \times \frac{1}{50}}{1000} = \left(\frac{w}{100} \right) \times 2$$

$$\therefore w_{\text{CaCO}_3} = 0.02 \text{ mg}$$

$$\therefore \text{Temporary hardness} = \frac{0.02}{50} \times 10^6 = 400 \text{ ppm CaCO}_3$$

Ex.29 100 ml sample of hard water containing only Ca^{2+} hardness is passed through a column of cation exchange resin ($\text{H}^+ - \text{resin}$). The water coming out the column require 20 ml of 0.02M NaOH for its titration. What is hardness of water as ppm of Ca^{2+}

Sol. $n_{\text{eq. NaOH}} = n_{\text{eq. H}^+} = n_{\text{eq. Ca}^{2+}}$

$$\text{or, } \left(\frac{20 \times 0.02}{1000} \right) \times 1 = \left(\frac{w}{40} \right) \times 2$$

$$\therefore w_{\text{Ca}^{2+}} = 8 \times 10^{-3} \text{ gm}$$

$$\text{Hence, hardness} = \frac{8 \times 10^{-3}}{100} \times 10^6 = 80 \text{ ppm Ca}^{2+}$$

Ex.30 When 50 gm hard water containing 40 ppm Ca^{2+} ions and 54 ppm Al^{3+} ions is passed through cation exchange resin (acidic resin), the final water sample requires 40 ml NaOH solution for titration. Calculate the normality of NaOH sample used.

Sol. Mass of Ca^{2+} ions in hard water = $\frac{40}{10^6} \times 50 = 2 \times 10^{-3} \text{ gm}$

$$\text{and mass of Al}^{3+} \text{ ions in hard water} = \frac{54}{10^6} \times 50 = 2.7 \times 10^{-3} \text{ gm}$$

$$\text{Hence, moles of H}^+ \text{ produced} = \left(\frac{2 \times 10^{-3}}{40} \right) \times 2 + \left(\frac{2.7 \times 10^{-3}}{27} \right) \times 3 = 4 \times 10^{-4}$$

$$\text{Now, } n_{\text{eq. H}^+} = n_{\text{eq. OH}^-}$$

$$\text{or, } 4 \times 10^{-4} \times 1 = \frac{40 \times N_{\text{OH}^-}}{1000}$$

$$\therefore \text{Normality of NaOH solution} = 0.01 \text{ N}$$

SOME MORE SOLVED EXAMPLES

Ex.31 A 3.00 g sample containing Fe_3O_4 , Fe_2O_3 and an inert impure substance, is treated with excess of KI solution in presence of dilute H_2SO_4 . The entire iron is converted into Fe^{2+} along with the liberation of iodine. The resulting solution is diluted to 100 mL. A 20 mL of the diluted solution require 11 mL of 0.5 M $\text{Na}_2\text{S}_2\text{O}_3$ solution to reduce the iodine present. A 50 mL of diluted solution after complete extraction of the iodine requires 12.80 mL of 0.25 KMnO_4 solution in dilute H_2SO_4 medium for the oxidation of Fe^{2+} . Calculate the percentages of Fe_2O_3 and Fe_3O_4 in the original sample.

Sol. Let number of moles of Fe_3O_4 is 'a' and no. of moles of Fe_2O_3 is 'b' in 3 gram sample.

Now, 'a' moles Fe_3O_4 means (a moles Fe_2O_3 + a mole of FeO)

So, in sample (a + b) moles Fe_2O_3 and 'a' mole FeO present.



No. of gram equivalent of Fe_2O_3 = no. of gram equivalent of I_2

$$n_{\text{Fe}_2\text{O}_3} \times n_{\text{f}(\text{Fe}_2\text{O}_3)} = n_{\text{I}_2} \times n_{\text{f}(\text{I}_2)}$$

$$(a + b) \times 2 = n_{\text{I}_2} \times 2$$

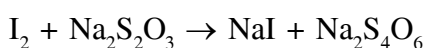
$$n_{\text{I}_2} = (a + b) \text{ moles.}$$

Now, solution is diluted to 100 ml

$$\text{So, } [\text{I}_2] = \frac{n}{V_{\text{litre}}} = \frac{(a + b)}{100/1000} = 10(a + b)$$

$$[\text{Fe}^{2+}] = \left(\frac{n(\text{Fe}^{2+})}{V_{\text{litre}}} \right) = \frac{(3a + 2b)}{100/1000} = 10(3a + 2b)$$

Now 20 ml of this solution titrated with 11 ml, 0.5 M $\text{Na}_2\text{S}_2\text{O}_3$



No. of gram equivalent of I_2 = no. of gram equivalent of $\text{Na}_2\text{S}_2\text{O}_3$

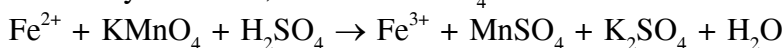
$$(N \times V)_{\text{I}_2} = (N \times V)_{\text{Na}_2\text{S}_2\text{O}_3}$$

$$(M \times n_f \times V)_{\text{I}_2} = (M \times n_f \times V)_{\text{Na}_2\text{S}_2\text{O}_3}$$

$$10(a + b) \times 2 \times 20 = 0.5 \times 1 \times 11$$

$$a + b = \frac{11}{800} \quad \dots\dots (i)$$

50 ml of solution after extraction of I_2 allowed to oxidation of Fe^{2+} by 12.80 ml, 0.25 M KMnO_4 .



no. of gram equivalent of Fe^{2+} = no. of gram equivalent of KMnO_4 .

$$(M \times n_f \times V)_{\text{Fe}^{2+}} = (M \times n_f \times V)_{\text{KMnO}_4}$$

$$10(3a + 2b) \times 1 \times 50 = 0.25 \times 5 \times 12.80$$

$$3a + 2b = \frac{16}{500} \quad \dots\dots (ii)$$

By (i) & (ii)

$$a = 0.0045 \text{ moles}$$

$$b = 0.00925 \text{ moles}$$

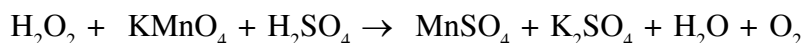
$$\text{so } \% \text{Fe}_2\text{O}_3 = \frac{\text{Wt}(\text{Fe}_2\text{O}_3)}{\text{Total wt.}} \times 100 = \frac{b \times 160}{3} \times 100 = \frac{0.00925 \times 160}{3} \times 100 = 49.33\%$$

$$\% \text{Fe}_2\text{O}_3 = \frac{\text{Wt}(\text{Fe}_2\text{O}_3)}{\text{Total wt.}} \times 100 = \frac{a \times 232}{3} \times 100 = \frac{0.0045 \times 232}{3} \times 100 = 34.8\%$$

Ex.32 Hydrogen peroxide solution (20 mL) reacts quantitatively with a solution of KMnO_4 (20 mL) acidified with dilute H_2SO_4 . The same volume of KMnO_4 solution is just decolorized by 10 mL of MnSO_4 in neutral medium simultaneously forming a dark brown precipitate of hydrated MnO_2 . The brown precipitate is dissolved in 10 mL of 0.2 M sodium oxalate under boiling condition in the presence of dilute H_2SO_4 . Write the equations involved in the reactions and calculate the molarity of H_2O_2 .

Sol. Let molarity of H_2O_2 solution is M_1

20 ml, M_1 molar H_2O_2 react with 20 ml KMnO_4 in H_2SO_4 solution

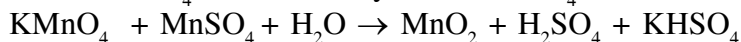


no. of gram equivalent of H_2O_2 = no. of gram equivalent of KMnO_4

$$(M \times n_f \times V)_{\text{H}_2\text{O}_2} = (M \times n_f \times V)_{\text{KMnO}_4}$$

$$M_1 \times 2 \times 20 = M_{\text{KMnO}_4} \times 5 \times 20 \quad \text{.....(i)}$$

20 ml KMnO_4 decolourise by 10 ml MnSO_4 in neutral medium.

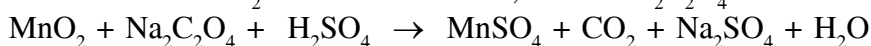


no. of gram equivalent of KMnO_4 = no. of gram equivalent of MnO_2

$$M_{\text{KMnO}_4} \times 3 \times 20 \text{ ml} = \text{no. of gram equivalent of } \text{MnO}_2$$

$$M_{\text{KMnO}_4} \times 3 \times 20 \times 10^{-3} = n_{\text{MnO}_2} \times 6/5 \quad \text{.....(ii)}$$

and obtained MnO_2 dissolved in 10 ml, 0.2 M $\text{Na}_2\text{C}_2\text{O}_4$



no. of gram equivalent of MnO_2 = no. of gram equivalent of $\text{Na}_2\text{C}_2\text{O}_4$

$$n_{\text{MnO}_2} \times 2 = 0.2 \times 2 \times 10 \times 10^{-3}$$

$$n_{\text{MnO}_2} = 2 \times 10^{-3} \text{ moles} \quad \text{.....(iii)}$$

From equation (ii)

$$M_{\text{KMnO}_4} \times 3 \times 20 \times 10^{-3} = 2 \times 10^{-3} \times \frac{6}{5}$$

$$M_{\text{KMnO}_4} = \frac{2}{50}$$

From equation (1)

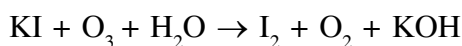
$$M_1 \times 2 \times 20 = \frac{2}{50} \times 5 \times 20$$

$$M_1 = 0.1 \text{ M}$$

Ex.33 One litre of a mixture of O_2 and O_3 ($O_3 \longrightarrow O_2 + O^{2-}$) at 1 atm and 273 K was allowed to react with an excess of acidified solution of KI. The iodine liberated required 40 ml of M/10 sodium thiosulphate solution for titration. What is the percent of ozone in the mixture ?

Sol. Let in 1 litre mixture x litre is O_2 and y litre is O_3 at NTP

excess KI reduce O_3 into O_2 and O^{2-}



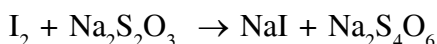
no. of gram equivalent of O_3 = no. of gram equivalent of I_2

$$(n \times n_f)_{O_3} = (n \times n_f)_{I_2}$$

$$\frac{y}{22.4} \times 2 = n_{I_2} \times 2$$

$$n_{I_2} = \frac{y}{22.4} \quad \dots\dots(1)$$

Now, released I_2 required 40 ml, $\frac{1}{10}$ M $Na_2S_2O_3$



No. of gram equivalent of I_2 = no. of gram equivalent of $Na_2S_2O_3$

$$n_{I_2} \times 2 = \frac{1}{10} \times 1 \times 40 \times 10^{-3}$$

$$n_{I_2} = 2 \times 10^{-3} \text{ moles}$$

Now from equation (1)

$$\frac{y}{22.4} = 2 \times 10^{-3}$$

$$y = (44.8 \times 10^{-3}) \text{ litre} = 0.0448 \text{ litre}$$

$$x = 1 - y = 1 - 0.0448 = 0.9552 \text{ litre}$$

$$\% \text{ (by wt)} = \frac{\text{wt}(O_3)}{\text{wt. (total)}} \times 100 = \frac{n(O_3) \times 48}{n(O_3) \times 48 + n(O_2) \times 32} \times 100$$

$$= \frac{\frac{y}{22.4} \times 48}{\frac{y}{22.4} \times 48 + \frac{x}{22.4} \times 32} \times 100 = \frac{3y}{3y + 2x} \times 100$$

$$= \frac{3 \times 0.0448}{3 \times 0.0448 + 2 \times 0.9552} \times 100 = 6.57 \%$$

Ex.34 A mixture of FeO and Fe₂O₃ is reacted with acidified KMnO₄ solution having a concentration of 2/5 M, 100 mL of which was used. The solution was then titrated with Zn dust which converted Fe³⁺ of the solution to Fe²⁺. The Fe²⁺ required 1000 mL of 2/15 M K₂Cr₂O₇ solution. Find the % by mol of FeO and Fe₂O₃.

Sol. Let no. of moles of FeO is x millimoles & Fe₂O₃ is y millimoles in mixture.



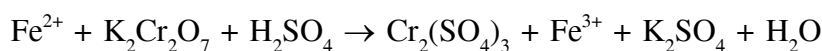
no. of milligram equivalent of FeO = no. of milli gram equivalent of KMnO₄.

$$x \times 1 = \frac{2}{5} \times 5 \times 100$$

$$x = 200 \text{ m moles.}$$

Zn - dust convert all Fe³⁺ to Fe²⁺

So no. of millimoles of Fe²⁺ in solution is = (x + 2y).



no. of milli gram equivalent of Fe²⁺ = no. of milligram equivalent of K₂Cr₂O₇

$$(x + 2y) \times 1 = \frac{2}{15} \times 6 \times 1000$$

$$(200 + 2y) = \frac{2}{15} \times 1000$$

$$y = 300 \text{ millimoles.}$$

$$\% \text{ FeO (by moles)} = \frac{x}{x+y} \times 100 = \frac{200}{500} \times 100 = 40\%$$

$$\% \text{ Fe}_2\text{O}_3 \text{ (by moles)} = 60\%$$

Ex.35 A substance of crude copper is boiled in H₂SO₄ till all the copper has reacted. The impurities are inert to the acid. The SO₂ liberated in the action is passed into 100 mL of 0.4 M acidified KMnO₄ (SO₂ → SO₄²⁻) The solution of KMnO₄ after passage of SO₂ is allowed to react with oxalic acid and requires 25 mL of 1 M oxalic acid. If the purity of copper is 95.25%, what was the weight of the sample.

Sol. let given sample is w gram and no. of moles of Cu in given sample is x millimoles.



x millimoles x millimoles



no. of milligram eq. of SO₂ = no. of milli gram eq. of KMnO₄

$$x \times 2 = \text{no. of gram eq. of KMnO}_4$$

left KMnO₄ reduced by H₂C₂O₄



no. of milligram eq. of KMnO_4 (left)

= no. of mill gram equ. of $\text{H}_2\text{C}_2\text{O}_4$

$$= 1 \times 2 \times 25$$

So, total milligram eq. of KMnO_4 = milligram eq. of SO_2 + milligram eq. of $\text{H}_2\text{C}_2\text{O}_4$

$$0.4 \times 5 \times 100 = x \times 2 + 1 \times 2 \times 25$$

$$x = 75 \text{ millimoles}$$

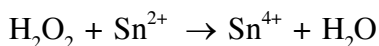
$$\text{wt}(\text{Cu}) = 95.25\% \text{ of } W$$

$$75 \times 10^{-3} \times 63.5 = \frac{95.25}{100} \times W$$

$$\Rightarrow W = 5 \text{ gram.}$$

Ex.36 H_2O_2 is reduced rapidly by Sn^{2+} , the products being Sn^{4+} and water. H_2O_2 decomposes slowly at room temperature to yield O_2 and water. Calculate the volume of O_2 produced at 273 K and 1.00 atm when 200 g of 10.0 % by mass H_2O_2 in water is treated with 88.2 mL of 1 M Sn^{2+} and then the mixture is allowed to stand until no further reaction occurs.

Sol. Let x millimoles of H_2O_2 react with 1M, 88.2 ml Sn^{2+}



no. of milligram equivalent of H_2O_2

= no. of milligram eq. of Sn^{2+}

$$x_{\text{millimole}} \times 2 = 1 \times 2 \times 88.2$$

$$x = 88.2 \text{ millimoles}$$

Total given moles of H_2O_2 is

$$n_{\text{total}} = \frac{W(\text{H}_2\text{O}_2)}{34} = \frac{10\% \text{ of } 200 \text{ gram}}{34}$$

$$= \frac{20}{34} \text{ moles}$$

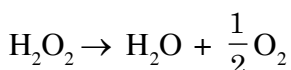
$$= 0.5882 \text{ moles}$$

$$= 588.2 \text{ millimoles.}$$

so. no. of moles of H_2O_2 that decomposed to O_2

$$= 588.2 - 88.2$$

$$= 500 \text{ millimoles}$$

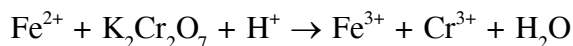


$$n(\text{O}_2) \text{ released} = \frac{1}{2} \times 500 = 250 \text{ millimoles}$$

$$V_{\text{O}_2} (\text{at } 1 \text{ atm, } 273 \text{ K}) = 250 \times 10^{-3} \times 22.4 = 5.6 \text{ litre.}$$

Ex.37 0.84 g iron ore containing x percent of iron was taken in a solution containing all the iron in ferrous condition. The solution required x ml of a dichromic solution for oxidizing the iron content to ferric state. Calculate the normality of dichromic solution.

Sol. Let Normality of $K_2Cr_2O_7$ solution is N



no. of gram eq. of Fe^{2+} = no. of gram eq. of $K_2Cr_2O_7$

$$n_{Fe^{2+}} \times 1 = N \times x \times 10^{-3}$$

$$\frac{W_{Fe^{2+}}}{56} \times 1 = N \times x \times 10^{-3}$$

$$\frac{\frac{x}{100} \times 0.84}{56} \times 1 = N \times x \times 10^{-3}$$

$$N = 0.15$$

Ex.38 5 g of pyrolusite (impure MnO_2) were heated with conc. HCl and Cl_2 evolved was passed through excess of KI solution. The iodine liberated required 40 mL of N/10 hypo solution. Find the % of MnO_2 in the pyrolusite.

Sol. Let moles of MnO_2 in 5 gram sample is x.



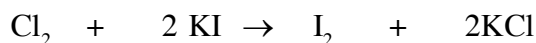
no. of gram eq. of MnO_2 = no. of gram equivalent of Cl_2

$$x \times 2 = n_{(Cl_2)} \times 2$$

$$x = n_{(Cl_2)}$$

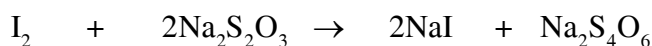
$$\text{so } n_{I_2} = n_{Cl_2} = x$$

evolved Cl_2 passes through KI that released I_2



x moles x moles

Evolved I_2 required 40 ml, $\frac{N}{10} Na_2S_2O_3$



no. of gram equivalent of I_2 = no. of gram equivalent of hypo

$$x \times 2 = \frac{1}{10} \times 40 \times 10^{-3}$$

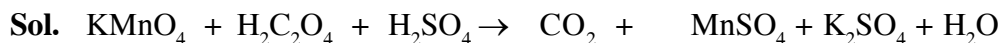
$$x = 2 \times 10^{-3} \text{ moles}$$

$$\% MnO_2 = \frac{\text{wt.}(MnO_2)}{\text{total wt.}} \times 100$$

$$= \frac{x \times 87}{5} \times 100 = \frac{2 \times 10^{-3} \times 87}{5} \times 100$$

$$= 3.48 \%$$

Ex.39 A 458 g sample containing Mn_3O_4 was dissolved and all manganese was converted to Mn^{2+} . In the presence of fluoride ion, Mn^{2+} is titrated with 3 lit of KMnO_4 solution (which was 1.25 N against oxalate in acidic medium), both reactants being converted to a complex of Mn(III), What was the % of Mn_3O_4 in the sample ?



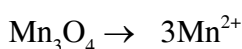
3 litre, M molar

$$\text{Normality (KMnO}_4) = \text{Molarity} \times n_f$$

$$1.25 = M \times 5$$

$$M = 0.25$$

Let x moles of Mn_3O_4 present in 458 gram sample



xmole 3xmoles



no. of gram eq. of Mn^{2+} = no. of gram eq. KMnO_4

$$3x \times 1 = (M \times n_f \times V)_{\text{KMnO}_4}$$

$$3x \times 1 = 0.25 \times 4 \times 3$$

$$x = 1 \text{ moles}$$

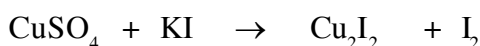
$$\% \text{ Mn}_3\text{O}_4 \text{ (by wt)} = \frac{w(\text{Mn}_3\text{O}_4)}{w_{\text{total}}} \times 100 = \frac{1 \times 229}{458} \times 100 = 50\%$$

Ex.40 80 gm of a sample of Anhydrous CuSO_4 was dissolved in water and made to 250 mL. 25 mL of this solution after taking usual precaution was treated with a little excess of KI solution. A white ppt. of Cu_2I_2 and iodine was evolved. The iodine so evolved required 40 mL of 1 M hypo solution. What is the approximate purity of CuSO_4 solution.

[Mol. wt. of $\text{CuSO}_4 = 160$]

Sol. Let 80 gram sample contains x moles of CuSO_4 that desolve in 250 ml solution

So 25 ml solution contain (x/10) moles CuSO_4

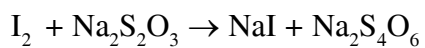


no. of gram eq. of CuSO_4 = no. of gram eq. of I_2

$$\frac{x}{10} \times 1 = n_{(\text{I}_2)} \times 2$$

$$n_{(\text{I}_2)} = \frac{x}{20}$$

Evolved I_2 required 40 ml, 1M hypo solution



no. of gram eq. of I_2 = no. of gram eq. of hypo

$$n(I_2) \times 2 = 1 \times 1 \times 40 \times 10^{-3}$$

$$\frac{x}{20} \times 2 = 40 \times 10^{-3}$$

$$x = 40 \times 10^{-2} \text{ moles}$$

$$\% \text{ CuSO}_4 \text{ (by mass)} = \frac{\text{wt}(\text{CuSO}_4)}{\text{wt total}} \times 100$$

$$= \frac{40 \times 10^{-2} \times 160}{80} \times 100 = 80\%$$

EXERCISE # S-I**OXIDATION NUMBER**

1. Calculate oxidation number of the underlined elements in the following compounds or ions

- | | | | |
|--|--|---|--|
| (a) $\text{H}_4\underline{\text{P}}_2\text{O}_7$ | (b) $\underline{\text{S}}_2\text{O}_3^{2-}$ | (c) $\text{Ca}\underline{\text{O}}_2$ | (d) $\underline{\text{Zn}}\text{O}_2^{2-}$ |
| (e) $\underline{\text{Cr}}\text{O}_2\text{Cl}_2$ | (f) $\text{Ca}(\underline{\text{Cl}}\text{O}_2)_2$ | (g) $\underline{\text{C}}_3\text{O}_2$ | (h) $\text{K}\underline{\text{I}}_3$ |
| (i) $\underline{\text{Fe}}_3\text{O}_4$ | (j) $\underline{\text{Br}}_3\text{O}_8$ | (k) $\text{CaO}\underline{\text{Cl}}_2$ | (l) $\underline{\text{O}}_2^+$ |
| (m) $\underline{\text{O}}_2^-$ | (n) $\underline{\text{C}}\text{N}^-$ | | |

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2. Calculate oxidation number of the underlined elements in the following compounds or ions

- | | |
|---|--|
| (a) $\text{K}[\underline{\text{Co}}(\text{C}_2\text{O}_4)_2 \cdot (\text{NH}_3)_2]$ | (b) $\text{K}\underline{\text{Al}}(\underline{\text{S}}\text{O}_4)_2 \cdot 12\text{H}_2\text{O}$ |
| (c) $[\underline{\text{Fe}}(\text{CN})_6]^{4-}$ | (d) $\underline{\text{Fe}}_{0.93}\text{O}$ |
| (e) $\text{H}_2\underline{\text{S}}_2\text{O}_8$ | (f) $\underline{\text{Cr}}\text{O}_5$ |
| (g) $\text{H}_2\underline{\text{S}}\text{O}_5$ | |

RR0180**OXIDATION - REDUCTION**

3. Classify each of the following unbalanced half-reactions as either an oxidation or a reduction :

RR0181

- | | |
|--|--|
| (a) $\text{O}_2(\text{g}) \longrightarrow \text{OH}^-(\text{aq})$ | (b) $\text{H}_2\text{O}_2(\text{aq}) \longrightarrow \text{O}_2(\text{g})$ |
| (c) $\text{MnO}_4^-(\text{aq}) \longrightarrow \text{MnO}_4^{2-}(\text{aq})$ | (d) $\text{CH}_3\text{OH}(\text{aq}) \longrightarrow \text{CH}_2\text{O}(\text{aq})$ |
| (e) $\text{MnO}_2(\text{s}) \longrightarrow \text{Mn}_3\text{O}_4(\text{s})$ | (f) $\text{FeO}(\text{s}) \longrightarrow \text{Fe}_3\text{O}_4(\text{s})$ |
| (g) $\text{CH}_3\text{COOH}(\text{aq.}) \longrightarrow \text{CH}_3\text{CH}_2\text{OH}(\text{aq.})$ | |

OXIDISING AND REDUCING AGENT

4. Identify the oxidant (oxidising agent) and reductant (reducing agent) in the following reactions

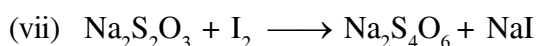
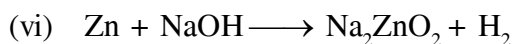
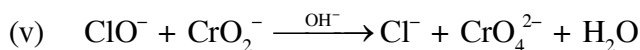
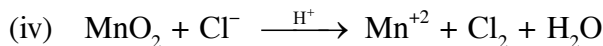
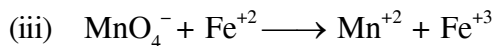
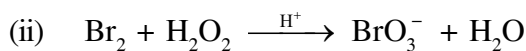
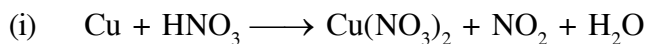
RR0182

- | |
|---|
| (a) $\text{Zn}(\text{s}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{ZnO}(\text{s})$ |
| (b) $\text{Zn}(\text{s}) + 2\text{H}^+(\text{aq}) \longrightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$ |
| (c) $10\text{H}^+(\text{aq}) + 4\text{Zn}(\text{s}) + \text{NO}_3^-(\text{aq}) \longrightarrow 4\text{Zn}^{2+}(\text{aq}) + \text{NH}_4^+(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$ |
| (d) $2\text{H}_2\text{S}(\text{aq}) + \text{SO}_2(\text{g}) \longrightarrow 3\text{S}(\text{s}) + 2\text{H}_2\text{O}(\text{l})$ |
| (e) $2\text{MnO}_4^-(\text{aq}) + 16\text{H}^+(\text{aq}) + 10\text{Cl}^-(\text{aq.}) \longrightarrow 2\text{Mn}^{2+}(\text{aq.}) + 8\text{H}_2\text{O}(\text{l}) + 5\text{Cl}_2(\text{g})$ |
| (f) $2\text{MnO}_4^-(\text{aq}) + 6\text{H}^+(\text{aq}) + 5\text{H}_2\text{O}_2(\text{aq.}) \longrightarrow 2\text{Mn}^{2+}(\text{aq.}) + 8\text{H}_2\text{O}(\text{l}) + 5\text{O}_2(\text{g})$ |

BALANCING OF REDOX

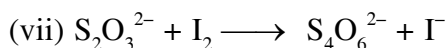
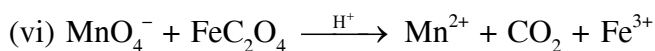
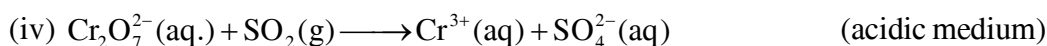
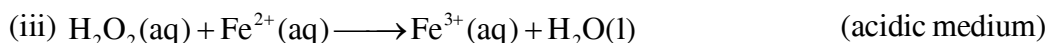
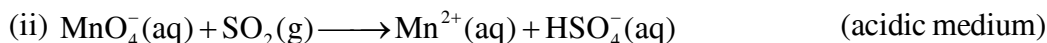
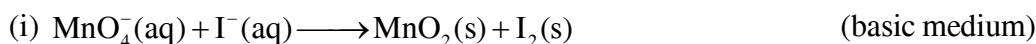
5. Balance the following redox reactions by oxidation number method :

RR0183

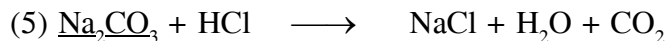
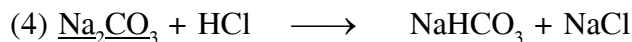
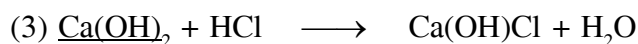
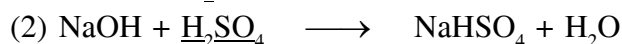
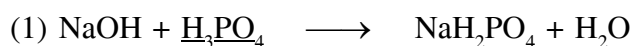


6. Balance the following redox reactions by half-reaction method (Ion electron method):

RR0184

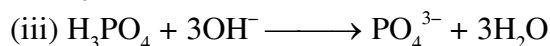
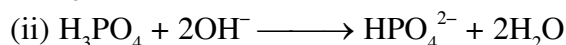
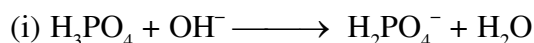
*n*-FACTOR & EQUIVALENT WEIGHT

7. Find the n-factor of underlined species in the following non redox reaction.



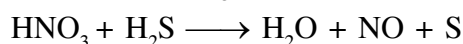
RR0014

8. What will be the equivalent weight of H_3PO_4 in each of the reaction.



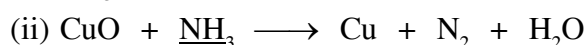
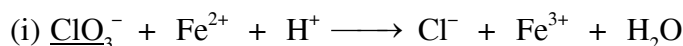
RR0015

9. What is the equivalent weight of HNO_3 in following reaction ?



RR0016

10. Find out the equivalent weight of the underlined species in the following reaction :



RR0017

11. Find out the n_{factor} of IO_3^- in the following disproportionation reaction.

$$\text{I}_2 \xrightarrow{\text{OH}^-} \text{IO}_3^- + \text{I}^-$$
 RR0019
12. What is the equivalent weight of the following salts ?
 (i) Na_2SO_4 (ii) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ (iii) AlCl_3 (iv) $\text{Al}_2(\text{SO}_4)_3$
 (v) $\text{Fe}_2(\text{SO}_4)_3 \cdot 9\text{H}_2\text{O}$ RR0185
13. What are the n-factor of underlined species in the following processes ?
 (i) $\underline{\text{Fe}_{0.93}\text{O}} \longrightarrow \text{Fe}_2\text{O}_3$ (ii) $\text{Fe}_{0.93}\text{O} \longrightarrow \underline{\text{Fe}_2\text{O}_3}$
 (iii) $\underline{\text{Fe}_2\text{O}_3} \longrightarrow \text{Fe}_3\text{O}_4$ (iv) $\text{Fe}_2\text{O}_3 \longrightarrow \underline{\text{Fe}_3\text{O}_4}$ RR0186
14. What are the n-factors of underlines species in the following process ?
 (i) $\underline{\text{FeC}_2\text{O}_4} \longrightarrow \text{Fe}_2\text{O}_3 + \text{CO}_2$ (ii) $\underline{\text{FeS}_2} \longrightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2$
 (iii) $\underline{\text{CrBr}_3} \longrightarrow \text{CrO}_4^{2-} + \text{BrO}_4^-$ (iv) $\underline{\text{Co}(\underline{\text{MnO}_4})_3} \longrightarrow \text{Co}^{2+} + \text{Mn}^{2+}$ RR0187
15. What are the equivalent weights of H_2SO_4 in the following processes/reactions ?
 (i) $\text{Mg} + \text{H}_2\text{SO}_4 \longrightarrow \text{MgSO}_4 + \text{SO}_2 + \text{H}_2\text{O}$
 (ii) $\text{NaOH} + \text{H}_2\text{SO}_4 \longrightarrow \text{NaHSO}_4 + \text{H}_2\text{O}$
 (iii) $\text{H}_2\text{SO}_4 \longrightarrow \text{H}_2\text{S}_2\text{O}_8$
 (iv) $\text{Pb} + \text{PbO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{PbSO}_4 + \text{H}_2\text{O}$ RR0188

ACID BASE TITRATION

16. How many millilitre of 0.5 M H_2SO_4 is needed to dissolve 4.94 g of copper II carbonate ?
 [Cu = 63.5] [JEE 1999]
RR0189
17. An aqueous solution of 6.3 g of oxalic acid dihydrate ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) is made upto 250 mL. The volume (in ml) of 0.1 N NaOH required to completely neutralise 10 mL of this solution is :
[JEE 2001]
RR0021
18. Calculate volume (in ml) of 1N H_3PO_4 required to react with 20 ml 2N $\text{Ca}(\text{OH})_2$ solution ?
RR0022
19. Calculate volume (in L) of 0.4 M NaOH required to react with following mixture
 $\text{HCl}(1 \text{ mol}) + \text{H}_2\text{SO}_4(2 \text{ mol})$ RR0024
20. How many litre of 0.1N HCl are required to react completely with 19 gm mixture of Na_2CO_3 and NaHCO_3 containing equimolar amount of these two ?
RR0026
21. H_3PO_4 is a tri basic acid and one of its salt is NaH_2PO_4 . What volume (in ml) of 1 M NaOH solution should be added to 12 g of NaH_2PO_4 to convert it into Na_3PO_4 ?
RR0027
22. A sample containing 4.2g of KOH and $\text{Ca}(\text{OH})_2$ is neutralized by an acid. It consumes 0.1 equivalent of acid. Calculate the mass percentage of KOH in the sample. (K = 39)
R00061
23. What volume (in ml) 0.4 M - HCl solution needed for complete reaction with 3.4 gm Al_2O_3 ?
RR0190

24. 20 ml of 2M - NaOH solution is diluted to 400 ml. 25 ml of the diluted solution exactly requires 50 ml of HNO_3 solution for titration. What is the molarity of HNO_3 solution ? **RR0191**
25. 85.8 gm washing soda, $\text{Na}_2\text{CO}_3 \cdot 10 \text{H}_2\text{O}$, is dissolved in sufficient water to get 600 ml solution. What volume (in ml) of this solution will neutralise completely, 40 ml of 0.2 M - H_2SO_4 solution ? **RR0192**

REDOX TITRATION

26. A 5.0 cm^3 solution of H_2O_2 liberates 0.508 g of iodine from an acidified KI solution. Calculate the strength of H_2O_2 solution in terms of volume strength at STP. ($I = 127$) **[JEE' 1995]**
RR0029
27. Calculate the normality of a solution containing 15.8 g of KMnO_4 in 50 mL acidic solution. ($K = 39$, $Mn = 55$) **RR0030**
28. 20 mL of H_2O_2 after acidification with dil H_2SO_4 required 30 mL of $\frac{N}{12}$ KMnO_4 for complete oxidation. The strength of H_2O_2 solution (in gm/litre) is [Molar mass of $\text{H}_2\text{O}_2 = 34$] **RR0031**
29. A volume of 10.0 ml of 1 M SeO_2 reacted with exactly 20 ml of 2 M CrSO_4 . In the reaction, Cr^{2+} was oxidized to Cr^{3+} . To what oxidation state was selenium converted by the reaction ? **RR0033**
30. Potassium acid oxalate $\text{K}_2\text{C}_2\text{O}_4 \cdot 3\text{H}_2\text{C}_2\text{O}_4 \cdot 4\text{H}_2\text{O}$ can be oxidized by MnO_4^- in acid medium. Calculate the volume (in ml) of 0.1 M KMnO_4 reacting in acid solution with 5.08 gm of the acid oxalate. ($K = 39$) **RR0034**
31. A 1 g sample of H_2O_2 solution containing x% H_2O_2 by mass requires x cm^3 of a KMnO_4 solution for complete oxidation under acidic condition. Calculate the normality of KMnO_4 solution. **RR0035**
32. Metallic tin in the presence of HCl is oxidized by $\text{K}_2\text{Cr}_2\text{O}_7$ to stannic chloride, SnCl_4 . What volume (in litre) of deci-normal dichromate solution would be reduced by 11.9 gm of tin [$\text{Sn} = 119$] **RR0036**
33. 520 gm mixture of Fe_2O_3 and FeO reacts completely with 158 gm KMnO_4 in acidic medium. Calculate the mole % of Fe_2O_3 in mixture. [$\text{Fe} = 56$, $K = 39$, $Mn = 55$] **RR0065**
34. Calculate the millimoles of Br_2 produced when 10 ml of 0.1 M BrO_3^- reacts with excess of Br^- . **RR0066**
35. What volume (in ml) of 0.5 M- $\text{Na}_2\text{S}_2\text{O}_3$ solution will react completely with 12.7 gm iodine ? ($I = 127$) **RR0193**

BACK TITRATION

36. 10 g CaCO_3 were dissolved in 250 ml of 1 M HCl. What volume (in ml) of 2 M KOH would be required to neutralise excess HCl ? **RR0041**
37. 20 ml of 2M H_2SO_4 solution is mixed with 30 ml of 3M - NaOH solution. Calculate the volume (in ml) of 4M-HCl solution needed to neutralise the resulting solution. **RR0194**
38. 50 ml of 0.02 M -acidified KMnO_4 solution is mixed with 40 ml of KI solution. If the resulting solution requires 20 ml of 0.1 M- $\text{K}_2\text{Cr}_2\text{O}_7$ solution for titration of excess KI, calculate the molarity of KI solution. **RR0195**
39. 10 ml of 0.8 M - HNO_3 solution is mixed with 20 ml of 0.2 M - NaOH solution and the mixture is diluted to 100 ml. If 25 ml of the diluted solution exactly require 8 ml of KOH solution for titration, what is the concentration (in gm/litre) of KOH solution used ? (K = 39) **RR0196**
40. 6.3 gm $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ is dissolved in sufficient water to get 500 ml solution. To 50 ml of this solution, 25 ml of (N/5) KMnO_4 solution is added in presence H_2SO_4 . Calculate the volume (in ml) of (N/6) $\text{K}_2\text{Cr}_2\text{O}_7$ solution needed for titration of excess oxalate. **RR0197**

HARDNESS OF WATER

41. One litre of a sample of hard water contains 11.1 mg of CaCl_2 & 9.5 mg of MgCl_2 . What is degree of hardness in terms of ppm of CaCO_3 **RR0045**
42. A sample of hard water contains 324 ppm of $\text{Ca}(\text{HCO}_3)_2$. How many gm of CaO is needed to remove all bicarbonates from 100 L hard water ? **RR0198**
43. The number of hardness creating salts among NaCl , K_2CO_3 , $\text{Mg}(\text{HCO}_3)_2$, $\text{Mg}(\text{NO}_3)_2$, Na_3PO_4 , $\text{Ca}(\text{HCO}_3)_2$, MgBr_2 , is **RR0199**
44. 50 L of a water sample contains 7.3 gm $\text{Mg}(\text{HCO}_3)_2$. Calculate temporary hardness of the water in terms of ppm of CaCO_3 . **RR0200**
45. A sample of tap water contains 300 ppm MgSO_4 , 648 ppm $\text{Ca}(\text{HCO}_3)_2$ and 585 ppm NaCl. Calculate permanent hardness of tap water in terms of ppm of CaCO_3 . **RR0201**

EXERCISE # S-II

- Find out the oxidation number of the underlined elements in the following compounds:
 (a) $\underline{\text{C}}\text{H}_3\underline{\text{C}}\text{H}_2\text{OH}$ (b) $\underline{\text{C}}\text{H}_3\underline{\text{C}}\text{OOH}$ (c) $\text{H}\underline{\text{Q}}\text{F}$ (d) $\text{C}_6\text{H}_5\underline{\text{C}}\text{HO}$
 (e) $\text{C}_6\text{H}_5\underline{\text{N}}\text{O}_2$ (f) $\text{C}_6\text{H}_5\underline{\text{N}}\text{H}_2$ (g) $(\underline{\text{C}}\text{N})_2$ (h) $\text{H}\underline{\text{C}}\text{N}$
 (i) $\text{H}\underline{\text{N}}\underline{\text{C}}$ (j) $\underline{\text{C}}\text{OCl}_2$ **RR0202**
- Balance the following redox reactions by half-reaction method
 (i) $\text{KMnO}_4(\text{aq.}) + \text{FeC}_2\text{O}_4(\text{aq.}) + \text{H}_2\text{SO}_4(\text{aq.}) \longrightarrow$
 $\text{MnSO}_4(\text{aq.}) + \text{Fe}_2(\text{SO}_4)_3(\text{aq.}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ (Acidic medium) **RR0054**
 (ii) $\text{HNO}_2(\text{aq.}) \rightarrow \text{NO}_3^- + \text{NO}(\text{g})$ (Acidic medium) **RR0057**
 (iii) $\text{IO}_3^- + \text{I}^- + \text{H}^+ \rightarrow \text{I}_2 + \text{H}_2\text{O}$ (Acidic medium) **RR0058**
 (iv) $\text{Cu}_2\text{S} + \text{MnO}_4^- + \text{H}^+ \rightarrow \text{Cu}^{2+} + \text{Mn}^{2+} + \text{SO}_2$ (Acidic medium) **RR0059**
 (v) $\text{HCHO}(\text{l}) + [\text{Ag}(\text{NH}_3)_2]^+(\text{aq.}) + \text{OH}^-(\text{aq.}) \rightarrow \text{Ag}(\text{s}) + \text{HCOO}^-(\text{aq.}) + \text{NH}_3(\text{aq.}) + \text{H}_2\text{O}(\text{l})$
 (Basic medium) **RR0060**
- Find out the n_{factor} of $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ in the following decomposition reaction.
 $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \longrightarrow \text{Cr}_2\text{O}_3 + \text{N}_2(\text{g}) + \text{H}_2\text{O}$ **RR0018**
- A mixture of H_2SO_4 and $\text{H}_2\text{C}_2\text{O}_4$ (oxalic acid) and some inert impurity weighing 3.185 g was dissolved in water and the solution made up to 1 litre, 10 mL of this solution required 3 mL of 0.1 N NaOH for complete neutralization. In another experiment 100 mL of the same solution in hot condition required 4 mL of 0.02M KMnO_4 solution for complete reaction. The wt. % of H_2SO_4 in the mixture is :- **RR0072**
- 25 ml of 0.4 M -acidified KMnO_4 solution is treated with excess of KI solution. The liberated I_2 exactly requires 20 ml of $\text{Na}_2\text{S}_2\text{O}_3$ solution for titration. What is the molarity of $\text{Na}_2\text{S}_2\text{O}_3$ solution? **RR0203**
- 25 ml of 0.4 M -acidified KIO_3 solution is treated with excess of KI solution. The liberated I_2 exactly requires 20 ml of $\text{Na}_2\text{S}_2\text{O}_3$ solution for titration. What is the molarity of $\text{Na}_2\text{S}_2\text{O}_3$ solution? **RR0204**
- 17.4 gm pyrolusite ore is dissolved in excess of HCl solution. The liberated Cl_2 gas is passed through excess of KI solution. If the liberated I_2 exactly requires 40 ml of decimolar hypo solution for titration, calculate the percentage of pure MnO_2 in the pyrolusite ore. ($\text{Mn} = 55$). **RR0205**
- 4.00 gm NaOH is dissolved in 100 ml HCl solution and the solution is diluted to 500 ml. If 25 ml of the diluted solution exactly requires 10 ml of 0.02 M - Na_2CO_3 solution for titration, calculate the concentration of original HCl solution (in gm/litre). **RR0206**
- 30 ml of '22.7V' H_2O_2 solution is added in 50 ml 0.3 M - KMnO_4 solution in presence of excess H_2SO_4 and the resulting solution is diluted to 225 ml. 25 ml of the diluted solution exactly require 20 ml of 0.05 M solution of an oxidising agent for titration. Calculate the moles of electron taken by each mole of oxidising agent. **RR0207**
- 12.7 gm I_2 is dissolved in 200 ml of 1.5 M - NaOH solution. Calculate the volume (in ml) of 0.8 M - HCl solution needed for complete neutralisation of excess NaOH. [$\text{I} = 127$] **RR0208**

EXERCISE # O-I

SINGLE CORRECT :

OXIDATION NUMBER

- The oxidation number of phosphorus in $\text{Ba}(\text{H}_2\text{PO}_2)_2$ is :
(A) +3 (B) +2 (C) +1 (D) -1 **RR0074**
- In the coordination compound, $\text{K}_4[\text{Ni}(\text{CN})_6]$, the oxidation state of nickel is **[AIEEE-03]**
(A) +1 (B) +2 (C) -1 (D) 0 **RR0076**
- The oxidation state of Cr in $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$ is - **[AIEEE-05]**
(A) +2 (B) +3 (C) 0 (D) +1 **RR0077**
- Oxidation number of Cl in CaOCl_2 (bleaching powder is) **[AIEEE-02]**
(A) Zero, since it contains Cl_2 (B) -1, since it contains Cl^-
(C) +1, since it contains ClO^- (D) +1 and -1 since it contains ClO^- and Cl^- **RR0078**
- The oxidation number of sulphur in S_8 , S_2F_2 and H_2S respectively are :
(A) 0, +1 and -2 (B) +2, +1 and -2
(C) 0, +1 and +2 (D) -2, +1 and -2 **RR0079**
- The incorrect order of decreasing oxidation number of S in compounds is :-
(A) $\text{H}_2\text{S}_2\text{O}_7 > \text{Na}_2\text{S}_4\text{O}_6 > \text{Na}_2\text{S}_2\text{O}_3 > \text{S}_8$ (B) $\text{H}_2\text{SO}_5 > \text{H}_2\text{SO}_3 > \text{SCl}_2 > \text{H}_2\text{S}$
(C) $\text{SO}_3 > \text{SO}_2 > \text{S}_8 > \text{H}_2\text{S}$ (D) $\text{H}_2\text{SO}_4 > \text{SO}_2 > \text{H}_2\text{S} > \text{H}_2\text{S}_2\text{O}_8$ **RR0080**
- Match List-I (Compounds) with List-II (Oxidation states of nitrogen) and select answer using the codes given below the lists :-

List-I

- (a) NaN_3
(B) N_2H_2
(C) NO
(D) N_2O_5

List-II

1. +5
2. +2
3. -1/3
4. -1

Code :

	(a)	(B)	(C)	(D)
(A)	3	4	2	1
(B)	4	3	2	1
(C)	3	4	1	2
(D)	4	3	1	2

RR0085

OXIDATION REDUCTION

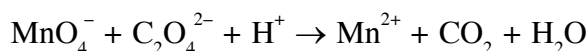
- Which of the following is a redox reaction ? **[AIEEE-02]**
(A) $2\text{NaAg}(\text{CN})_2 + \text{Zn} \longrightarrow \text{Na}_2\text{Zn}(\text{CN})_4 + 2\text{Ag}$
(B) $\text{BaO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$
(C) $\text{N}_2\text{O}_5 + \text{H}_2\text{O} \longrightarrow 2\text{HNO}_3$
(D) $\text{AgNO}_3 + \text{KI} \longrightarrow \text{AgI} + \text{KNO}_3$ **RR0081**

9. Which reaction does not represent a disproportionation reaction ?
 (A) $\text{Cl}_2 + \text{OH}^- \longrightarrow \text{Cl}^- + \text{ClO}_3^- + \text{H}_2\text{O}$ (B) $2\text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{O}_2$
 (C) $2\text{Cu}^+ \longrightarrow \text{Cu}^{+2} + \text{Cu}$ (D) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \longrightarrow \text{N}_2 + \text{Cr}_2\text{O}_3 + 4\text{H}_2\text{O}$ **RR0082**
10. Which of the following is not a redox reaction?
 (A) $\text{BaO}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$ (B) $2\text{BaO} + \text{O}_2 \rightarrow 2\text{BaO}_2$
 (C) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$ (D) $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 2\text{H}_2\text{O} + 3\text{S}$ **RR0083**
11. The molecular weight of compounds ?
 (a) Na_2SO_4 (b) $\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}$ (c) $\text{Ca}_3(\text{PO}_4)_2$ respectively are X, Y and Z.
 The correct set of equivalent weight will be :
 (A) $\frac{X}{2}, \frac{Y}{3}, \frac{Z}{6}$ (B) $\frac{X}{2}, Y, \frac{Z}{3}$ (C) $X, \frac{Y}{3}, \frac{Z}{3}$ (D) X, Y, Z **RR0245**
12. H_2O_2 acts as a reducing agent in:
 (A) $\text{FeCl}_2 + \text{HCl} + \text{H}_2\text{O}_2 \longrightarrow \text{FeCl}_3 + \text{H}_2\text{O}$ (B) $\text{Cl}_2 + \text{H}_2\text{O}_2 \longrightarrow \text{HCl} + \text{O}_2$
 (C) $\text{HI} + \text{H}_2\text{O}_2 \longrightarrow \text{I}_2 + \text{H}_2\text{O}$ (D) $\text{H}_2\text{SO}_3 + \text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$ **RR0084**
13. MnO_4^- is good oxidising agent in different medium changing to - **[AIEEE-02]**
 $\text{MnO}_4^- \longrightarrow \text{Mn}^{2+}$
 $\longrightarrow \text{MnO}_4^{2-}$
 $\longrightarrow \text{MnO}_2$
 $\longrightarrow \text{Mn}_2\text{O}_3$
 Changes in oxidation number respectively are -
 (A) 1, 3, 4, 5 (B) 5, 4, 3, 2 (C) 5, 1, 3, 4 (D) 2, 6, 4, 3 **RR0086**
14. The oxidation state of chromium in the final product formed by the reaction between KI and acidified potassium dichromate solution is - **[AIEEE-05]**
 (A) +6 (B) +4 (C) +3 (D) +2 **RR0104**

BALANCING OF REDOX REACTIONS

15. During the disproportionation of iodine to iodide and iodate ions, the ratio of iodate and iodide ions formed in alkaline medium is :-
 (A) 1 : 5 (B) 5 : 1 (C) 3 : 1 (D) 1 : 3 **RR0124**
16. In the reaction,
 $x\text{HI} + y\text{HNO}_3 \longrightarrow \text{NO} + \text{I}_2 + \text{H}_2\text{O}$
 (A) $x = 3, y = 2$ (B) $x = 2, y = 3$ (C) $x = 6, y = 2$ (D) $x = 6, y = 1$ **RR0087**
17. The number of electrons to balance the following equation :-
 $\text{NO}_3^- + 4\text{H}^+ + e^- \rightarrow 2\text{H}_2\text{O} + \text{NO}$ is
 (A) 5 (B) 4 (C) 3 (D) 2 **RR0088**

18. For the redox reaction,

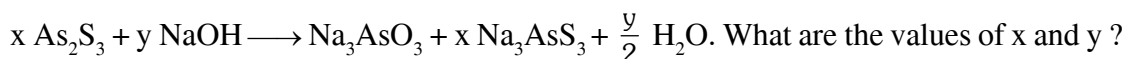


the correct coefficients of the reactants for the balanced reaction are :

MnO_4^-	$\text{C}_2\text{O}_4^{2-}$	H^+
(A) 2	5	16
(B) 16	5	2
(C) 5	16	2
(D) 2	16	5

RR0090

19. When arsenic sulphide is boiled with NaOH, sodium arsenite and sodium thioarsenite are formed



What are the values of x and y ?

- (A) 1, 6 (B) 2, 8 (C) 2, 6 (D) 1, 4 **RR0125**

20. Number of moles of electrons taken up when 1 mole of NO_3^- ions is reduced to 1 mole of NH_2OH is

- (A) 2 (B) 4 (C) 5 (D) 6 **RR0089**

n-FACTOR AND EQUIVALENT WEIGHT

21. When N_2 is converted into NH_3 , the equivalent weight of nitrogen will be :

- (A) 1.67 (B) 2.67 (C) 2.33 (D) 4.67 **RR0092**

22. The equivalent mass of MnSO_4 is half its molecular mass when it is converted to : **[JEE 1998]**

- (A) Mn_2O_3 (B) MnO_2 (C) MnO_4^- (D) MnO_4^{2-} **RR0093**

23. When KBrO_3 ion reacts with Br^- ion in acid solution Br_2 is liberated. The equivalent weight of KBrO_3 in this reaction is

- (A) $M/8$ (B) $M/3$ (C) $M/5$ (D) $M/6$ **RR0094**

24. A solution of KMnO_4 is reduced to MnO_2 . The normality of KMnO_4 solution is 0.6. The molarity is

- (A) 1.8M (B) 0.6M (C) 0.1M (D) 0.2M **RR0096**

25. 0.52 g of a dibasic acid required 100 mL of 0.2 N NaOH for complete neutralization. The equivalent weight of acid is

- (A) 26 (B) 52 (C) 104 (D) 156 **RR0098**

26. What is the equivalent weight of H_2SO_4 in the reaction ?



- (A) 12.25 (B) 49 (C) 61.25 (D) 98.00 **RR0099**

27. The mass of oxalic acid crystals ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) required to prepare 50 mL of a 0.2 N solution is:-

- (A) 4.5 g (B) 6.3 g (C) 0.63 g (D) 0.45 g **RR0101**

ACID - BASE TITRATION

28. If 25 mL of a H_2SO_4 solution reacts completely with 1.06 g of pure Na_2CO_3 , what is the normality of this acid solution :
 (A) 1 N (B) 0.5 N (C) 1.8 N (D) 0.8 N **RR0102**
29. 100 ml of 0.1M $\text{NaAl}(\text{OH})_2\text{CO}_3$ is neutralised by 0.25 N HCl to form NaCl, AlCl_3 and CO_2 . Volume of HCl required is
 (A) 10 mL (B) 40 mL (C) 100 mL (D) 160 mL **RR0116**

REDOX TITRATION

30. The number of mole of KMnO_4 that will need to react completely with one mole ferrous oxalate in acidic solution is : **[JEE 1997]**
 (A) 2/5 (B) 3/5 (C) 4/5 (D) 1 **RR0105**
31. The number of mole of KMnO_4 that will be needed to react with one mole of sulphite ion in acidic solution is : **[JEE 1997]**
 (A) 2/5 (B) 3/5 (C) 4/5 (D) 1 **RR0106**
32. A mixture of NaHC_2O_4 and $\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{C}_2\text{O}_4$ required equal volumes of 0.2 N KMnO_4 and 0.12 N NaOH separately. What is the molar ratio of NaHC_2O_4 and $\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{C}_2\text{O}_4$ in the mixture ?
 (A) 6 : 1 (B) 1 : 2 (C) 3 : 1 (D) 4 : 1 **RR0112**
33. Volume V_1 mL of 0.1 M $\text{K}_2\text{Cr}_2\text{O}_7$ is needed for complete oxidation of 0.678 g N_2H_4 in acidic medium. The volume of 0.3 M KMnO_4 needed for same oxidation in acidic medium will be :-
 (A) $\frac{2}{5} V_1$ (B) $\frac{5}{2} V_1$ (C) $\frac{1}{3} V_1$ (D) $\frac{5}{6} V_1$ **RR0113**
34. As_2O_3 is oxidised to H_3AsO_4 by KMnO_4 in acidic medium. Volume of 0.02M KMnO_4 required for this purpose by 1mmol of As_2O_3 will be
 (A) 10 mL (B) 20 mL (C) 40 mL (D) 80 mL **RR0114**
35. x mmol of XeF_4 quantitatively oxidized KI to I_2 and liberated Xe, alongwith formation of KF. This iodine required 20 ml of decinormal hypo solution for exact titration. The value of x is
 (A) 0.5 (B) 1.0 (C) 2.0 (D) 5.0 **RR0117**
36. 0.3 g of an oxalate salt was dissolved in 100 mL solution. The solution required 90 mL of N/20 KMnO_4 for complete oxidation. The % of oxalate ion in salt is :-
 (A) 33% (B) 66% (C) 70% (D) 40% **RR0119**

37. The number of moles of $\text{Cr}_2\text{O}_7^{2-}$ needed to oxidize 0.136 equivalents of N_2H_5^+ by the reaction $\text{N}_2\text{H}_5^+ + \text{Cr}_2\text{O}_7^{2-} \rightarrow \text{N}_2 + \text{Cr}^{3+} + \text{H}_2\text{O}$ is
(A) 0.136 (B) 0.068 (C) 0.0227 (D) 0.272 **RR0121**
38. HNO_3 oxidises NH_4^+ ions to nitrogen and itself gets reduced to NO_2 . The moles of HNO_3 required by 1 mol of $(\text{NH}_4)_2\text{SO}_4$ is :-
(A) 4 (B) 5 (C) 6 (D) 2 **RR0123**

HARDNESS OF WATER

39. Which of the following salt does not cause hardness in water ?
(A) $\text{Ca}(\text{NO}_3)_2$ (B) K_2SO_4 (C) MgSO_4 (D) MgCl_2 **RR0209**
40. Distilled water may be obtained from hard water by -
(A) Calgon process (B) Zeolite process
(C) Clark process (D) Ion-exchange resin process **RR0210**

EXERCISE # O-II

SINGLE CORRECT :

- If 10 g of V_2O_5 is dissolved in acid and is reduced to V^{2+} by zinc metal, how many mole I_2 could be reduced by the resulting solution if it is further oxidised to VO^{2+} ions ?
[Assume no change in state of Zn^{2+} ions] ($V = 51, O = 16, I = 127$) :
(A) 0.11 mole of I_2 (B) 0.22 mole of I_2
(C) 0.055 mole of I_2 (D) 0.44 mole of I_2 **RR0120**
- Dichloroacetic acid ($CHCl_2COOH$) is oxidized to CO_2 and Cl_2 by 600 meq of an oxidizing agent. Same amount of acid can neutralize how many moles of ammonia to form ammonium dichloroacetate?
(A) 0.0167 (B) 0.1 (C) 0.3 (D) 0.6 **RR0122**
- An element A in a compound ABD has oxidation number A^{n-} . It is oxidised by $Cr_2O_7^{2-}$ in acid medium. In the experiment 1.68×10^{-3} moles of $K_2Cr_2O_7$ were used for 3.36×10^{-3} moles of ABD. The new oxidation number of A after oxidation is :-
(A) 3 (B) $3 - n$ (C) $n - 3$ (D) $+n3$ **RR0127**
- A sample of $KMnO_4$ solution required 50 ml when titrated against 3 mmol of oxalic acid. The normality of same solution in reaction with alkaline H_2O_2 is
(A) 0.120 N (B) 0.060 N (C) 0.072 N (D) 0.036 N **RR0211**
- In the reaction $CrO_5 + H_2SO_4 \rightarrow Cr_2(SO_4)_3 + H_2O + O_2$ one mole of CrO_5 will liberate how many moles of O_2 ?
(A) 5/2 (B) 5/4 (C) 9/2 (D) 7/4 **RR0095**

ONE OR MORE THAN ONE CORRECT :

- Which of the following process(es) is/are either oxidation or reduction ?
(A) $S_2O_3^{2-} \rightarrow S_4O_6^{2-}$ (B) $C_4H_6 \rightarrow CO_2$
(C) $Cr_2O_7^{2-} \rightarrow CrO_4^{2-}$ (D) $H_2SO_4 \rightarrow H_2S_2O_8$ **RR0212**
- The correct information(s) about the n-factor of given species in the reaction :
 $KMnO_4 + H_2SO_4 + HCl \rightarrow K_2SO_4 + MnSO_4 + Cl_2 + H_2O$ is/are -
(A) $KMnO_4 \rightarrow 5$ (B) $H_2SO_4 \rightarrow 10/3$
(C) $HCl \rightarrow 1$ (D) $K_2SO_4 \rightarrow 2$ **RR0213**
- 0.2 mole of Cl_2 is reacted with 500 ml of 1.2 M - NaOH solution
 $Cl_2 + OH^- \rightarrow ClO_3^- + Cl^- + H_2O$
The incorrect information(s) related is/are -
(A) Cl_2 is limiting reagent (B) 0.4 mole NaOH will left unreacted
(C) $[ClO_3^-]_{\text{final solution}} = 0.133 \text{ M}$ (D) $[Cl^-]_{\text{final solution}} = 0.267 \text{ M}$ **RR0214**
- 100 ml of 0.4 M -acidified $KMnO_4$ solution may be decolourised completely by
(A) 200 ml 1N - $K_2Cr_2O_7$ solution (B) 300 ml 0.5M - H_2O_2 solution
(C) 100 ml 0.8N - KI solution (D) 75 ml 1.4 N - $H_2C_2O_4$ solution **RR0215**

10. Statement-1 :- Moles of KMnO_4 required for oxidation of Fe^{2+} in acidic and basic medium will be different.

Statement-2 :- Final oxidation state to which Mn^{7+} will be reduced will be different in case of acidic and basic medium.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
- RR0131**

RR0131

- 11. Statement-1 :-** Mass of a particular substance that combine with 8 gm of oxygen is said to be equivalent weight of substance.

Statement-2 :- x gm of metal gave y gm of its oxide, so equivalent weight of metal is $\left(\frac{x}{y-x} \right) \times 8$

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.
- RR0132**

RR0132

- 12. Statement-1 :-** Degree of hardness of water is measured in terms of ppm of CaCO_3 .

Statement-2 :- If water contains 120 ppm of MgSO_4 , and 2ppm NaCl , its hardness in terms of $\text{CaCO}_3 > 100$ ppm.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
- RR0133**

RR0133

MATCH THE COLUMN

- 13. Column-I**

- (A) $\text{P}_2\text{H}_4 \longrightarrow \text{PH}_3 + \text{P}_4\text{H}_2$

- (B) $\text{I}_2 \longrightarrow \text{I}^- + \text{IO}_3^-$

- (C) $\text{MnO}_4^- + \text{Mn}^{2+} + \text{H}_2\text{O} \longrightarrow \underline{\text{Mn}_3\text{O}_4} + \text{H}^+$

- (D) $\underline{\text{H}_3\text{PO}_2} \longrightarrow \text{PH}_3 + \text{H}_3\text{PO}_3$

Column-II

- (p) $E = \frac{3M}{4}$

- (q) $E = \frac{3M}{5}$

- $$(r) \quad E = \frac{15M}{26}$$

- (s) $E = \frac{5M}{6}$

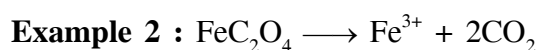
RR0134

Paragraph for Q.14 to Q.17

$$\text{Equivalent weight} = \frac{\text{Molecular weight / Atomic weight}}{n - \text{factor}}$$

n-factor is very important in redox as well as non-redox reactions. With the help of n-factor we can predict the molar ratio of the reactant species taking part in reactions. The reciprocal of n-factor's ratio of the reactants is the molar ratio of the reactants.

In general n-factor of acid/base is number of moles of H^+ / OH^- furnished per mole of acid/base. n-factor of a reactant is no. of moles of electrons lost or gained per mole of reactant.

Example 1 :

Total no. of moles of e^- lost by 1 mole of $\text{FeC}_2\text{O}_4 = 1 + 1 \times 2 \Rightarrow 3$

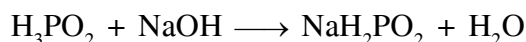
$\therefore$ n-factor of $\text{FeC}_2\text{O}_4 = 3$

14. n-factor of $\text{Ba}(\text{MnO}_4)_2$ in acidic medium is :

(A) 2 (B) 6 (C) 10 (D) none of these

RR0135

15. For the reaction,



What is the equivalent weight of H_3PO_2 ? (mol. wt. is M)

(A) M (B) M/2 (C) M/3 (D) none of these

RR0136

16. For the reaction, $\text{Fe}_{0.95}\text{O}$ (molar mass : M) $\longrightarrow \text{Fe}_2\text{O}_3$. What is the eq. wt. of $\text{Fe}_{0.95}\text{O}$?

(A) $\frac{M}{0.85}$ (B) $\frac{M}{0.95}$ (C) $\frac{M}{0.8075}$ (D) none of these

RR0137

17. In the reaction, $x\text{VO} + y\text{Fe}_2\text{O}_3 \longrightarrow \text{FeO} + \text{V}_2\text{O}_5$. What is the value of x and y respectively ?

(A) 1, 1 (B) 2, 3 (C) 3, 2 (D) none of these

RR0138

TABLE TYPE QUESTION :

Column-I (1 mole of each oxidizing agent is taken)	Column-II (oxidation number)	Column-III (Reducing agents)	
(A) KMnO_4 (acidic)	(P) O.N of Mn = 4	(1) 3 mole of FeSO_4	
(B) KMnO_4 (neutral)	(Q) O.N of Mn = 7	(2) 0.5 mole of I_2 changes to HIO_3	
(C) MnO_2 (acidic)	(R) O.N of Cr = 6	(3) 1 mole of $\text{K}_2\text{C}_2\text{O}_4$	
(D) K_2CrO_4 (acidic)	(S) O.N of Cr = 7	(4) 1.5 mole K_2SO_3	
18. Which of the following is correct ?			
(A) A ; P ; 2	(B) A ; Q ; 4	(C) B ; Q ; 1	(D) B ; Q ; 3
19. Which of the following is correct ?			
(A) A ; Q ; 3	(B) C ; P ; 3	(C) C ; P ; 4	(D) C ; Q ; 1
20. Which of the following is/are correct ? (one or more than one correct)			
(A) D ; R ; 1	(B) D ; R ; 2	(C) D ; S ; 1	(D) D ; R ; 4

RR0139

RR0139

RR0139

EXERCISE # J-MAINS

1. Consider the reaction [JEE(Main-online)-2014]
- $$\text{H}_2\text{SO}_{3(\text{aq})} + \text{Sn}_{(\text{aq})}^{4+} + \text{H}_2\text{O}_{(\text{l})} \rightarrow \text{Sn}_{(\text{aq})}^{2+} + \text{HSO}_4^{-}(\text{aq}) + 3\text{H}^{+}(\text{aq})$$
- Which of the following statements is correct?
- (1) H_2SO_3 is the reducing agent because it undergoes oxidation
 (2) H_2SO_3 is the reducing agent because it undergoes reduction
 (3) Sn^{4+} is the reducing agent because it undergoes oxidation
 (4) Sn^{4+} is the oxidizing agent because it undergoes oxidation RR0143
2. In which of the following reaction H_2O_2 acts as a reducing agent ? [JEE(Main)-2014]
- (a) $\text{H}_2\text{O}_2 + 2\text{H}^{+} + 2\text{e}^{-} \rightarrow 2\text{H}_2\text{O}$ (b) $\text{H}_2\text{O}_2 - 2\text{e}^{-} \rightarrow \text{O}_2 + 2\text{H}^{+}$
 (c) $\text{H}_2\text{O}_2 + 2\text{e}^{-} \rightarrow 2\text{OH}^{-}$ (d) $\text{H}_2\text{O}_2 + 2\text{OH}^{-} - 2\text{e}^{-} \rightarrow \text{O}_2 + 2\text{H}_2\text{O}$
 (1) (a), (c) (2) (b), (d) (3) (a), (b) (4) (c), (d) RR0144
3. The molecular formula of a commercial resin used for exchanging ions in water softening is $\text{C}_8\text{H}_7\text{SO}_3\text{Na}$ (Mol. w.t 206). What would be the maximum uptake of Ca^{2+} ions by the resin when expressed in mole per gram resin ? [JEE(Main)-2015]
- (1) $\frac{2}{309}$ (2) $\frac{1}{412}$ (3) $\frac{1}{103}$ (4) $\frac{1}{206}$ RR0145
4. The volume of 0.1N dibasic acid sufficient to neutralize 1 g of a base that furnishes 0.04 mole of OH^{-} in aqueous solution is : [JEE(Main)-OnLine-2016]
- (1) 400 mL (2) 200 mL (3) 600 mL (4) 800 mL RR0146
5. Which of the following reactions is an example of a redox reaction ? [JEE(Main)-2017]
- (1) $\text{XeF}_4 + \text{O}_2\text{F}_2 \rightarrow \text{XeF}_6 + \text{O}_2$ (2) $\text{XeF}_2 + \text{PF}_5 \rightarrow [\text{XeF}]^{+}\text{PF}_6^{-}$
 (3) $\text{XeF}_6 + \text{H}_2\text{O} \rightarrow \text{XeOF}_4 + 2\text{HF}$ (4) $\text{XeF}_6 + 2\text{H}_2\text{O} \rightarrow \text{XeO}_2\text{F}_2 + 4\text{HF}$ RR0147
6. In KO_2 , the nature of oxygen species and the oxidation state of oxygen atom are, respectively
- (1) Superoxide and $-1/2$ (2) Oxide and -2 [JEE(Main)-OnLine-2018]
 (3) Peroxide and $-1/2$ (4) Superoxide and -1 RR0150
7. The hardness of a water sample (in terms of equivalents of CaCO_3) containing 10^{-3} M CaSO_4 is: (molar mass of $\text{CaSO}_4 = 136 \text{ g mol}^{-1}$) [JEE(Main)-(Jan.)-2019]
- (1) 100 ppm (2) 50 ppm (3) 10 ppm (4) 90 ppm RR0153
8. 50 mL of 0.5 M oxalic acid is needed to neutralize 25 mL of sodium hydroxide solution. The amount of NaOH in 50 mL of the given sodium hydroxide solution is : [JEE(Main)-(Jan.)-2019]
- (1) 4 g (2) 2 g (3) 8 g (4) 1 g RR0154
9. In order to oxidise a mixture one mole of each of FeC_2O_4 , $\text{Fe}_2(\text{C}_2\text{O}_4)_3$, FeSO_4 and $\text{Fe}_2(\text{SO}_4)_3$ in acidic medium, the number of moles of KMnO_4 required is - [JEE(Main)-(April)-2019]
- (1) 3 (2) 2 (3) 1 (4) 1.5 RR0155

10. An example of a disproportionation reaction is : [JEE(Main)-(April)-2019]
 (1) $2\text{KMnO}_4 \rightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$ (2) $2\text{MnO}_4^- + 10\text{I}^- + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 5\text{I}_2 + 8\text{H}_2\text{O}$
 (3) $2\text{CuBr} \rightarrow \text{CuBr}_2 + \text{Cu}$ (4) $2\text{NaBr} + \text{Cl}_2 \rightarrow 2\text{NaCl} + \text{Br}_2$ **RR0158**
11. 100 mL of a water sample contains 0.81 g of calcium bicarbonate and 0.73 of magnesium bicarbonate. The hardness of this water sample expressed in terms of equivalents of CaCO_3 is:
 (molar mass of calcium bicarbonate is 162 g mol^{-1} and magnesium bicarbonate is 146 g mol^{-1})
 (1) 1,000 ppm (2) 10,000 ppm [JEE(Main)-(April)-2019]
 (3) 100 ppm (4) 5,000 ppm **RR0159**
12. 25 ml of the given HCl solution requires 30 mL of 0.1 M sodium carbonate solution. What is the volume of this HCl solution required to titrate 30 mL of 0.2 M aqueous NaOH solution?
 (1) 25 mL (2) 50 mL [JEE(Main)-(Jan)-2019]
 (3) 12.5 mL (4) 75 mL **RR0160**
13. Oxidation number of potassium in K_2O , K_2O_2 and KO_2 , respectively, is : [JEE(Main)-(Jan)-2020]
 (1) +1, +4 and +2 (2) +1, +2 and +4 (3) +1, +1 and +1 (4) +2, +1 and $+\frac{1}{2}$
RR0167
14. The compound that cannot act both as oxidising and reducing agent is : [JEE(Main)-(Jan)-2020]
 (1) H_2O_2 (2) H_2SO_3 (3) HNO_2 (4) H_3PO_4
RR0168
15. The strength of an aqueous NaOH solution is most accurately determined by titrating :
 (Note : consider that an appropriate indicator is used) [JEE(Main)-(Jan)-2020]
 (1) Aq. NaOH in a volumetric flask and concentrated H_2SO_4 in a conical flask
 (2) Aq. NaOH in a pipette and aqueous oxalic acid in a burette
 (3) Aq. NaOH in a burette and concentrated H_2SO_4 in a conical flask
 (4) Aq. NaOH in a burette and aqueous oxalic acid in a conical flask **RR0217**
16. A 20.0 mL solution containing 0.2 g impure H_2O_2 reacts completely with 0.316 g of KMnO_4 in acid solution. The purity of H_2O_2 (in %) is _____ (mol. wt. of $\text{H}_2\text{O}_2 = 34$; mol. wt. of $\text{KMnO}_4 = 158$) [JEE(Main)-(Jan)-2020]
RR0218
17. A 100 mL solution was made by adding 1.43 g of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$. The normality of the solution is 0.1 N. The value of x is _____. [JEE(Main)-(Jan)-2020]
 (The atomic mass of Na is 23 g mol^{-1}) :- **RR0219**
18. The volume, in mL, of 0.02 M $\text{K}_2\text{Cr}_2\text{O}_7$ solution required to react with 0.288 g of ferrous oxalate in acidic medium is _____. [JEE(Main)-(Jan)-2020]
 (Molar mass of Fe = 56 g mol^{-1}) **RR0220**

19. The reaction of sulphur in alkaline medium is the below: [JEE(Main)-(Jan)-2021]

$$\text{S}_{8(s)} + a \text{OH}^-_{(aq)} \rightarrow b \text{S}^{2-}_{(aq)} + c \text{S}_2\text{O}_3^{2-}_{(aq)} + d \text{H}_2\text{O}_{(l)}$$

 The values of 'a' is _____. (Integer answer) **RR0222**
20. In basic medium CrO_4^{2-} oxidises $\text{S}_2\text{O}_3^{2-}$ to form SO_4^{2-} and itself changes into $\text{Cr}(\text{OH})_4^-$. The volume of 0.154 M CrO_4^{2-} required to react with 40 mL of 0.25 M $\text{S}_2\text{O}_3^{2-}$ is _____ mL. (Rounded-off to the nearest integer) [JEE(Main)-(Jan)-2021]
RR0223
21. 10.0 ml of Na_2CO_3 solution is titrated against 0.2 M HCl solution. The following titre values were obtained in 5 readings.
 4.8 ml, 4.9 ml, 5.0 ml, 5.0 ml and 5.0 ml [JEE(Main)-(Jan)-2021]
 Based on these readings, and convention of titrimetric estimation of concentration of Na_2CO_3 solution is _____ mM. (Round off to the Nearest integer) **RR0224**
22. 250 mL of 0.5 M NaOH was added to 500 mL of 1 M HCl. The number of unreacted HCl molecules in the solution after complete reaction is _____ $\times 10^{21}$. (Nearest integer)
 $(N_A = 6.022 \times 10^{23})$ [JEE(Main)-(Jan)-2021]
RR0225
23. When 10 mL of an aqueous solution of Fe^{2+} ions was titrated in the presence of dil H_2SO_4 using diphenylamine indicator, 15 mL of 0.02 M solution of $\text{K}_2\text{Cr}_2\text{O}_7$ was required to get the end point. The molarity of the solution containing Fe^{2+} ions is $x \times 10^{-2}$ M. The value of x is _____. (Nearest integer) [JEE(Main)-(Jan)-2021]
RR0226
24. When 10 mL of an aqueous solution of KMnO_4 was titrated in acidic medium, equal volume of 0.1 M of an aqueous solution of ferrous sulphate was required for complete discharge of colour. The strength of KMnO_4 in grams per litre is _____ $\times 10^{-2}$. (Nearest integer)
 $[\text{Atomic mass of K} = 39, \text{Mn} = 55, \text{O} = 16]$ [JEE(Main)-(Jan)-2021]
RR0227
25. 0.01 M KMnO_4 solution was added to 20.0 mL of 0.05 M Mohr's salt solution through a burette. The initial reading of 50 mL burette is zero. The volume of KMnO_4 solution left in the burette after the end point is _____ mL. (nearest integer) [JEE(Main)-(June)-2022]
RR0246
26. A 2.0 g sample containing MnO_2 is treated with HCl liberating Cl_2 . The Cl_2 gas is passed into a solution of KI and 60.0 mL of 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$ is required to titrate the liberated iodine. The percentage of MnO_2 in the sample is _____. (Nearest integer) [Atomic masses (in u) Mn = 55; Cl = 35.5; O = 16, I = 127, Na = 23, K = 39, S = 32] [JEE(Main)-(June)-2022]
RR0247

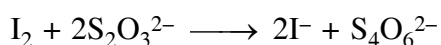
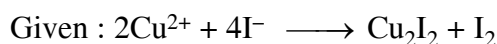
27. In neutral or faintly alkaline medium, KMnO_4 being a powerful oxidant can oxidize, thiosulphate almost quantitatively, to sulphate. In this reaction overall change in oxidation state of manganese will be :

[JEE(Main)-(July)-2022]

- (A) 5 (B) 1 (C) 0 (D) 3

RR0248

28. 20 mL of 0.02 M hypo solution is used for the titration of 10 mL of copper sulphate solution, in the presence of excess of KI using starch as an indicator. The molarity of Cu^{2+} is found to be $\text{_____} \times 10^{-2}$ M [nearest integer]



[JEE(Main)-(July)-2022]

RR0249

29. In the titration of KMnO_4 and oxalic acid in acidic medium, the change in oxidation number of carbon at the end point is _____

[JEE(Main)-(July)-2022]

RR0250

30. 20 mL of 0.02 M $\text{K}_2\text{Cr}_2\text{O}_7$ solution is used for the titration of 10 mL of Fe^{2+} solution in the acidic medium. The molarity of Fe^{2+} solution is $\text{_____} \times 10^{-2}$ M. (Nearest Integer) [JEE(Main)-(July)-2022]

RR0251

EXERCISE # J-ADVANCED

1. Reduction of the metal centre in aqueous permanganate ion involves - [JEE-2011]
 (A) 3 electrons in neutral medium (B) 5 electrons in neutral medium
 (C) 3 electrons in alkaline medium (D) 5 electrons in acidic medium **RR0161**
2. Reaction of Br_2 with Na_2CO_3 in aqueous solution gives sodium bromide and sodium bromate with evolution of CO_2 gas. The number of sodium bromide molecules involved in the balanced chemical equation is. [JEE- 2011]
RR0162
3. Which ordering of compounds is according to the decreasing order of the oxidation state of nitrogen-
 (A) HNO_3 , NO , NH_4Cl , N_2 (B) HNO_3 , NO , N_2 , NH_4Cl
 (C) HNO_3 , NH_4Cl , NO , N_2 (D) NO , HNO_3 , NH_4Cl , N_2 [JEE- 2012]
RR0163
4. 25 mL of household bleach solution was mixed with 30 mL of 0.50 M KI and 10 mL of 4 N acetic acid. In the titration of the liberated iodine, 48 mL of 0.25 N $\text{Na}_2\text{S}_2\text{O}_3$ was used to reach the end point. The molarity of the household bleach solution is [JEE- 2012]
 (A) 0.48 M (B) 0.96 M (C) 0.24 M (D) 0.024 M **RR0164**
5. For the reaction [JEE- 2014]
 $\text{I}^- + \text{ClO}_3^- + \text{H}_2\text{SO}_4 \rightarrow \text{Cl}^- + \text{HSO}_4^- + \text{I}_2$
 The correct statement(s) in the balanced equation is / are :
 (A) Stoichiometric coefficient of HSO_4^- is 6
 (B) Iodide is oxidized
 (C) Sulphur is reduced
 (D) H_2O is one of the products **RR0228**
6. In neutral or faintly alkaline solution, 8 moles permanganate anion quantitatively oxidize thiosulphate anions to produce X moles of a sulphur containing product. the magnitude of X is [JEE- 2016]
RR0165
7. The order of the oxidation state of the phosphorus atom in H_3PO_2 , H_3PO_4 , H_3PO_3 and $\text{H}_4\text{P}_2\text{O}_6$ is [JEE- 2017]
 (A) $\text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2$ (B) $\text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6$
 (C) $\text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_4$ (D) $\text{H}_3\text{PO}_4 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6$ **RR0229**
8. To measure the quantity of MnCl_2 dissolved in an aqueous solution, it was completely converted to KMnO_4 using the reaction, [JEE- 2018]
 $\text{MnCl}_2 + \text{K}_2\text{S}_2\text{O}_8 + \text{H}_2\text{O} \rightarrow \text{KMnO}_4 + \text{H}_2\text{SO}_4 + \text{HCl}$ (equation not balanced).
 Few drops of concentrated HCl were added to this solution and gently warmed. Further, oxalic acid (225 mg) was added in portions till the colour of the permanganate ion disappeared. The quantity of MnCl_2 (in mg) present in the initial solution is _____.
 (Atomic weights in g mol^{-1} : Mn = 55, Cl = 35.5) **RR0166**

9. The amount of water produced (in g) in the oxidation of 1 mole of rhombic sulphur by conc. HNO_3 to a compound with the highest oxidation state of sulphur is ____

(Given data : Molar mass of water = 18 g mol^{-1})

RR0230

10. 5.00 mL of 0.10 M oxalic acid solution taken in a conical flask is titrated against NaOH from a burette using phenolphthalein indicator. The volume of NaOH required for the appearance of permanent faint pink color is tabulated below for five experiments. What is the concentration, in molarity, of the NaOH solution ? [JEE- 2020]

Exp. No.	Vol. of NaOH (mL)
1	12.5
2	10.5
3	9
4	9
5	9

RR0231

11. In the chemical reaction between stoichiometric quantities of KMnO_4 and KI in weakly basic solution, what is the number of moles of I_2 released for 4 moles of KMnO_4 consumed ? [JEE- 2020]

RR0232

Question Stem for Question Nos. 12 and 13

Question Stem

A sample (5.6 g) containing iron is completely dissolved in cold dilute HCl to prepare a 250 mL of solution. Titration of 25.0 mL of this solution requires 12.5 mL of 0.03 M KMnO_4 solution to reach the end point. Number of moles of Fe^{2+} present in 250 mL solution is $x \times 10^{-2}$ (consider complete dissolution of FeCl_2). The amount of iron present in the sample of y% by weight.

(Assume : KMnO_4 reacts only with Fe^{2+} in the solution

[JEE- 2021]

Use : Molar mass of iron as 56 g mol^{-1})

12. The value of x is ____.

RR0233

13. The value of y is ____.

RR0234

ANSWER KEY**EXERCISE # S-I**

1. (a) +5 (b) +2 (c) -1 (d) +2
 (e) +6 (f) +3 (g) +4/3 (h) -1/3
 (i) +8/3 (j) +16/3 (k) 0 (l) +1/2
 (m) -1/2 (n) +2
2. (a) +3 (b) +3, +6 (c) +2 (d) $+\frac{200}{93}$
 (e) +6 (f) +6 (g) +6
3. (a) Reduction (b) Oxidation (c) Reduction (d) Oxidation
 (e) Reduction (f) Oxidation (g) Reduction
4. (a) Oxidant : O_2 ; Reductant : Zn
 (b) Oxidant : H^+ ; Reductant : Zn
 (c) Oxidant : NO_3^- ; Reductant : Zn
 (d) Oxidant : SO_2 ; Reductant : H_2S
 (e) Oxidant : MnO_4^- ; Reductant : Cl^-
 (f) Oxidant : MnO_4^- ; Reductant : H_2O_2
5. (i) $Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$
 (ii) $Br_2 + 5H_2O_2 \longrightarrow 2BrO_3^- + 4H_2O + 2H^+$
 (iii) $MnO_4^- + 5Fe^{+2} + 8H^+ \longrightarrow Mn^{+2} + 5Fe^{+3} + 4H_2O$
 (iv) $MnO_2 + 2Cl^- + 4H^+ \longrightarrow Mn^{+2} + Cl_2 + 2H_2O$
 (v) $3ClO^- + 2CrO_2^- + 2OH^- \longrightarrow 3Cl^- + 2CrO_4^{2-} + H_2O$
 (vi) $Zn + 2NaOH \longrightarrow Na_2ZnO_2 + H_2$
 (vii) $2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$
6. (i) $2MnO_4^- + 6I^- + 4H_2O \longrightarrow 2MnO_2 + 3I_2 + 8OH^-$
 (ii) $H^+ + 2MnO_4^- + 5SO_2 + 2H_2O \longrightarrow 2Mn^{2+} + 5HSO_4^-$
 (iii) $2Fe^{2+} + H_2O_2 + 2H^+ \longrightarrow 2Fe^{3+} + 2H_2O$
 (iv) $Cr_2O_7^{2-} + 3SO_2 + 2H^+ \longrightarrow 2Cr^{3+} + 3SO_4^{2-} + H_2O$
 (v) $P_4 + 3OH^- + 3H_2O \longrightarrow PH_3 + 3H_2PO_2^-$
 (vi) $3MnO_4^- + 5FeC_2O_4 + 24H^+ \longrightarrow 3Mn^{2+} + 5Fe^{3+} + 10CO_2 + 12H_2O$
 (vii) $2S_2O_3^{2-} + I_2 \longrightarrow S_4O_6^{2-} + 2I^-$

- | | | | | | |
|-----|---|---|--------------------|------------------|-------|
| 7. | (1) 1 | (2) 1 | (3) 1 | (4) 1 | (5) 2 |
| 8. | (i) 98, (ii) 49, (iii) 32.66 or 32.67 | | | | |
| 9. | Ans. 21.00 | | | | |
| 10. | Ans. (i) 13.91 or 13.92 (ii) 5.66 or 5.67 | | | | |
| 11. | Ans. 5 | | | | |
| 12. | (i) 71.00 | (ii) 143.00 | (iii) 44.50 | (iv) 57.00 | |
| | (v) 93.66 or 93.67 | | | | |
| 13. | (i) 0.79 | (ii) $\frac{1.58}{0.93} = 1.69$ or 1.70 | (iii) 0.66 or 0.67 | (iv) 1.00 | |
| 14. | (i) 3 | (ii) 11 | (iii) 27 | (iv) 16 | |
| 15. | (i) 98.00 | (ii) 98.00 | (iii) 98.00 | (iv) 98.00 | |
| 16. | 80.00 | 17. 40.00 | 18. 40.00 | 19. 12.50 | |
| 20. | 3.00 | 21. 200.00 | 22. 35.08 or 35.09 | 23. 500.00 | |
| 24. | 0.05 | 25. 16.00 | 26. 4.54 | 27. 10.00 | |
| 28. | 2.12 | 29. 0 | 30. 160.00 | 31. 0.58 or 0.59 | |
| 32. | 4.00 | 33. 16.66 or 16.67 | 34. 3.00 | 35. 200.00 | |
| 36. | 25.00 | 37. 2.50 | 38. 0.42 | 39. 7.00 | |
| 40. | 30.00 | 41. 20.00 | 42. 11.20 | 43. 4.00 | |
| 44. | 100.00 | 45. 250.00 | | | |

EXERCISE # S-II

- | | | | | |
|----|---|------------|---------|---------|
| 1. | (a) -3, -1 | (b) -3, +3 | (c) 0 | (d) +1 |
| | (e) +3 | (f) -3 | (g) +3 | (h) +2 |
| | (i) +2 | (j) +4 | | |
| 2. | (i) $6\text{KMnO}_4(\text{aq.}) + 10\text{FeC}_2\text{O}_4(\text{aq.}) + 24\text{H}_2\text{SO}_4(\text{aq.}) \longrightarrow$
$6\text{MnSO}_4(\text{aq.}) + 5\text{Fe}_2(\text{SO}_4)_3(\text{aq.}) + 20\text{CO}_2(\text{g}) + 3\text{K}_2\text{SO}_4 + 24\text{H}_2\text{O}(\text{l})$ | | | |
| | (ii) $3\text{HNO}_2(\text{aq.}) \rightarrow \text{NO}_3^- + \text{H}^+ + 2\text{NO}(\text{g}) + \text{H}_2\text{O}$ | | | |
| | (iii) $\text{IO}_3^- + 5\text{I}^- + 6\text{H}^+ \rightarrow 3\text{I}_2 + 3\text{H}_2\text{O}$ | | | |
| | (iv) $5\text{Cu}_2\text{S} + 8\text{MnO}_4^- + 44\text{H}^+ \rightarrow 10\text{Cu}^{2+} + 5\text{SO}_2 + 8\text{Mn}^{2+} + 22\text{H}_2\text{O}$ | | | |
| | (v) $\text{HCHO}(\text{l}) + 2[\text{Ag}(\text{NH}_3)_2]^+(\text{aq.}) + 3\text{OH}^-(\text{aq.}) \rightarrow 2\text{Ag}(\text{s}) + 4\text{NH}_3(\text{aq.}) + \text{HCOO}^-(\text{aq.}) + 2\text{H}_2\text{O}(\text{l})$ | | | |
| 3. | 6 | 4. 40.00 | 5. 2.50 | 6. 3.00 |
| 7. | 1.00 | 8. 39.42 | 9. 5.00 | 10. 250 |

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	B	D	A	D	A	A	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	C	C	A	C	C	A	A	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	C	D	A	C	C	D	D	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	C	A	C	A	B	C	C	B	D

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	B	C	D	A,B,D	A,B,C	B,D	B	A
Que.	11	12	13				14	15	16	17
Ans.	B	C	A-s	B-q	C-r	D-P	C	A	A	B
Que.	18	19	20							
Ans.	C	B	A,D							

EXERCISE # J-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	2	1	1	1	1	1	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	3	4	4	85	10	50.00	12	173
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	50	226	18	316	30	13	D	4	1	24

EXERCISE # J-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	5	B	C	A,B,D	6	A	126	288.00	0.11
Que.	11	12	13							
Ans.	6	1.87 or 1.88	18.75							

Important Notes